ZOO302 – MIDTERM MERGED PPT
(LECTURE 01 TO 85)
Animal Form & Function-II

Prepared By:

All About Biotech

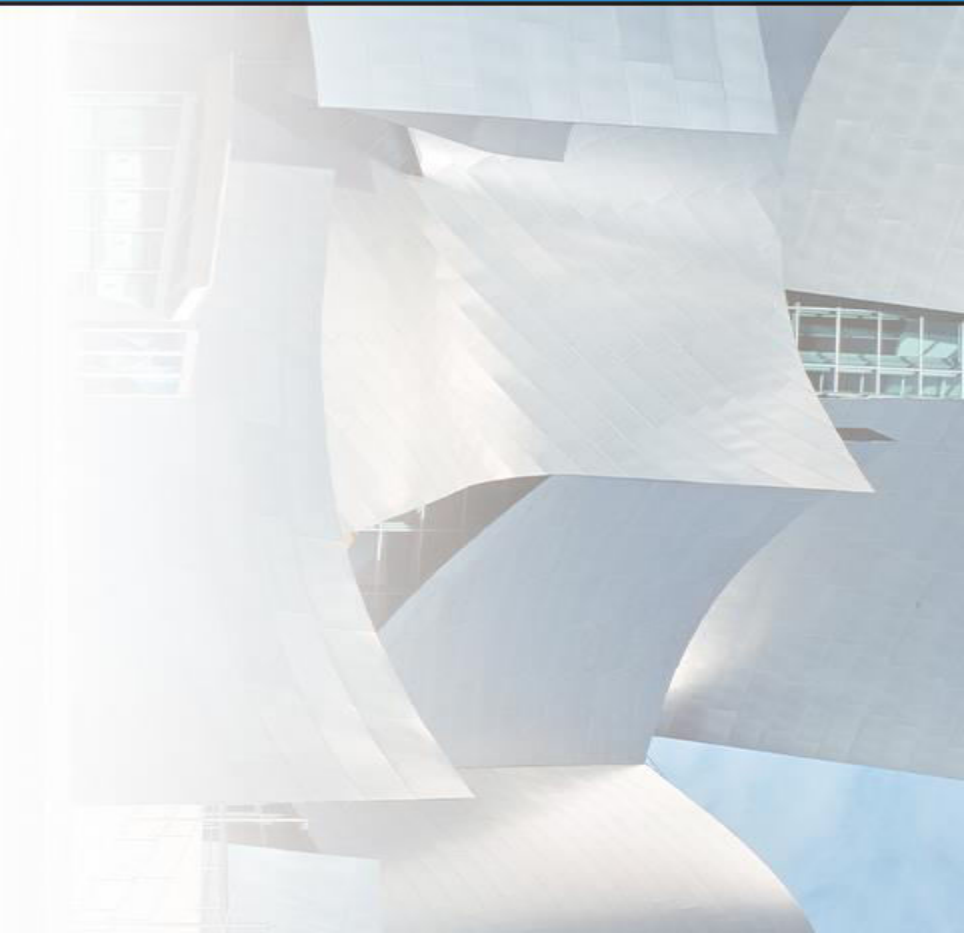
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Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

Animal Form and Function II

Module
01



Animal Form and Function I

Nutrition and Digestion

Nutrition and Digestion

- Nutrition includes all of those processes by which an animal takes in, digests, absorbs, stores, and uses food (nutrients) to meet its metabolic needs.

Nutrition and Digestion

In this chapter we shall discuss:

- Animal nutrition,
- Different strategies animals use for consuming and using food, and
- Various animal digestive systems.

Nutrition and Digestion

Nutrients:

- A nutrient is a component in foods that an organism uses to survive and grow.
- Nutrients may be organic or Inorganic.
- Organic nutrients consist of carbohydrates, fats, proteins (or their building blocks, amino acids), and vitamins.

Nutrition and Digestion

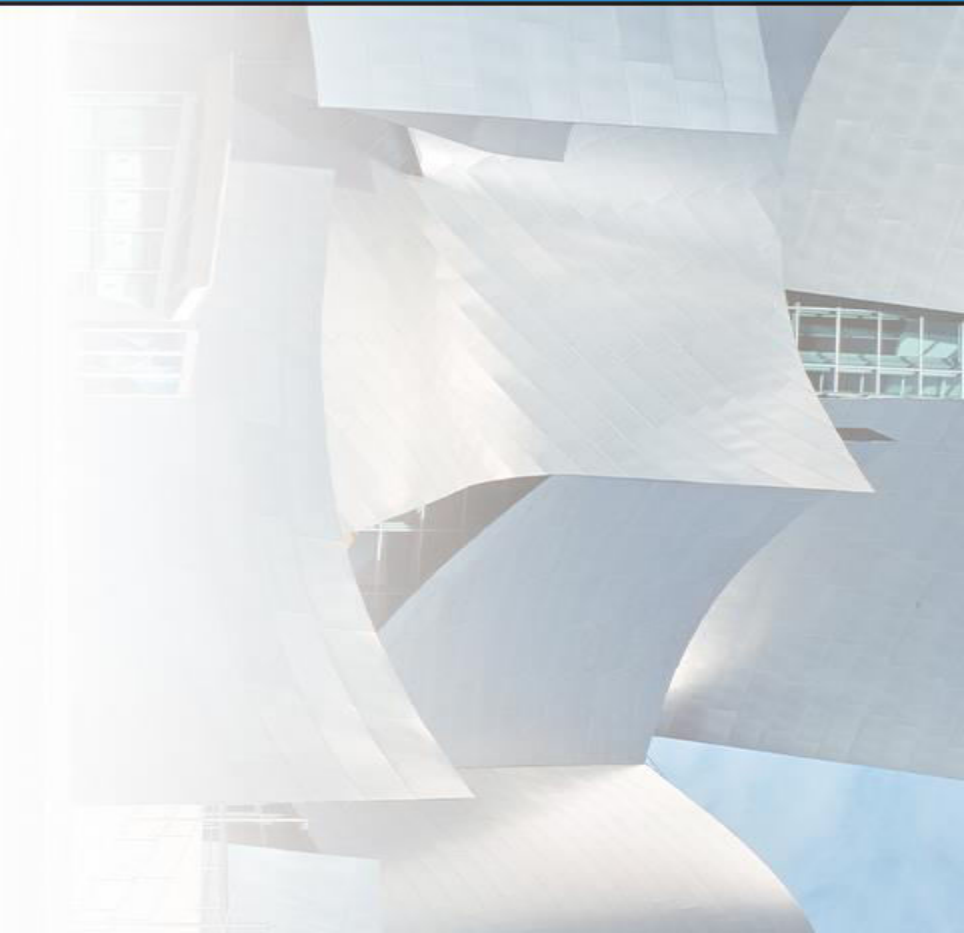
- Inorganic nutrients: These are dietary minerals, water and oxygen.

Nutrition and Digestion

- Digestion is the chemical and/or mechanical breakdown of complex food into diffusible molecules i.e. particles that individual cells of an animal can absorb.
- Starch into glucose.
- Proteins into amino acids.
- Fats into fatty acids & glycerol.

Animal Form and Function II

Module 02



Animal Form and Function 2

Evolution of Nutrition

Evolution of Nutrition

- Green plants and photosynthetic protists have the fewest such nutritional requirements because they can synthesize all their own complex molecules from simpler inorganic substances; they are autotrophs.

Evolution of Nutrition

- Animals, fungi, and bacteria that cannot synthesize many of their own organic molecules and must obtain them by consuming other organisms or their products are heterotrophs

Evolution of Nutrition

- Overall, the nutritional requirements of an animal are inversely related to its ability to synthesize molecules essential for life.
- The fewer such biosynthetic abilities an animal has, the more kinds of nutrients it must obtain from its environment.

Evolution of Nutrition





- Animals such as rabbits that survive entirely on plant material are herbivores.
- Carnivores such as hawks, are animals that eat only meat.
- Omnivores such as humans, bears, raccoons, and pigs, eat both plant and animal matter.

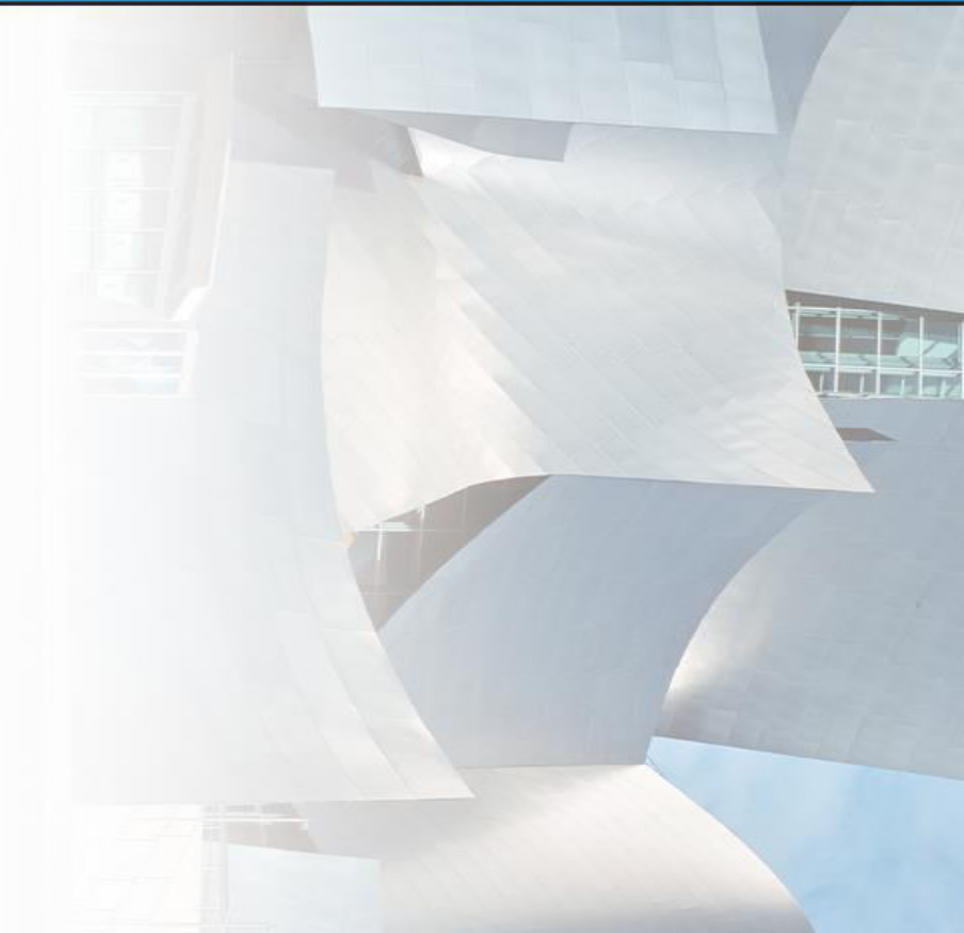
Evolution of Nutrition



- Insectivores, such as bats, eat primarily arthropods.

Animal Form and Function II

Module 03



Animal Form and Function II

Metabolic Fates of Nutrients in Heterotrophs

Metabolic Fates of Nutrients in Heterotrophs

The nutrients that a heterotroph ingests can be divided into:

- Macronutrients and
- Micronutrients.

Metabolic Fates of Nutrients in Heterotrophs

Macronutrients

Carbohydrates
Fats
proteins

- Monosaccharides e.g. *Glucose, Fructose, Galactose*
- Disaccharides e.g. *Sucrose, , Lactose*
- Polysaccharides e.g. *Starches, Glycog*en

Metabolic Fates of Nutrients in Heterotrophs

Fats:

- Fatty acids
- Oils

Metabolic Fates of Nutrients in Heterotrophs

Micronutrients

Micronutrients are needed in small quantities and include:

- Organic vitamins and
- Inorganic minerals.

Besides these nutrients, animals require water.

Animal Form and Function II



Module 04

Animal Form and Function 2

Calories and Energy

Calories and Energy

- **Calorie:** The energy value of food is measured in terms of calories or Calories.
- A calorie is the amount of energy required to raise the temperature of 1 g of water to 1° C.

Calories and Energy

- A calorie, with a small c, is also called a gram calorie.
- A kilocalorie, also known as a Calorie or kilogram calorie (kcal), is equal to 1,000 calories.

Calories and Energy

- Calories in food provide energy in the form of heat so that our bodies can function.
- Our bodies store and "burn" calories as fuel.
- Many dieters count calories and try to decrease caloric intake to lose weight.

Calories and Energy

AVERAGE CALORIC VALUES OF MACRONUTRIENTS

Macronutrients	calories per gram
Carbohydrates	4.1
Lipids	9.3
Proteins	4.3

One calorie contains 4.184 joule (SI unit of energy) International Steam Table

- *One joule of work is done on an object when a force of one newton is applied over a distance of one meter.r.*

Calories and Energy

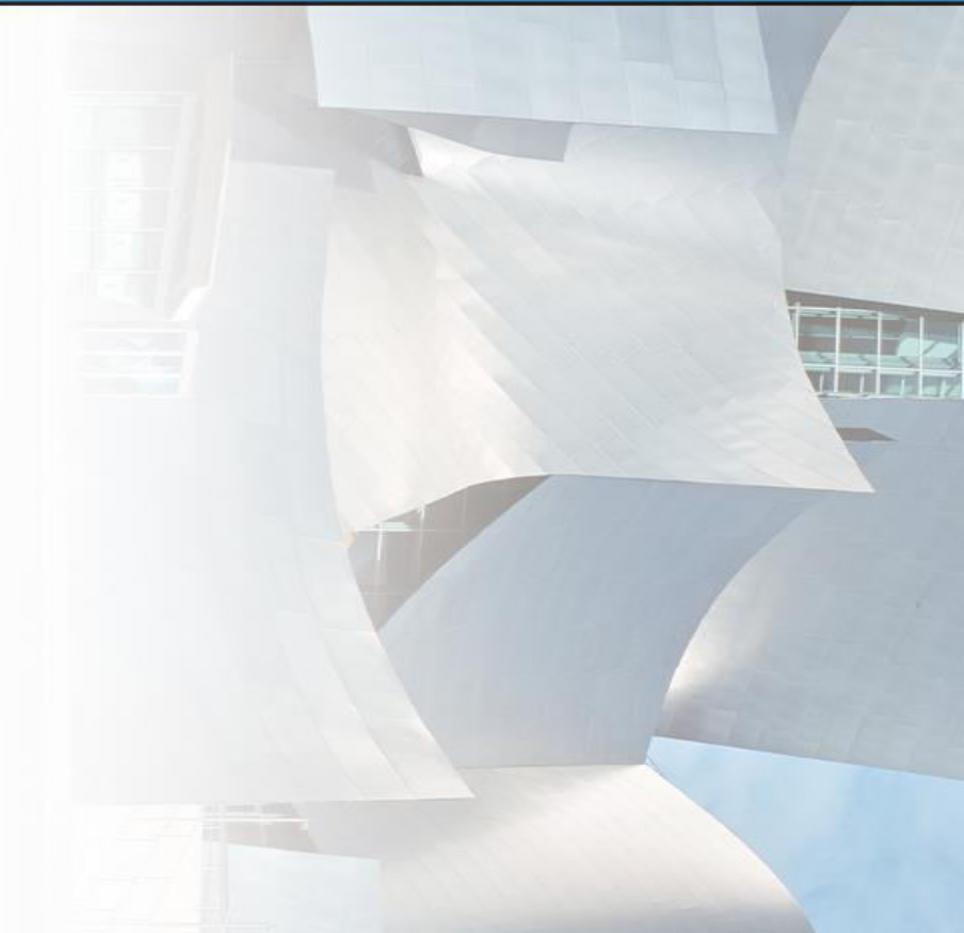




- Healthy foods that are high in calories include:
- One avocados (227 calories)
- 1 cup pea nuts (828 calories)
- 1 tablespoon of olive oil (119 calories)
- I cup of skimmed milk (86 calories).
- 1 cup low fat yogurt (225 cals)

Animal Form and Function II

Module 05



Animal Form and Function 2

Macronutrients
(Carbohydrates)

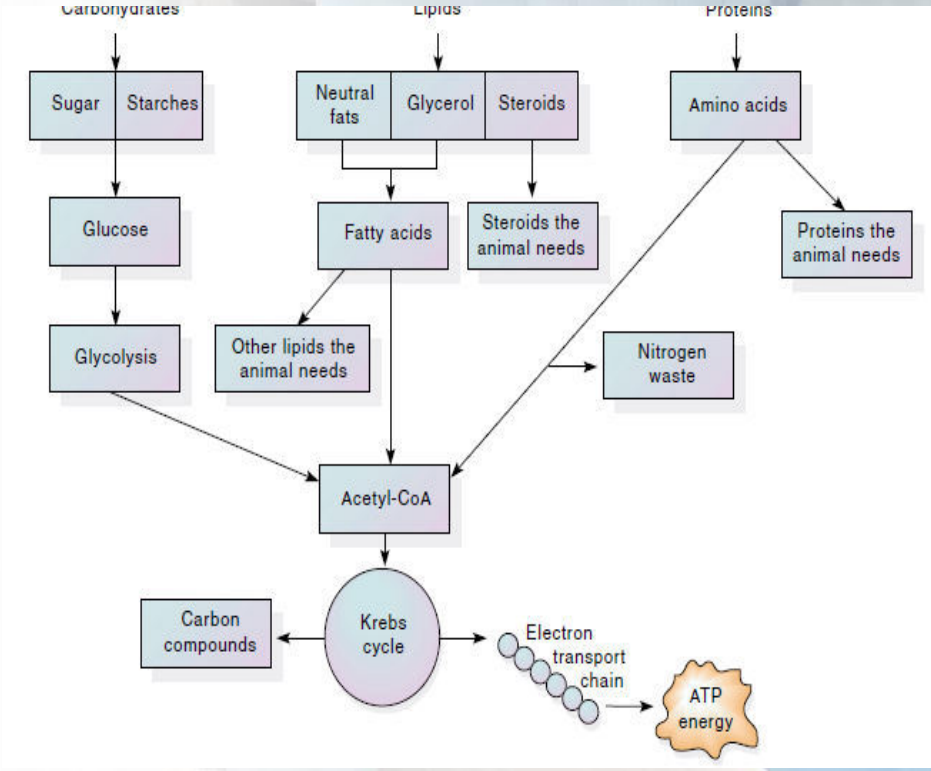
Macronutrients (Carbohydrates)

- Complex carbohydrates are the major dietary source of energy for heterotrophs.
- Most carbohydrates originally come from plant sources.
- Various polysaccharides, disaccharides, or any of monosaccharides (simple sugars) can meet this dietary need.

Macronutrients (Carbohydrates)

- Carbohydrate foods break down to their constituent sugars and starches, and ultimately into glucose.
- Individual cells use this sugar in glycolysis and aerobic respiration to create new carbon compounds or ATP energy

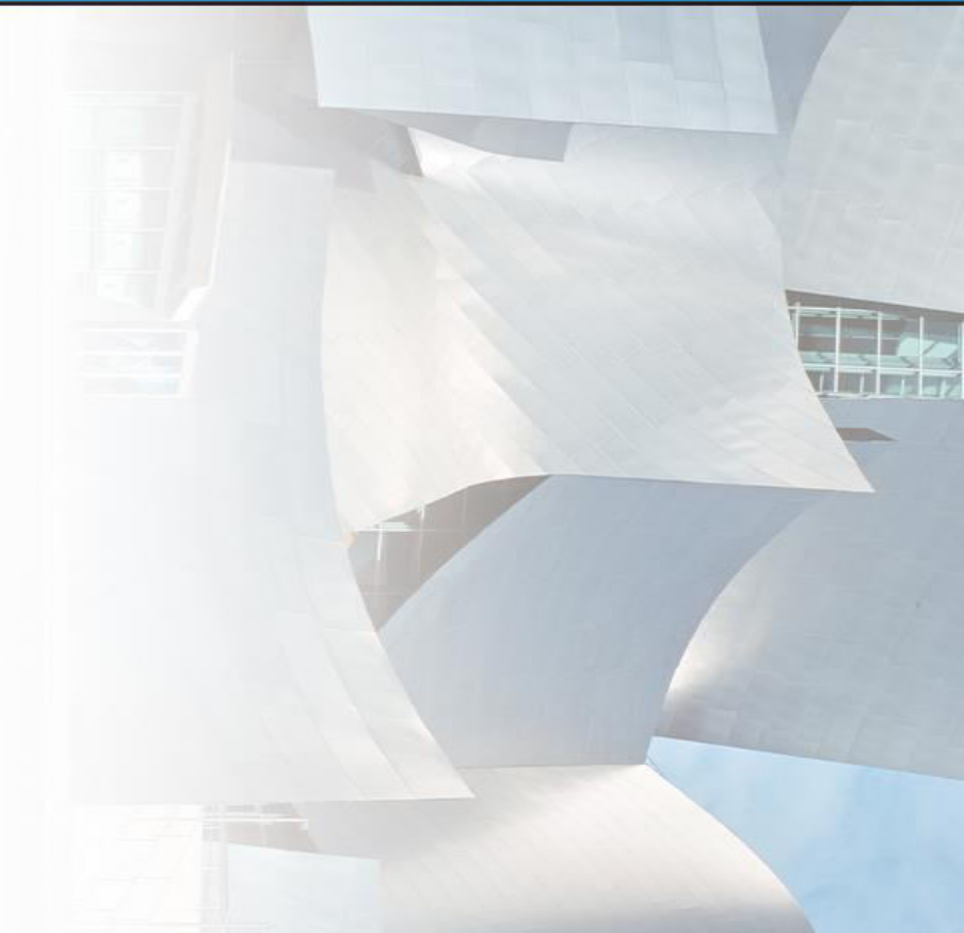
Macronutrients (Carbohydrates)



END

Animal Form and Function II

Module 6



Animal Form and Function 2

Macronutrients (Lipids)

Macromolecules (Lipids)

- **Lipids: Highly Compact Energy Storage Nutrients**
- **Neutral lipids (fats) or triacylglycerols are contained in fats and oils, meat and dairy products e.g. Nuts, Fruits and Vegetables**

Macromolecules (Lipids)

- **Lipids are the most concentrated source of food energy.**
- **They produce about 9 Calories (kcal) of usable energy per gram, more than twice the energy available from an equal mass of carbohydrate or protein.**

Macromolecules (Lipids)

- Many heterotrophs have an absolute dietary requirement for lipids, sometimes for specific types. For example, many animals require unsaturated fatty acids (e.g., linoleic acid, linolenic acid, and arachidonic acid.
- Linolenic acid $C_{18}H_{30}O_2$
- Linoleic acid $C_{18}H_{32}O_2$
- Arachidonic $C_{20}H_{32}O_2$

Macromolecules (Lipids)

- These fatty acids act as precursor molecules for the synthesis of sterols, the most common of which is cholesterol.
- The sterols are also required for the synthesis of steroid hormones and cholesterol, which is incorporated into cell membranes.

Macromolecules (Lipids)

- **Other lipids insulate the bodies of some vertebrates and help maintain a constant temperature.**

Animal Form and Function II



Module 07

Animal Form and Function 2

Macromolecules (Proteins)

Macromolecules (Proteins)

- **Proteins: Basic to the Structure and Function of Cells.**
- **Animal sources of protein include other animals and milk.**
- **Plant sources include beans, peas, and nuts.**

Macromolecules (Proteins)

- **Proteins are needed for their amino acids, which heterotrophs use to build their own body proteins.**

Macromolecules (Proteins)

An overview of Protein Functions

- **Structural proteins**
  Support
  e.g. Collagen, Keratin
- **Storage Proteins**
  Storage of amino acids
  e.g. Albumin, Casein
- **Transport proteins**
  Transport of other proteins
  e.g. Hemoglobin,

Macromolecules (Proteins)

- **Hormonal proteins**
  Coordination of activities
  e.g. Insulin
- **Contractile Proteins**
  Movement
  e.g. Actin and myosin
- **Defensive proteins**
  Protection against disease
  e.g. Antibodies
-

Macromolecules (Proteins)

- **Enzymatic**
  Acceleration of chemical reactions.
  e.g. Pepsin, trypsin,

Animal Form and Function II



Module 8

Animal Form and Function 2

Micronutrients (Part I)

Micronutrients (Part I)

Micronutrients are usually:

- **i) Inorganic minerals.**
- **ii) Vitamins,**
- **They are needed in small amounts i.e. in milligrams.**
- **Animals cannot synthesize them rapidly, thus, they must be obtained from the diet.**

Micronutrients (Part I)

- Inorganic Minerals:
- i) Macrominerals
- ii) Micromineals
- Minerals needed in relatively large amounts as compared to microminerals are called macrominerals or essential minerals.

Micronutrients (Part I)

Macrominerals:

- **Examples are:**
  Calcium,
  Chlorine,
  Magnesium,
  Potassium,
  Phosphorus,
  Sodium,
  Sulfur etc.

Micronutrients (Part I)

- **Calcium**
- **Most adults need about 1,000 milligrams of calcium per day.**
- **Ca is the chief component of bone and teeth; essential for normal blood clotting; needed for normal muscle, neuron, and cellular functioning.**

Micronutrients (Part I)

- **Chlorine:**
- **Cl is the principal negative ion in extracellular fluid; important in acid-base and fluid balance; needed to produce stomach HCl.**

Micronutrients (Part I)

- **Magnesium**
- **Mg is the component of many coenzymes; needed for normal neuron and muscle function. as well as carbohydrate and protein metabolism .**

Micronutrients (Part I)

- **Potassium (K)**
- **Major constituent of bones, blood plasma; needed for energy metabolism.**
- **Phosphorus (P)**
- **Major positive ion in cells; influences muscle contraction and neuron excitability; part of DNA, RNA, ATP, energy metabolism.**

Micronutrients (Part I)

- **Sodium (Na)**
- **Principal positive ion in extracellular fluid; important in fluid balance; essential for conduction of action potentials, active transport.**
- **Sulfur (S)**
- **Protein structure; detoxification reactions and other metabolic activity.**

Micronutrients (Part I)

- Animals excrete large quantities of these minerals, especially sodium, in the urine every day.
- Animals that sweat to help regulate body temperature lose sodium in their sweat.

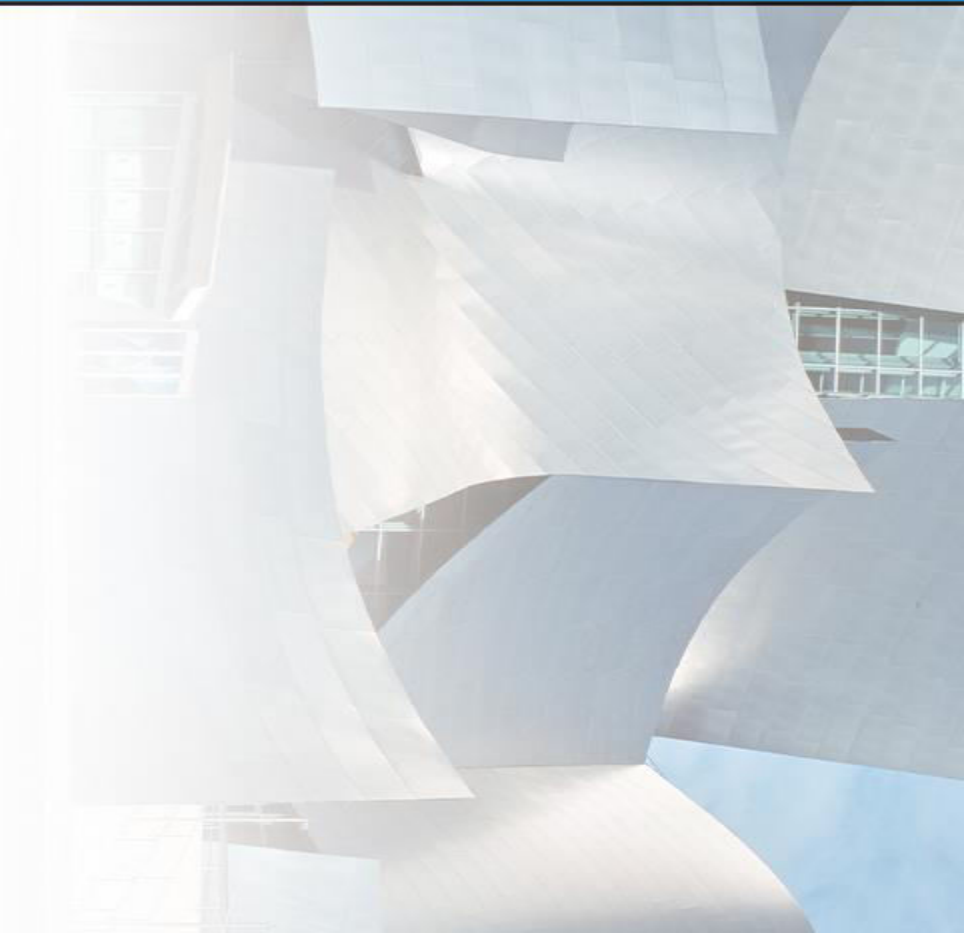
Micronutrients (Part I)

PHYSIOLOGICAL ROLES OF THE ESSENTIAL MINERALS (MACROMINERALS) ANIMALS REQUIRE IN LARGE AMOUNTS

MINERAL	MAJOR PHYSIOLOGICAL ROLES
Calcium (Ca)	Component of bone and teeth; essential for normal blood clotting; needed for normal muscle, neuron, and cellular function
Chlorine (Cl)	Principal negative ion in extracellular fluid; important in acid-base and fluid balance; needed to produce stomach HCl
Magnesium (Mg)	Component of many coenzymes; needed for normal neuron and muscle function, as well as carbohydrate and protein metabolism
Potassium (K)	Major constituent of bones, blood plasma; needed for energy metabolism
Phosphorus (P)	Major positive ion in cells; influences muscle contraction and neuron excitability; part of DNA, RNA, ATP, energy metabolism
Sodium (Na)	Principal positive ion in extracellular fluid; important in fluid balance; essential for conduction of action potentials, active transport
Sulfur (S)	Protein structure; detoxification reactions and other metabolic activity

Animal Form and Function II

Module 9



Animal Form and Function 2

Micronutrients (Part II)

Micronutrients (Part II)

Micronutrients

- **i) Inorganic minerals.**
- **ii) Vitamins,**

Micronutrients

- **needed in small amounts i.e. in micro milligrams also called trace minerals.**
- **Like macronutrients animals cannot synthesize them thus, they also must be obtained from the diet.**

Micronutrients (Part II)

Micronutrients:

- Copper
- Fluorine
- Cobalt
- Iodine
- Iron
- Manganese
- Molybdinum
- Selenium
- Zinc

Micronutrients (Part II)

Examples of Trace Minerals:

- **Copper (Cu)**
- **Component of many enzymes; essential for melanin and hemoglobin synthesis; part of cytochromes.**
- **Fluorine (F)**
- **Component of bone and teeth; prevents tooth decay.**

Micronutrients (Part II)

- Cobalt (Co)
- Component of vitamin B12; essential for red blood cell production.
- Iodine (I)
- Component of thyroid hormones
- Iron (Fe)
- Component of hemoglobin, myoglobin, enzymes, and cytochromes.

Micronutrients (Part II)

Manganese (Mn)

- Activates many enzymes; an enzyme essential for urea formation and parts of the Krebs cycle

Molybdenum (Mo)

- Constituent of some enzymes.

Selenium (Se)

- Needed in fat metabolism

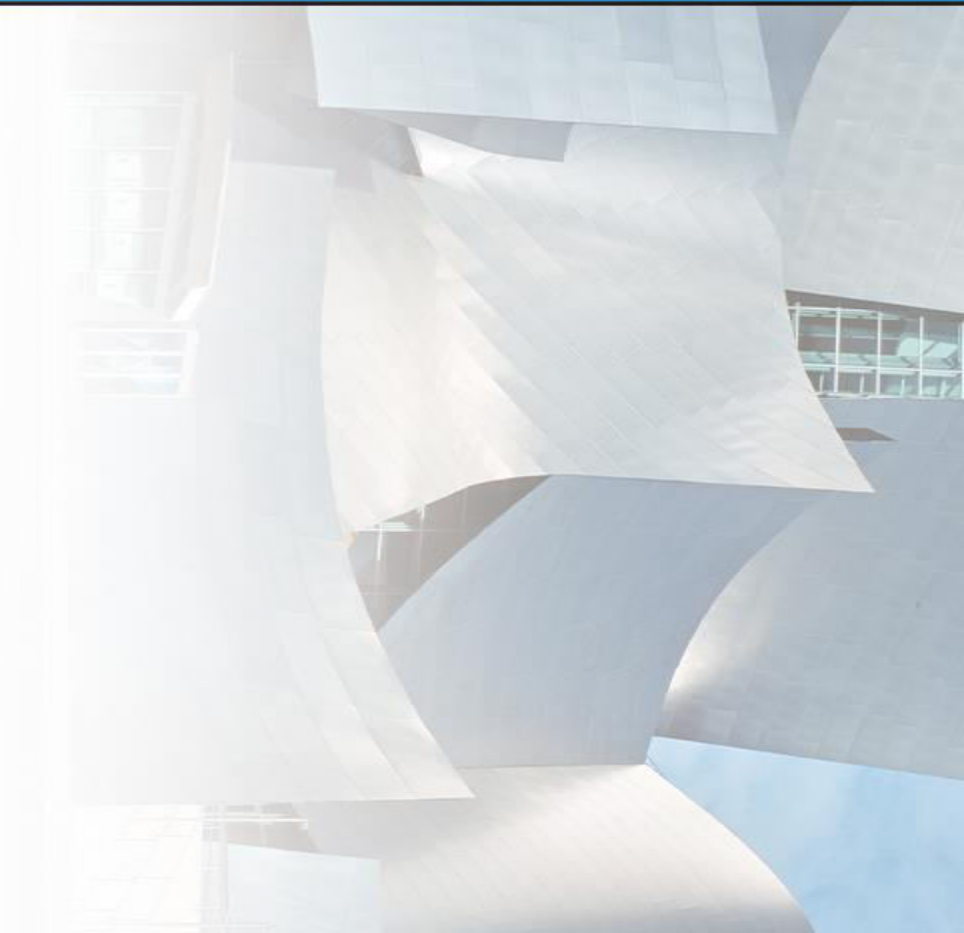
Micronutrients (Part II)

Zinc (Zn)

- Component of at least 70 enzymes.
- Needed for wound healing and fertilization.

Animal Form and Function II

Module 10



Animal Form and Function 2

Vitamins
(Part I)

Vitamins (Part I)

Vitamins are the organic substances that occur in many foods in small amounts and are necessary for normal metabolic functioning.

- Vitamins may be:
- i) water soluble or
- ii) fat soluble.
- Most water-soluble vitamins, are B vitamins and vitamin C.

Vitamins (Part I)

- Fat soluble vitamins include vitamins A, D, E and K

Water Soluble B-Vitamins

- Vitamin B1 (thiamine)
- Vitamin B2 (riboflavin)
- Vitamin B3 (niacin)
- Vitamin B5 (pantothenic acid)
- Vitamin B6 (pyridoxine
- Vitamin B7 (biotin)
- Vitamin B9 (folic acid)
- Vitamin B12 (cobalamin)

Vitamins (Part I)

VITAMIN A

Characteristics:

- Occurs in several forms; synthesized from carotenes.
- Stored in liver.
- Stable in heat, acids, and alkalis;
- Unstable in light.

Vitamins (Part I)

Functions:

- Necessary for synthesis of visual pigments, mucoproteins, and mucopolysaccharides
- Necessary for normal development of bones and teeth.
- Necessary for the maintenance of epithelial cells.

Vitamins (Part I)

Source

- **Liver, fish, whole milk, butter, eggs, leafy green vegetables, and yellow and orange vegetables and fruits.**

Vitamins (Part I)

VITAMIN D

Characteristics:

- A group of sterols;
- resistant to heat, oxidation, acids and alkalis.
- stored in liver, skin, brain, spleen and bone.

Functions:

- Promotes absorption of Ca and P.
- Promotes development of teeth and bones.

Vitamins (Part I)

Sources

- Produced in skin exposed to ultraviolet light.
- Milk, fish-liver oils, egg yolk, fortified foods (enriched food)

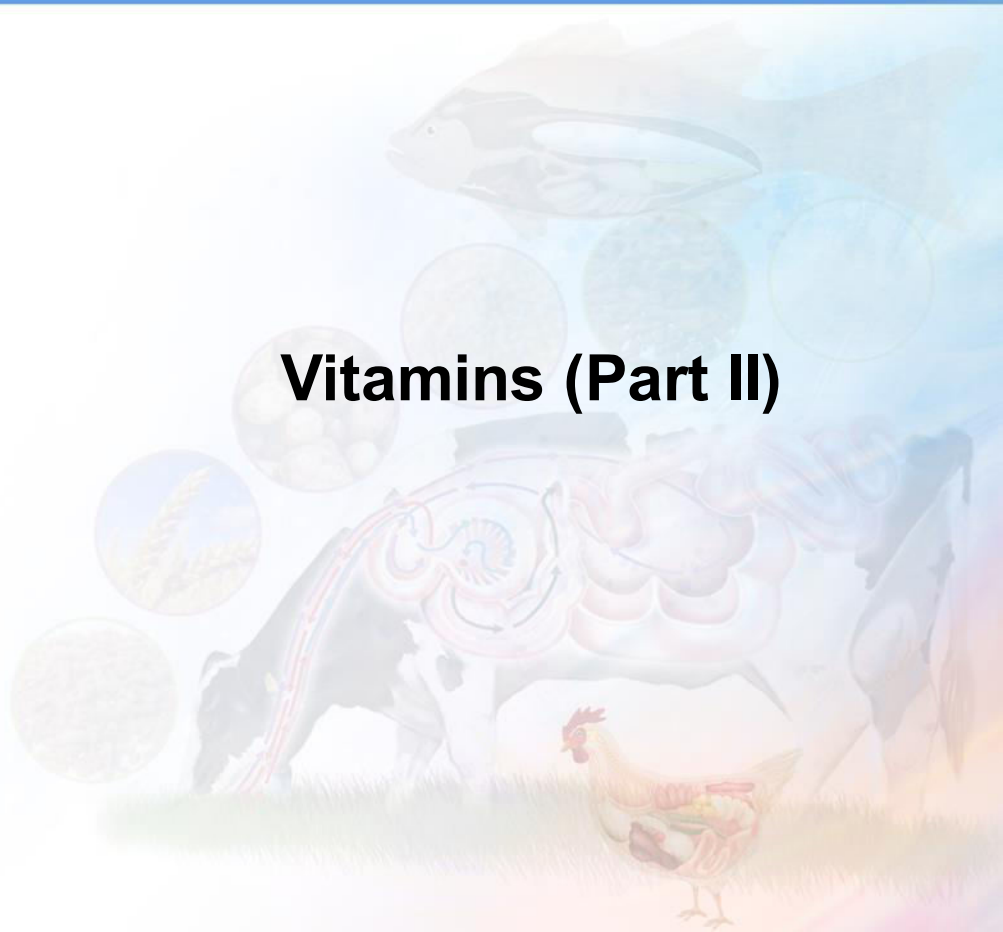
Animal Form & Function II

Module 21



Animal Form & Function II

Vitamins (Part II)

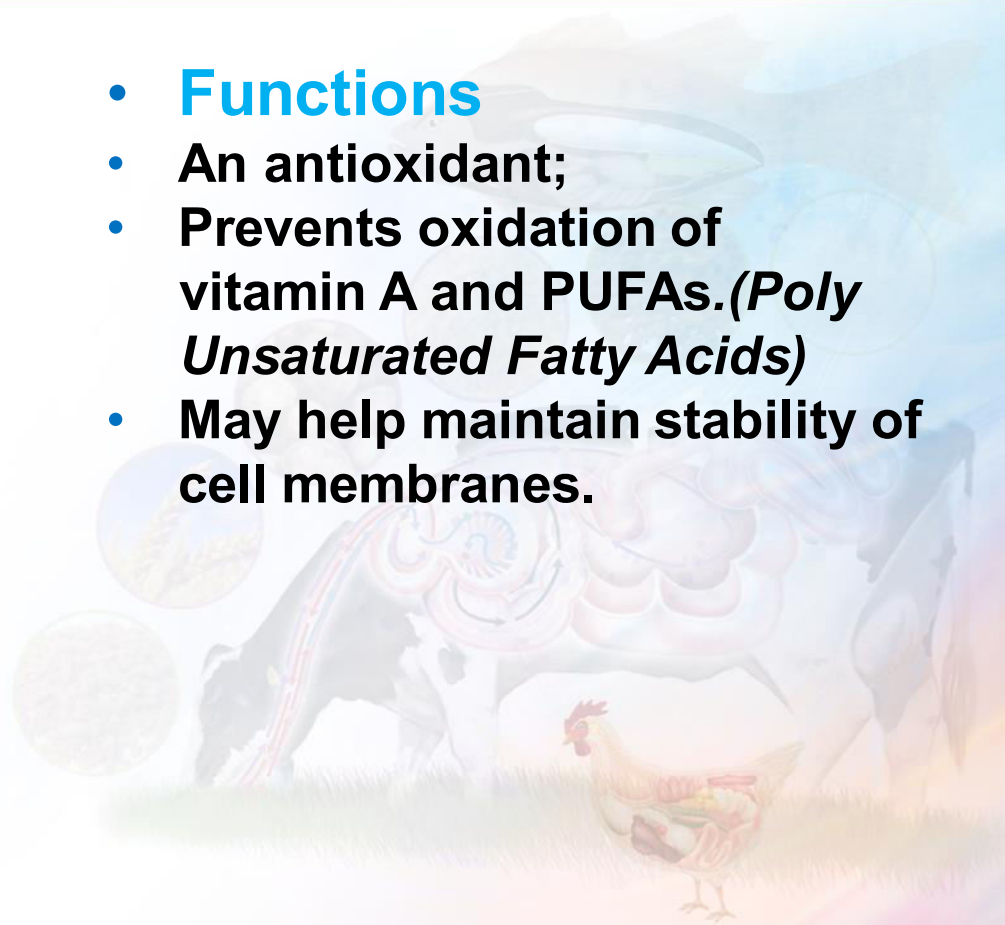


Vitamins - Part II

- **Vitamin E**
- **Characteristics**
- **A group of compounds; resistant to heat and visible light;**
- **Unstable in presence of oxygen and UV light**
- **Stored in muscles and adipose tissue.**

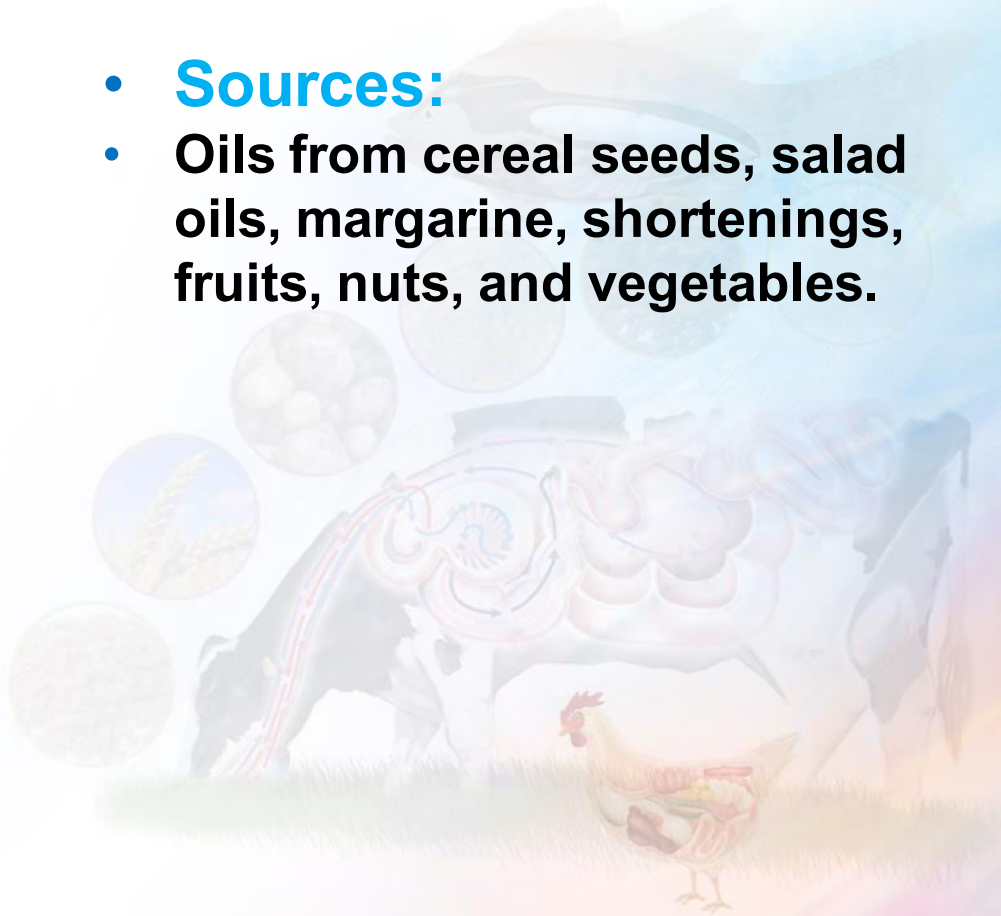
Vitamins - Part II

- **Functions**
- **An antioxidant;**
- **Prevents oxidation of vitamin A and PUFAs.*(Poly Unsaturated Fatty Acids)***
- **May help maintain stability of cell membranes.**



Vitamins - Part II

- **Sources:**
- **Oils from cereal seeds, salad oils, margarine, shortenings, fruits, nuts, and vegetables.**



Vitamins - Part II

Vitamin K

Characteristics

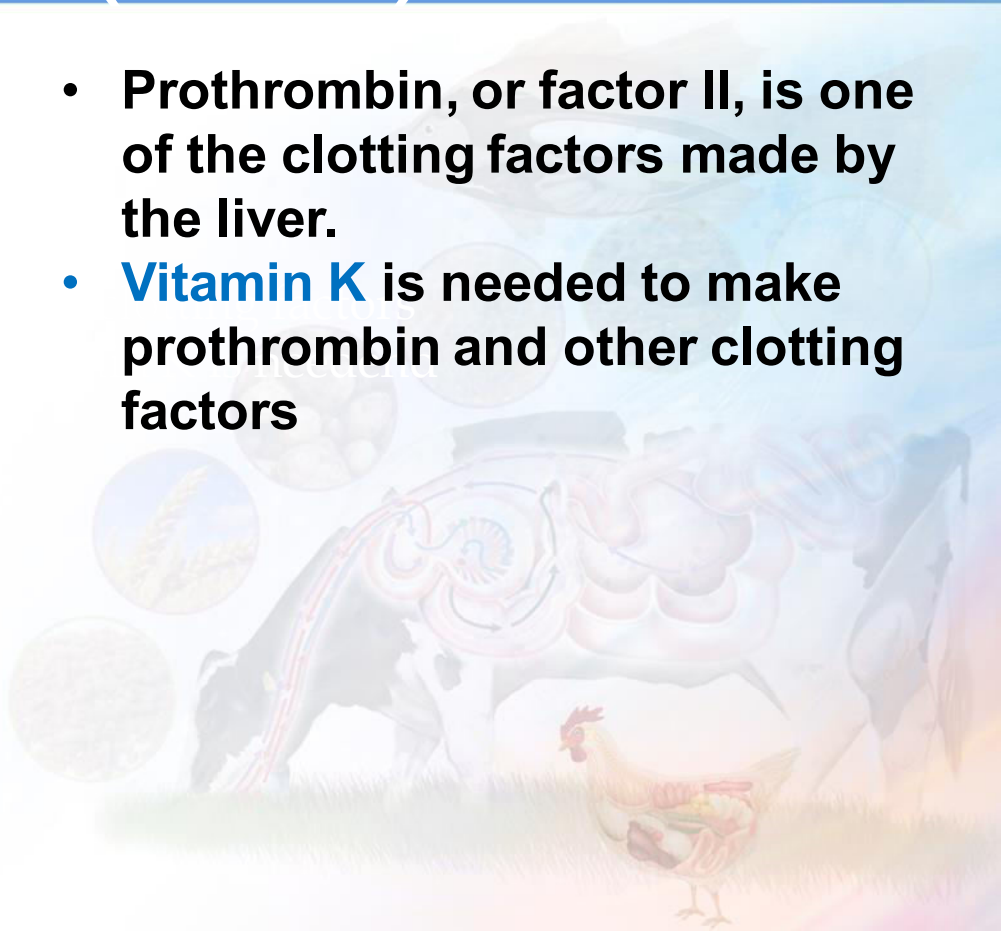
- Occurs in several forms;
- Resistant to heat,
- Destroys by acids, alkalis and light.
- Stored in liver.

Functions

- Needed for the synthesis of prothrombin.
- Needed for blood clotting

Vitamins - Part II

- Prothrombin, or factor II, is one of the clotting factors made by the liver.
- Vitamin K is needed to make prothrombin and other clotting factors



Vitamins - Part II



END

Animal Form and Function II

Module 12



Animal form and Function II



Vitamins Part III

Vitamins Part III

Water Soluble Vitamins

Vitamin B1 (thiamin)

Characteristics:

- Destroyed by heat and oxygen especially in alkaline environment

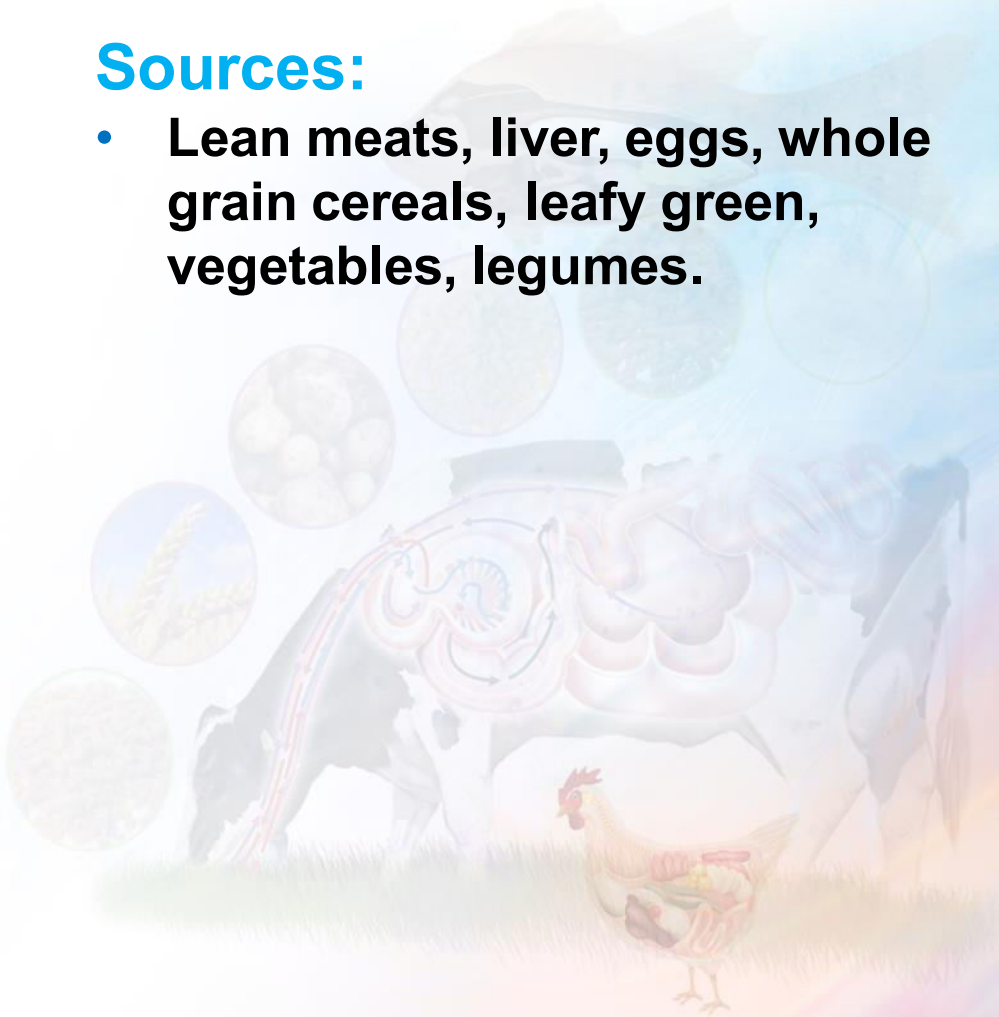
Functions:

- Part of coenzyme needed for oxidation of carbohydrates and in synthesis of ribose.

Vitamins Part III

Sources:

- **Lean meats, liver, eggs, whole grain cereals, leafy green, vegetables, legumes.**



Vitamins Part III

Vitamin B2 (Riboflavin)

Characteristics:

- Stable to heat, acids, and oxidation; destroyed by alkalis and light.

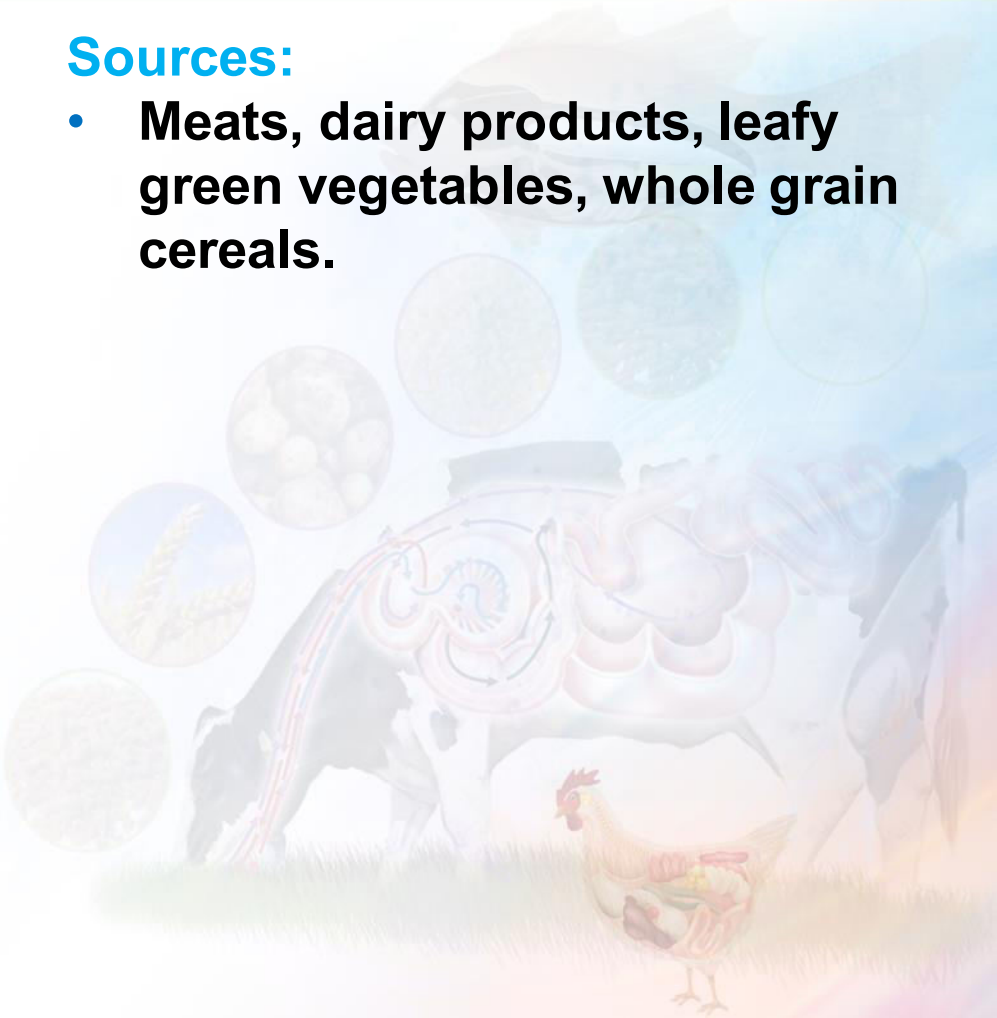
Functions:

- Part of enzymes and coenzymes.
- Needed for oxidation of glucose and fatty acids and for cellular growth.

Vitamins Part III

Sources:

- **Meats, dairy products, leafy green vegetables, whole grain cereals.**



Vitamins Part III

Vitamin B3 (Niacin)

Characteristics:

- Stable to heat, acids, and alkalis;
- Converted to niacinamide by cells;
- Synthesized from tryptophan.

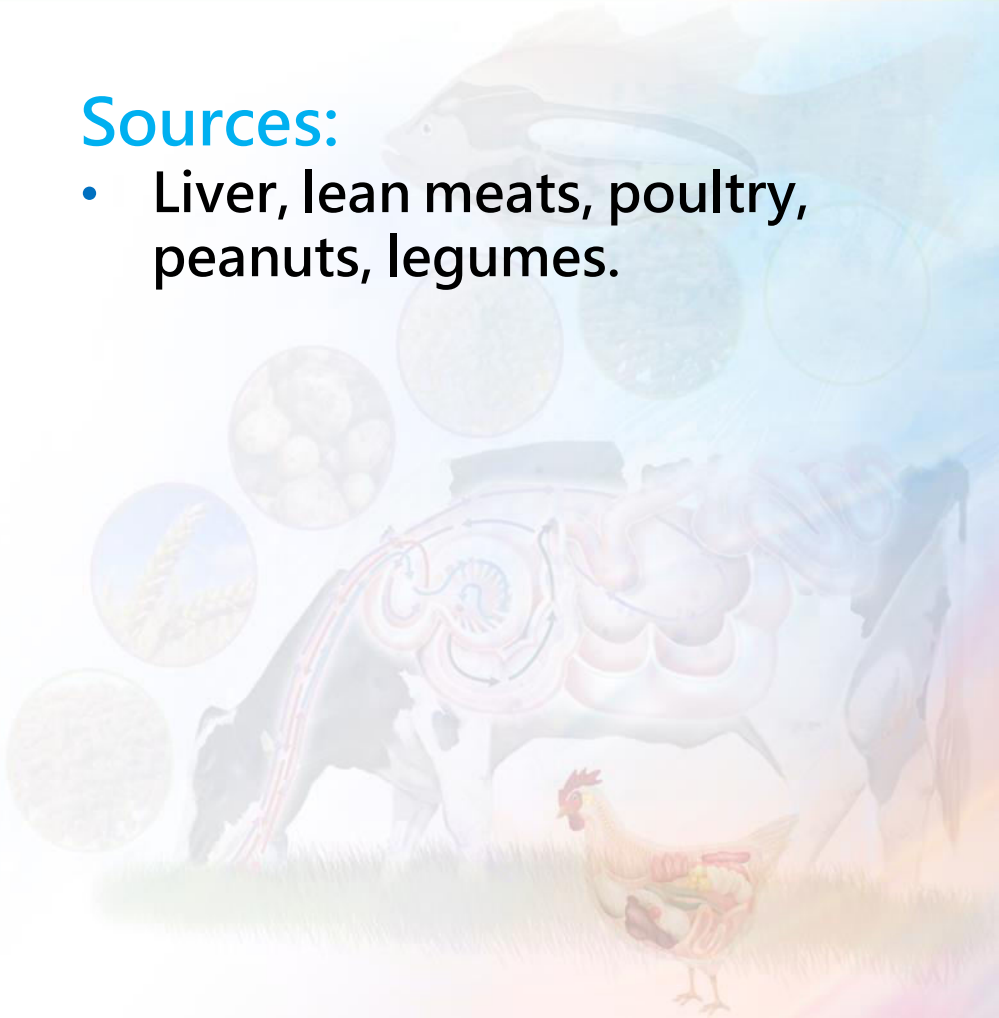
Functions:

- Part of coenzymes needed for oxidation of glucose and synthesis of
- proteins, fats, and nucleic acids.

Vitamins Part III

Sources:

- Liver, lean meats, poultry, peanuts, legumes.



Vitamins Part III

Vitamin B5 (Pantothenic acid)

Characteristics:

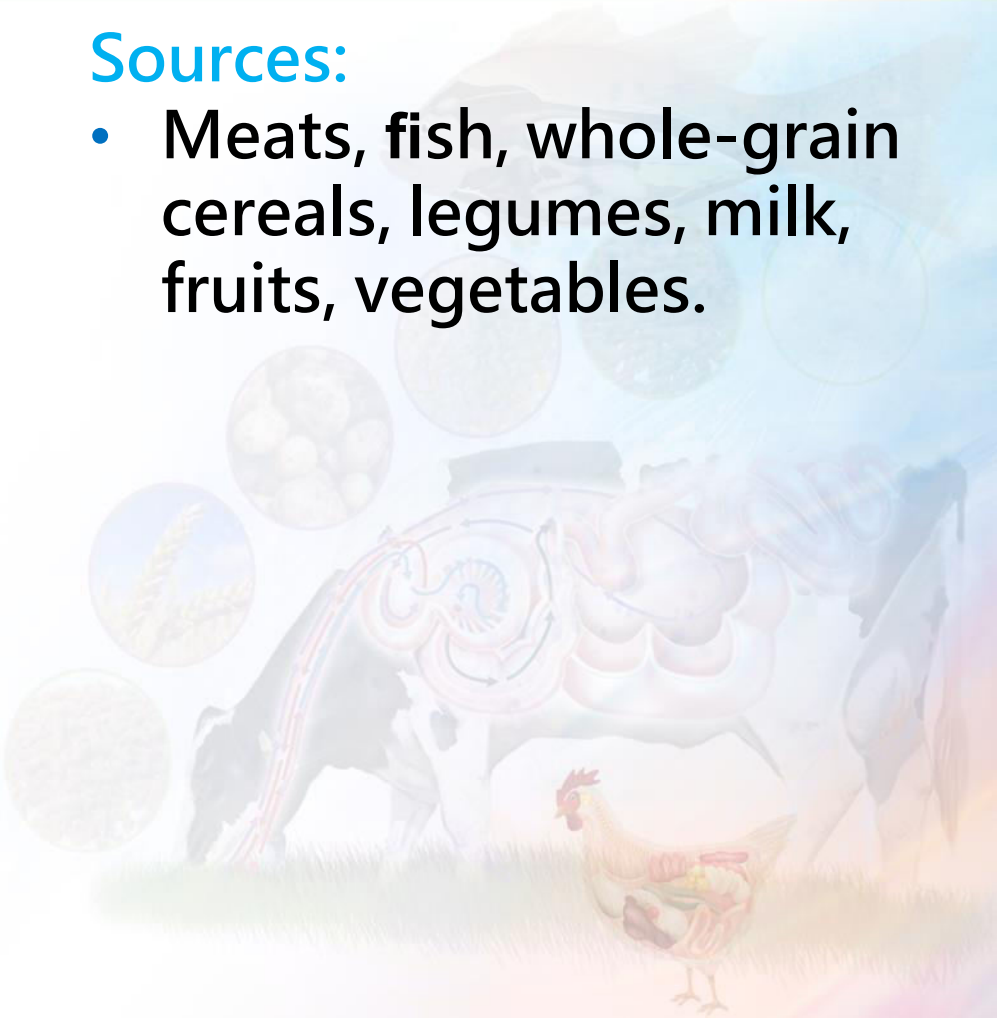
- Destroyed by heat, acids, and alkalis.

Functions: Part of coenzyme needed for oxidation of carbohydrates and fats.

Vitamins Part III

Sources:

- Meats, fish, whole-grain cereals, legumes, milk, fruits, vegetables.



Vitamins Part III

Vitamin B7 (Biotin)

Characteristics:

- Stable to heat, acids, and light destroyed by oxidation and alkalis.

Functions:

- Coenzyme needed for metabolism of amino acids and fatty acids.
- Also important for the synthesis of nucleic acid.

Vitamins Part III

Sources:

- Liver, egg yolk, nuts, legumes, mushrooms.

Deficiency:

- No appetite, anemia, scaly skin, weight loss, hair loss.

Vitamins Part III

Vitamin B9 (Folic acid)

Characteristics:

- Destroyed by oxidation in acid or by heat in alkaline environment.
- Stored in liver.

Resource:

- Liver, leafy green vegetables, whole-grain cereals, legume.

Vitamins Part III

Functions:

- Coenzyme needed for metabolism of certain amino acids and for the synthesis of DNA
- Promotes production of normal RBCs.

Deficiency:

- Anemia, diarrhea, miscarriage, birth defects,

Vitamins Part III

Vitamin B12 (Cyanocobalamine)

Characteristics:

- Complex, cobalt-containing compound.
- Stable to heat.
- Inactivated by light, strong acids and strong alkalis.
- Absorption regulated by intrinsic factor from gastric glands.
- Stored in liver.

Vitamins Part III

Functions

- Part of coenzyme needed for the synthesis of Nucleic acids and for the metabolism of carbohydrates.
- Plays role in synthesis of myelin.

Source

- Liver, meats, poultry, fish, milk, chees, eggs.

Deficiency:

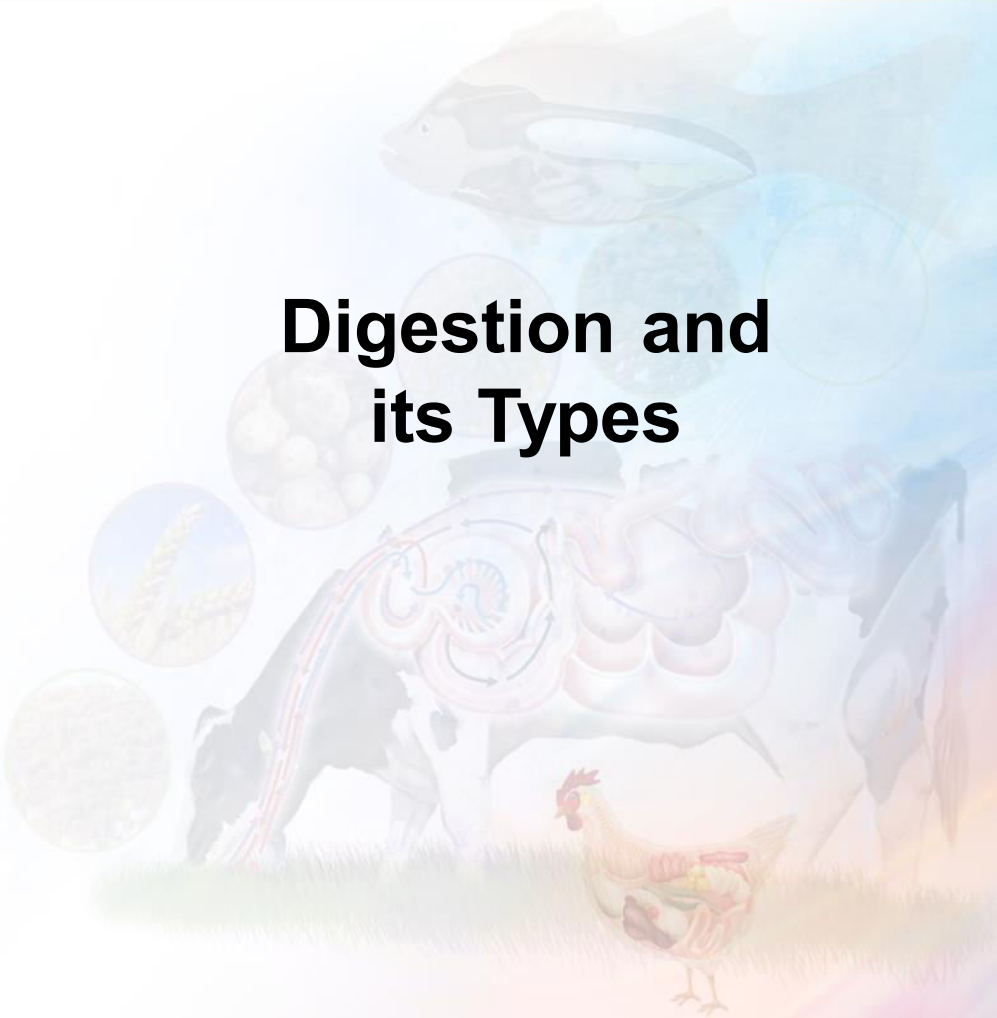
- Anemia, degenerative changes in NS.

Module 13



Animal Form & Function II

Digestion and its Types



Digestion and Its Types

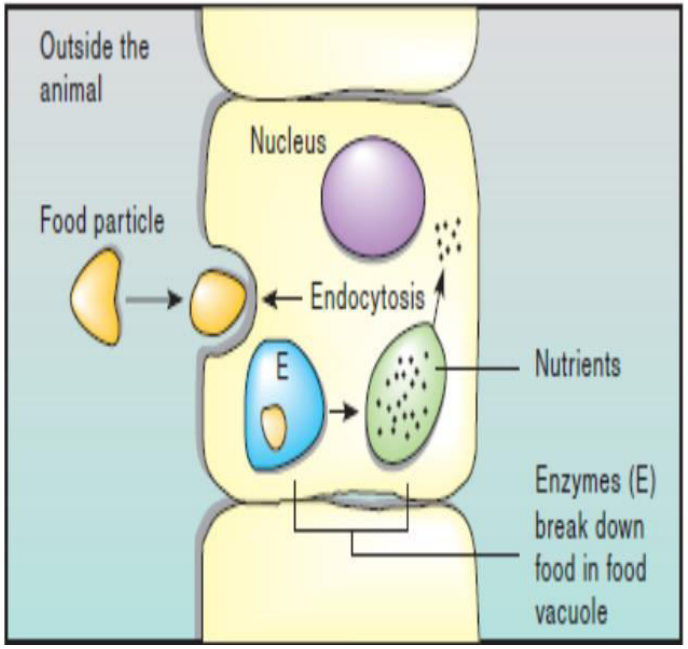
Digestion:

- The breakdown of complex organic compounds of food into simpler diffusible molecules by the action of enzymes. i.e.
- Proteins into amino acids.
- Carbohydrates into simple sugars.
- Fats into fatty acids and glycerol.

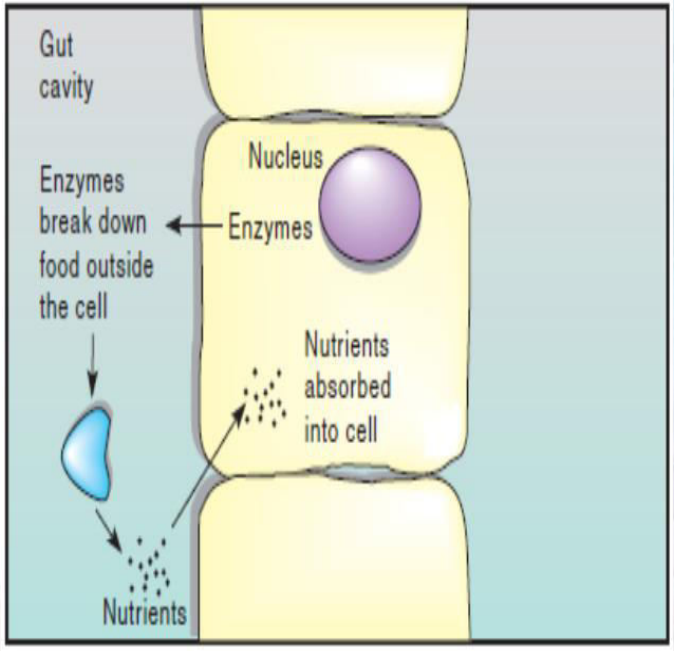
Digestion and Its Types

- **Digestion is of two main types:**
  Intracellular and Extracellular
  In intracellular digestion breakdown of food occurs within the cells.
- **First food in taken into the cell vacuole.**
- **The cytoplasm secretes digestive enzymes into the food vacuole.**

Digestion and Its Types



Intracellular Digestion



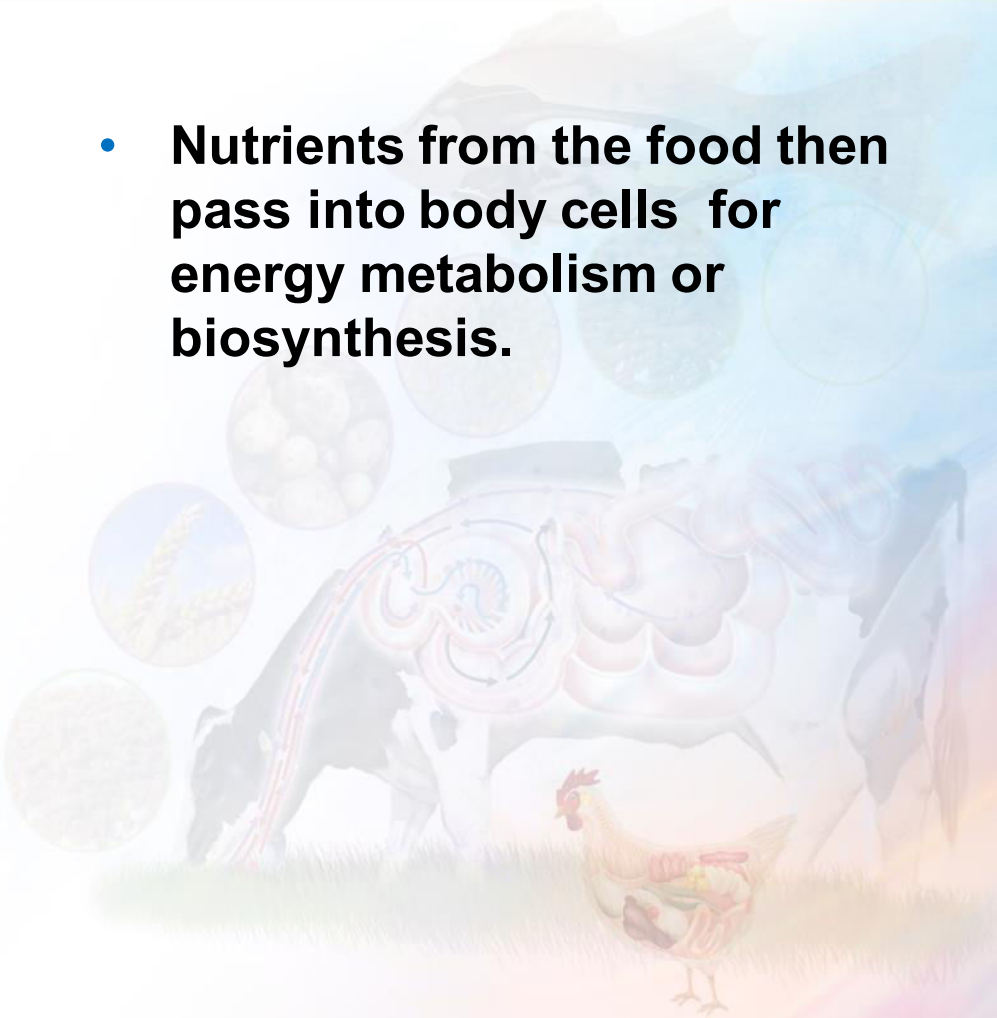
Extracellular Digestion

Digestion and Its Types

- Enzymes break larger molecules into smaller ones, which enter the cytoplasm.
- In extra cellular digestion, the enzymes are secreted outside the cell into the gut cavity or lumen where digestion occurs.

Digestion and Its Types

- Nutrients from the food then pass into body cells for energy metabolism or biosynthesis.



Animal Form & Function II

Module 14



Animal Form & Function II

Animal Strategies for Getting and Using Food



Animal Strategies for Getting and Using Food

Mostly protists and a few animals can absorb nutrients directly from their external environment (i.e. intracellular digestion).

Most animals work for their nutrients.

. Following are the specializations that have evolved for feeding and extracellular digestion.

Animal Strategies for Getting and Using Food

Various Specialization for Getting Food

1. Continuous feeders
2. Discontinuous feeders
3. Suspension feeders
4. Deposit feeders
5. Herbivory
6. Predation
7. Surface Nutrient Absorption
8. Fluid feeders

Animal Strategies for Getting and Using Food

Continuous and Discontinuous Feeders

- Continuous feeders are slow-moving or completely sessile animals.
- For example,
- Tube worms and barnacles, that remain in one place and continuously “strain” small food particles from the water.

Animal Strategies for Getting and Using Food

Discontinuous feeders:

i) Carnivores

ii) Herbivores

- They tend to be active, sometimes highly mobile, animals.
- Typically, discontinuous feeders have more specialized digestive systems than continuous feeders.
- This is because discontinuous feeders take in large meals.

Animal Strategies for Getting and Using Food

- For example, Lion, Tigers, cat, dogs
- They capture relatively large prey.
- Spend their time in the continuous pursuit of prey.
- Thus, carnivores have digestive systems that permit the storage and gradual digestion of large, relatively infrequent meals.

Animal Strategies for Getting and Using Food

Herbivores:

- They spend more time eating than carnivores do.
- They need to move from area to area.
- When food is exhausted in natural environments, they limit their grazing time to avoid excessive exposure to predators.



Animal Strategies for Getting and Using Food

- **Thus, their digestive systems permit relatively rapid food gathering and gradual digestion.**



Animal Form & Function II

Module 15



Animal Form & Function II

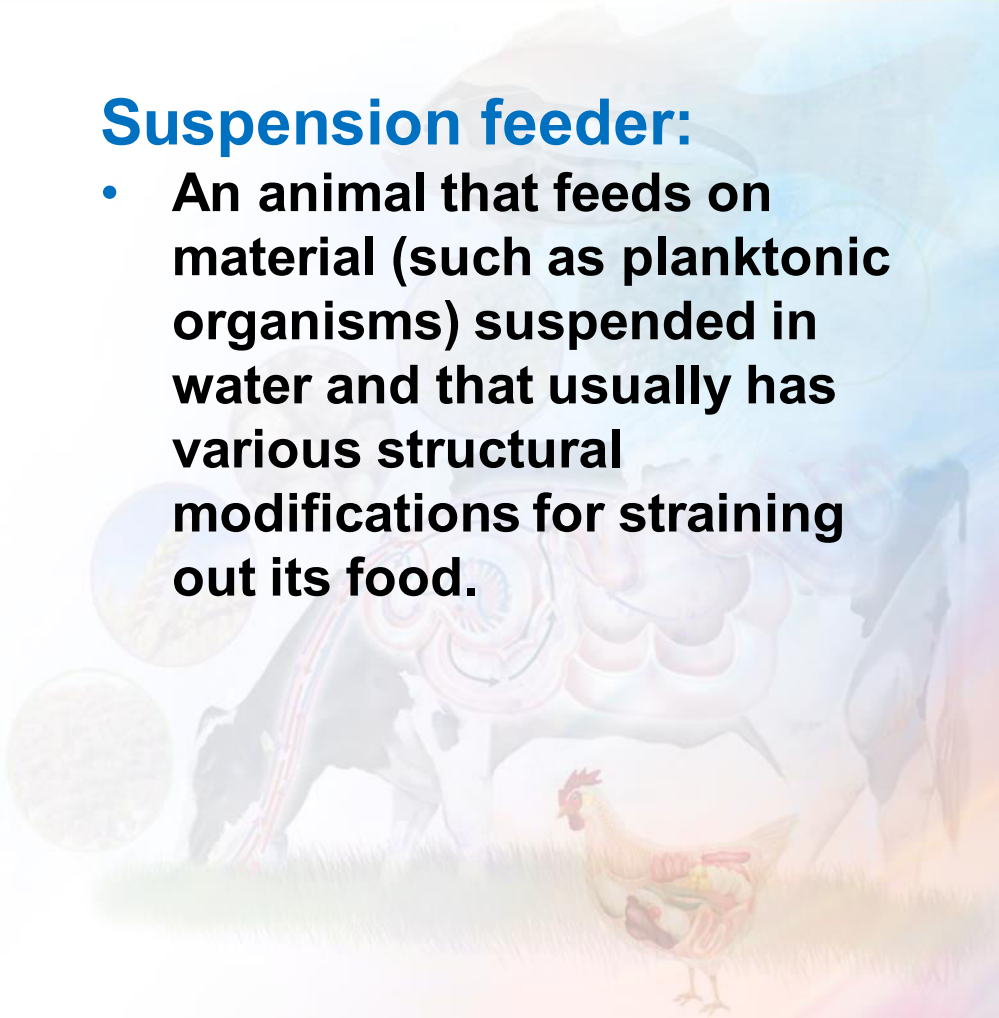
Suspension Feeders



Suspension Feeders

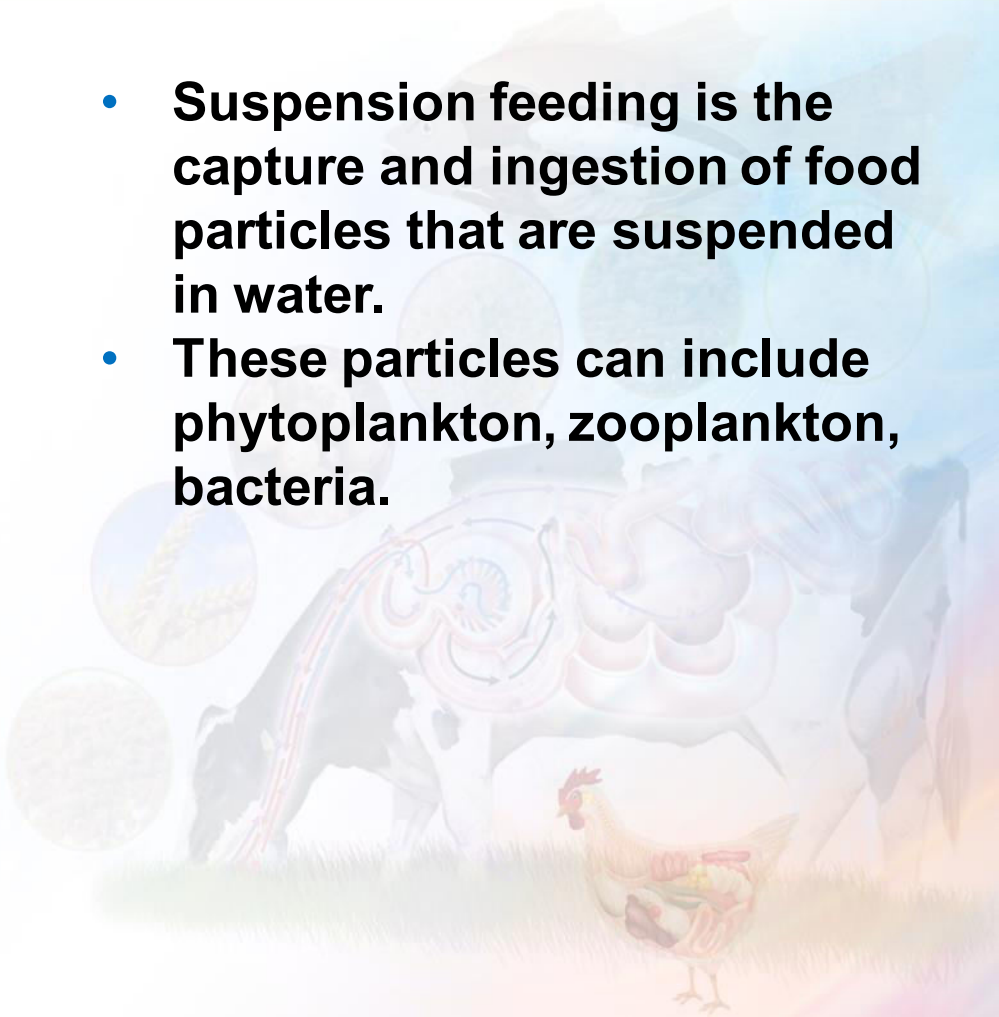
Suspension feeder:

- An animal that feeds on material (such as planktonic organisms) suspended in water and that usually has various structural modifications for straining out its food.



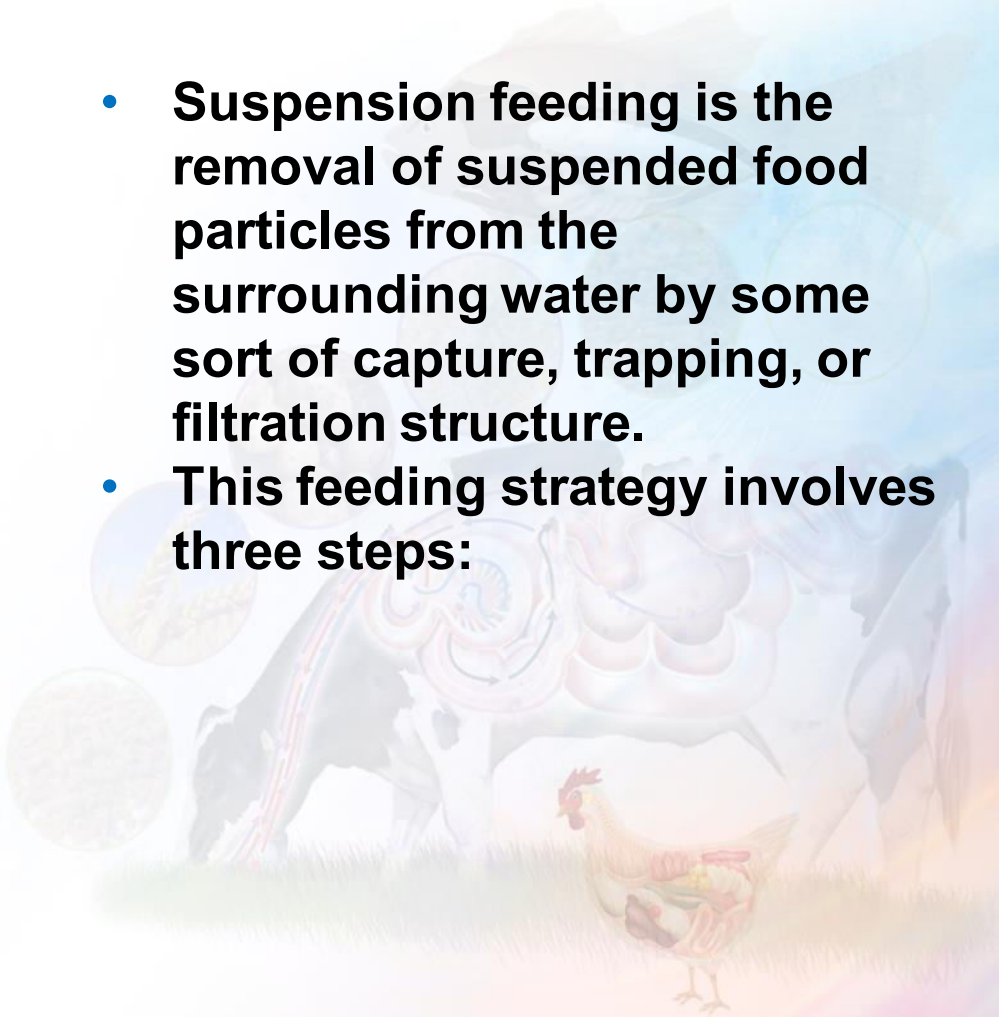
Suspension Feeders

- Suspension feeding is the capture and ingestion of food particles that are suspended in water.
- These particles can include phytoplankton, zooplankton, bacteria.



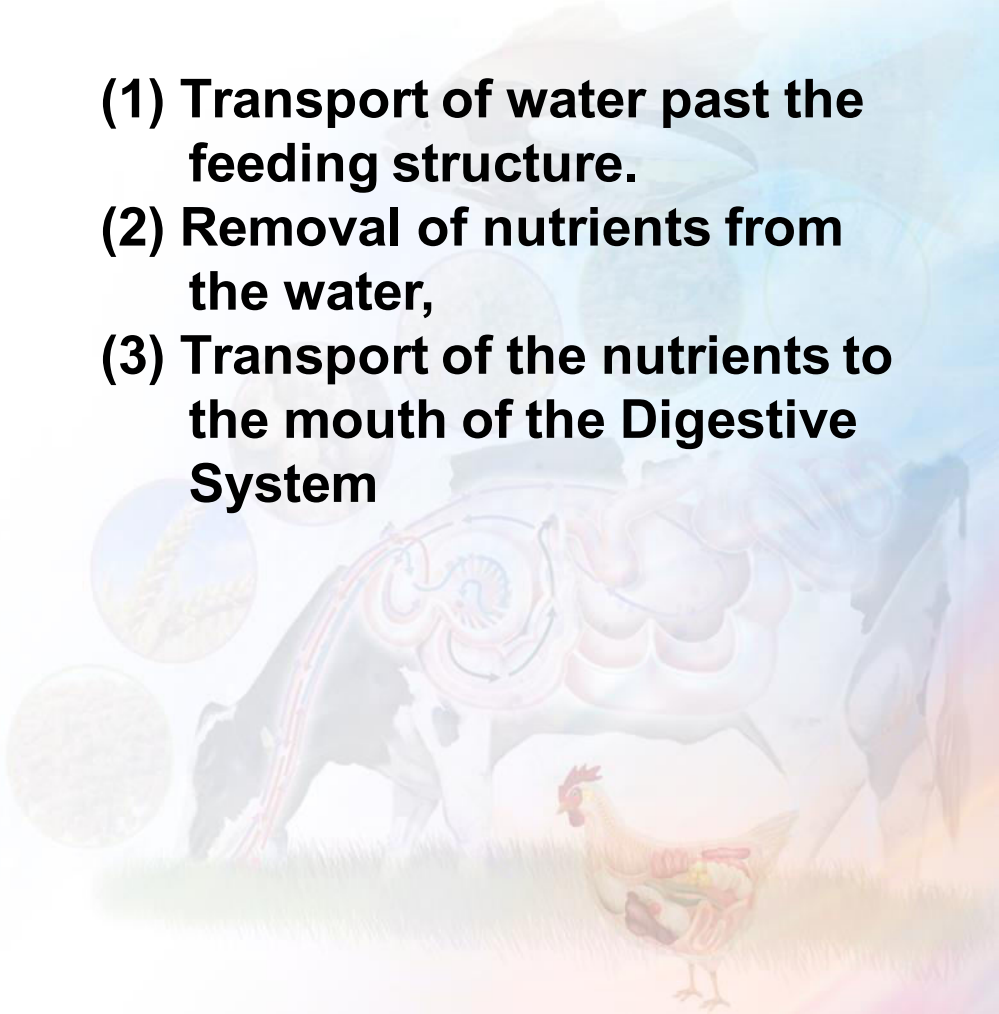
Suspension Feeders

- Suspension feeding is the removal of suspended food particles from the surrounding water by some sort of capture, trapping, or filtration structure.
- This feeding strategy involves three steps:



Suspension Feeders

(1) Transport of water past the feeding structure.
(2) Removal of nutrients from the water,
(3) Transport of the nutrients to the mouth of the Digestive System



Suspension Feeders





Sponges



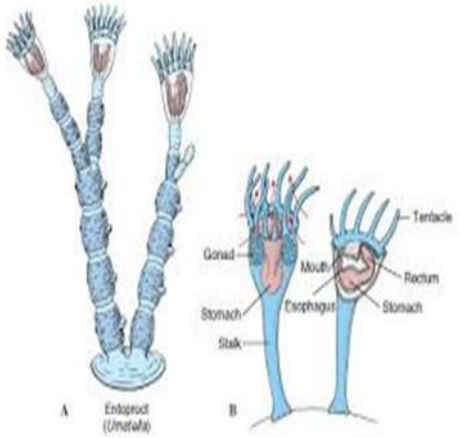


Brachiopods

Suspension Feeders

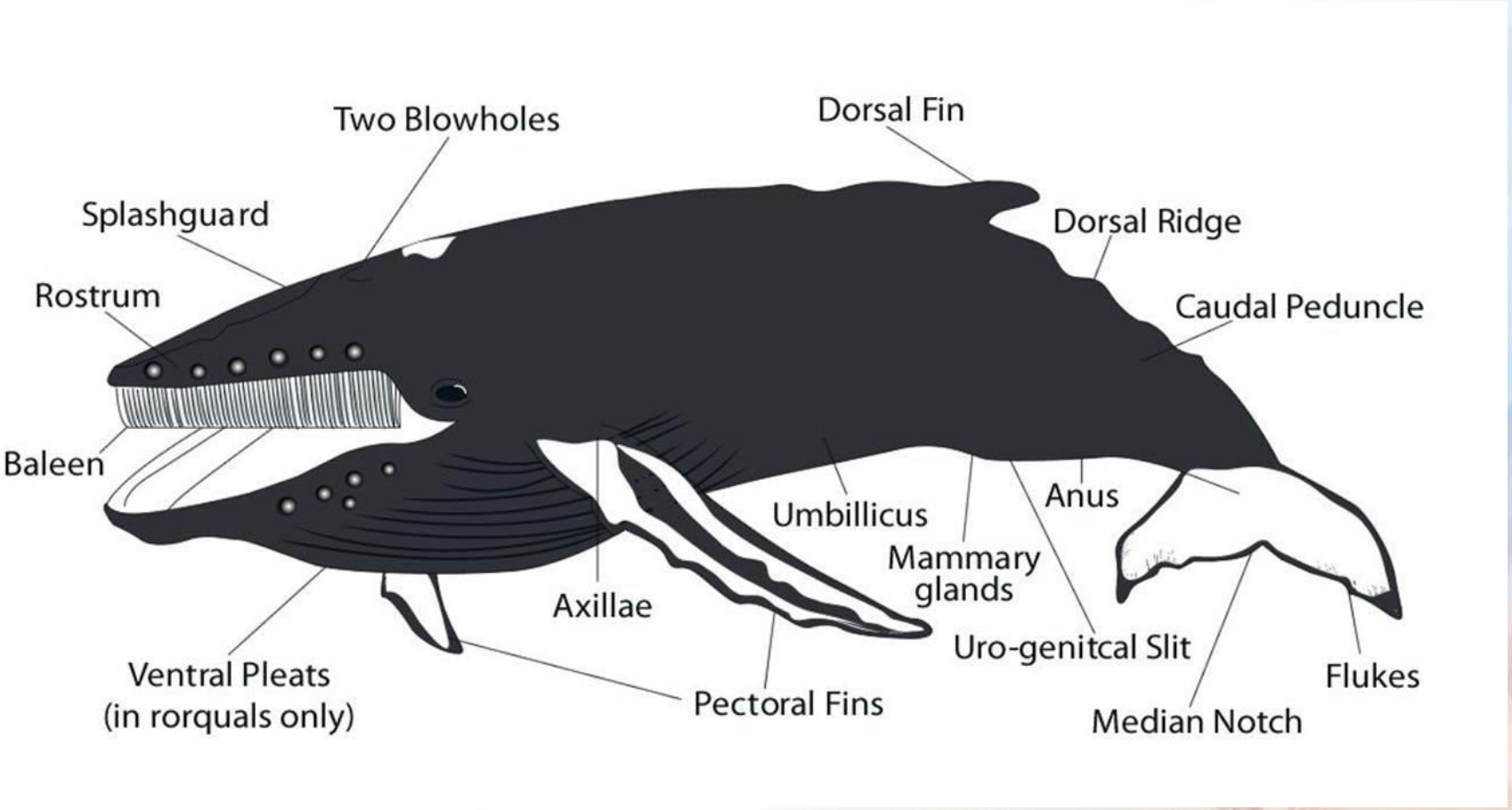






Most bivalves are suspension feeders

Suspension Feeders



Module 16



Deposit Feeders



Deposit Feeders

- **Deposit feeders** are the animals which are primarily the omnivorous.
- These animals obtain their nutrients from muds and sands or terrestrial soils.
- Deposit feeders simply swallow large quantities of sediment (mud, soil, sand, organic matter).

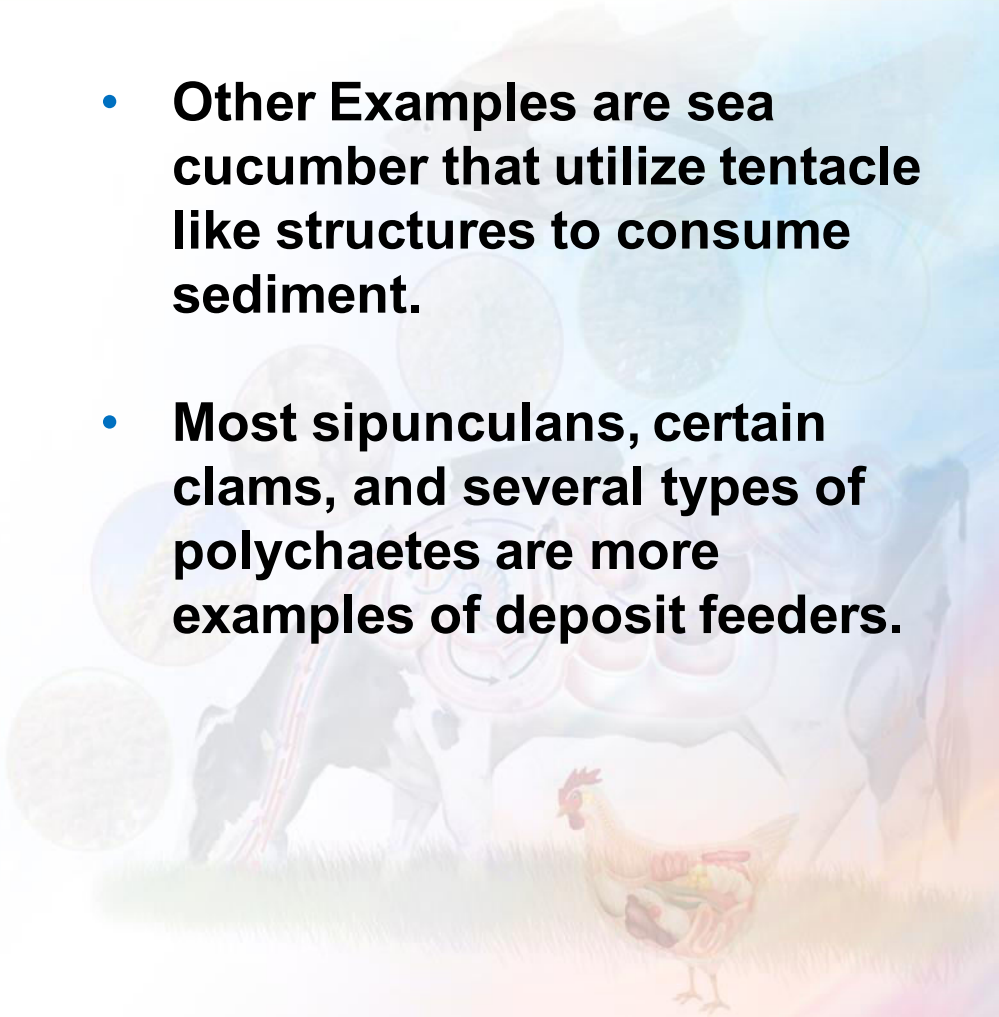


Deposit Feeders

- The usable nutrients are digested, and the remains pass out the anus.
- Deposit feeding occurs in:
- Many polychaete
- Annelids,
- Some snails,
- Some sea urchins and
- Most earthworms.

Deposit Feeders

- Other Examples are sea cucumber that utilize tentacle like structures to consume sediment.
- Most sipunculans, certain clams, and several types of polychaetes are more examples of deposit feeders.



Deposit Feeders







Sea cucumber



Sea Urchin

Deposit Feeders



Sipunculans



Clams



Polychaetes



Earthworm

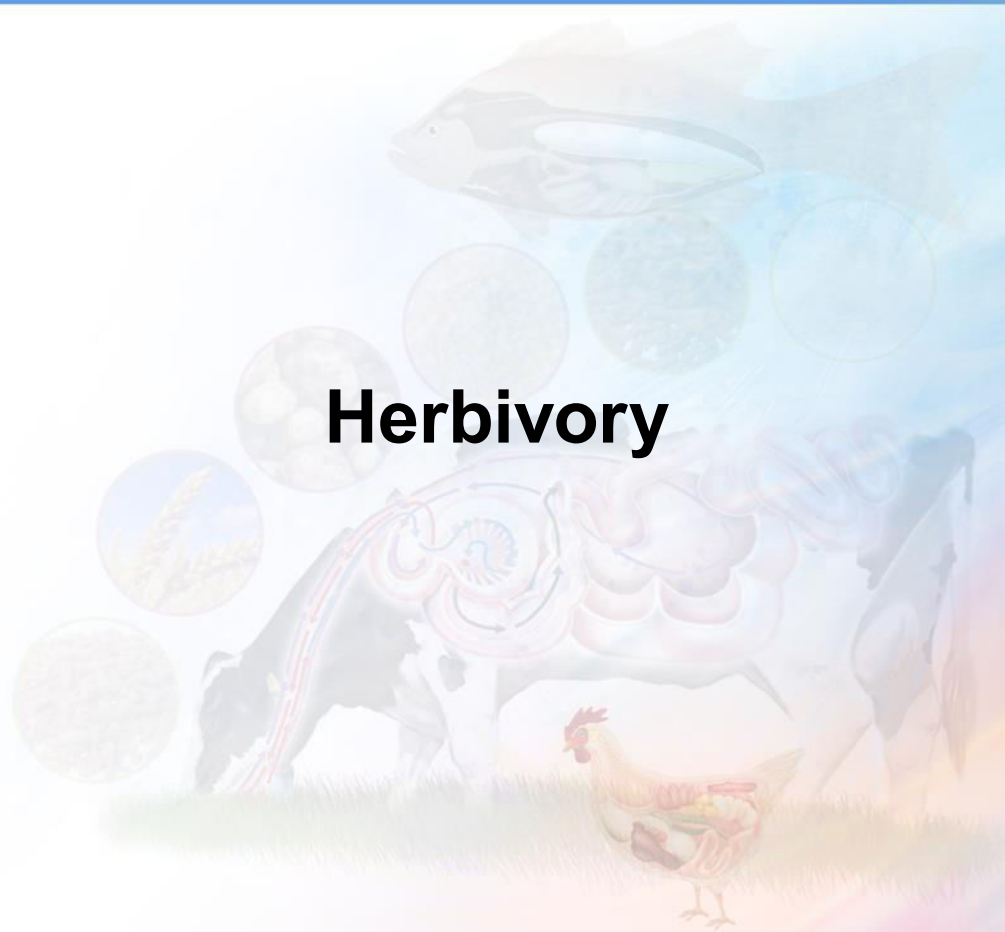
Animal Form & Function II

Module 17



Animal Form & Function II

Herbivory



Herbivory

Herbivory is the consumption of macroscopic plants.

- This common feeding strategy requires the ability to "bite and chew" large pieces of plant matter (macroherbivory).
- Invertebrates that evolved macroherbivory include molluscs, polychaete worms, arthropods, and sea urchins.

Herbivory

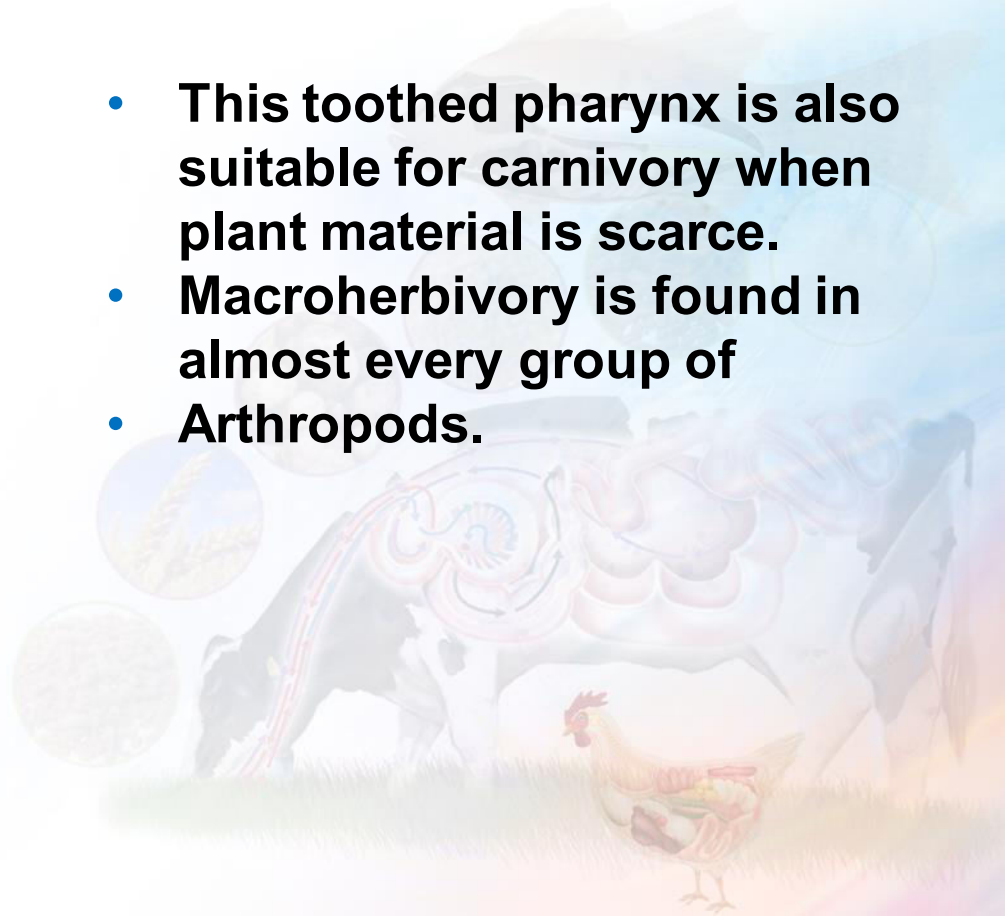
Macroherbivory is found in almost every group of arthropods. For example, insects and crustaceans have large, powerful mandibles capable of biting off plant material and subsequently grinding and chewing it before passing the plant material to the mouth

Herbivory

- Many molluscs have a radula.
- A radula is a muscularized, belt-like rasp armed with chitinous teeth.
- Molluscs use the radula to scrape algae off rocks or to tear the leaves off terrestrial plants.
- Polychaetes have sets of large chitinous teeth on an eversible proboscis or pharynx that is used to scrape off algae.

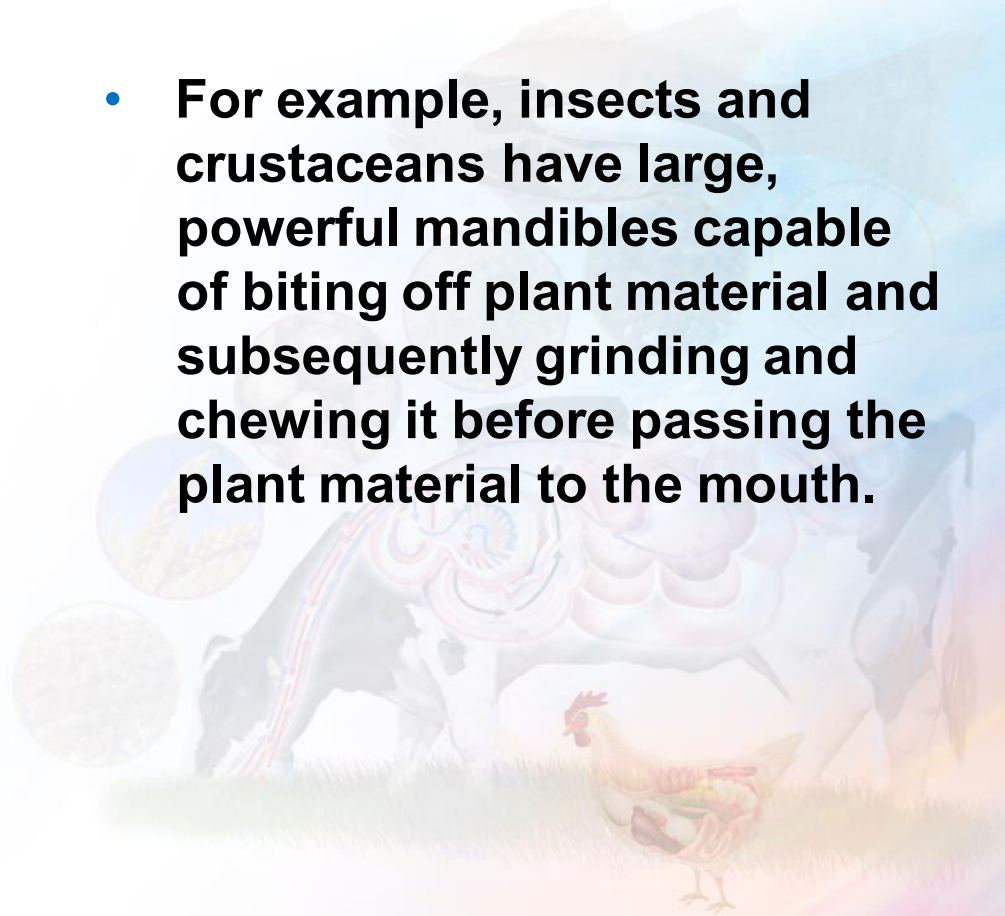
Herbivory

- This toothed pharynx is also suitable for carnivory when plant material is scarce.
- Macroherbivory is found in almost every group of
- Arthropods.



Herbivory

- For example, insects and crustaceans have large, powerful mandibles capable of biting off plant material and subsequently grinding and chewing it before passing the plant material to the mouth.



Animal Form & Function II

Module 18



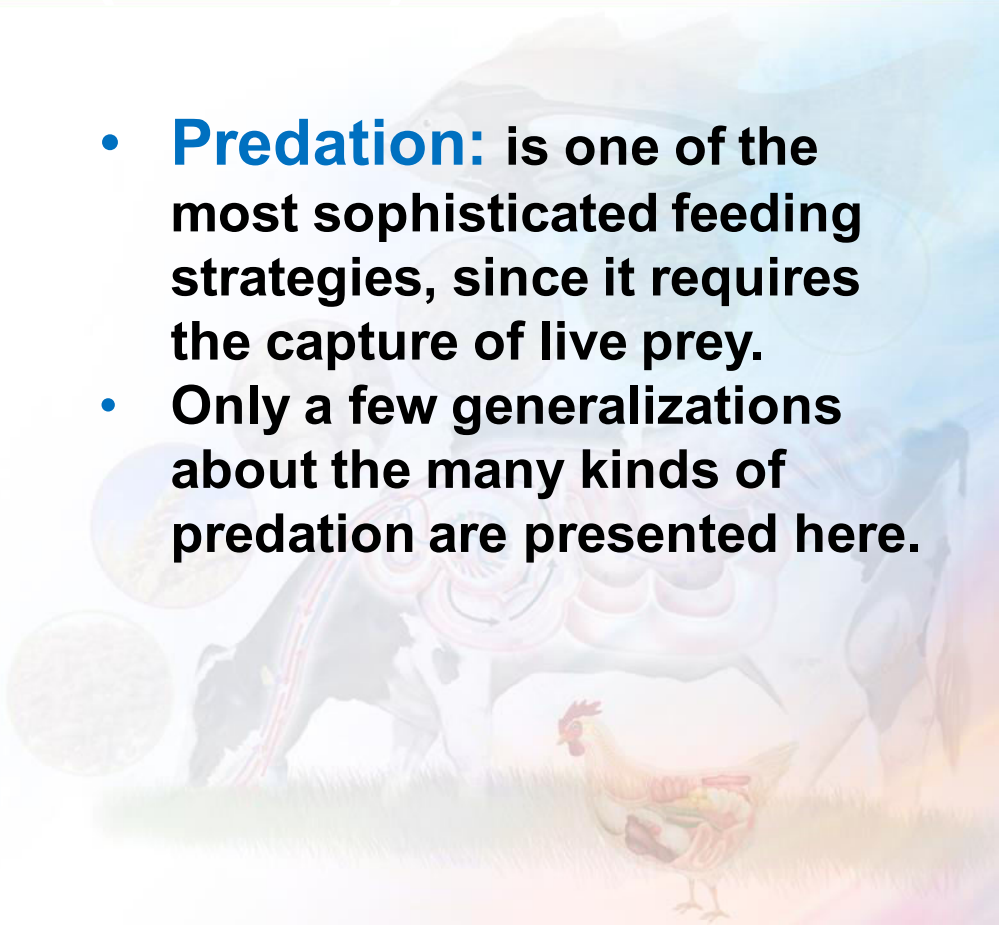
Animal Form & Function II

Predation



Predation

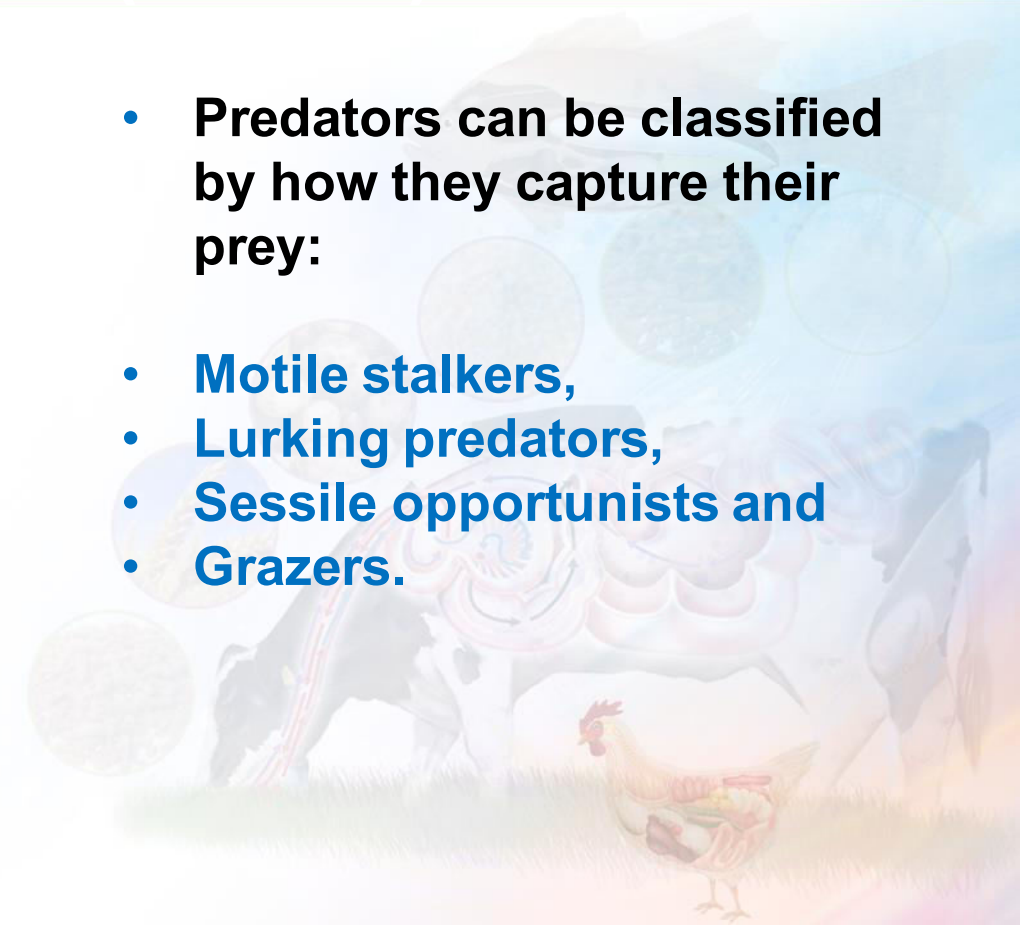
- **Predation:** is one of the most sophisticated feeding strategies, since it requires the capture of live prey.
- Only a few generalizations about the many kinds of predation are presented here.



Predation

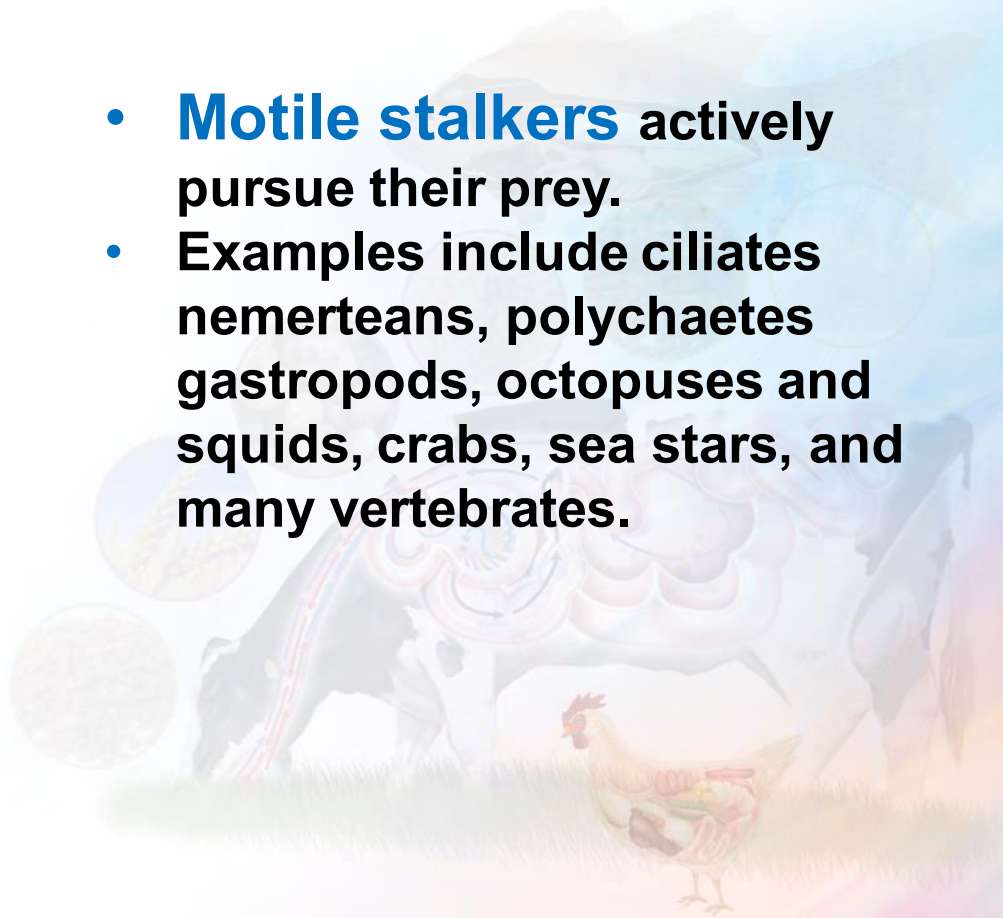
- **Predators can be classified by how they capture their prey:**

- **Motile stalkers,**
- **Lurking predators,**
- **Sessile opportunists and**
- **Grazers.**



Predation

- **Motile stalkers** actively pursue their prey.
- Examples include ciliates nemerteans, polychaetes gastropods, octopuses and squids, crabs, sea stars, and many vertebrates.



Predation



Praying Mantis



Shrimp

- **Lurking predators** sit and wait for their prey to come within seizing distance. Examples include certain species of praying mantises, shrimp, crabs, spiders, polychaetes, and many vertebrates.
- **Sessile opportunists** usually are not very mobile. They can only capture prey when the prey organism comes into contact with them.

Predation



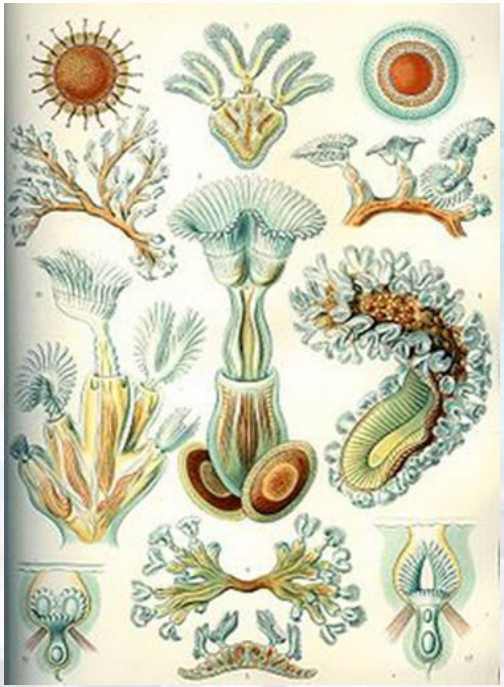


- Examples include certain protozoa, barnacles, and cnidarians.
- Grazing carnivores move about the substrate picking up small organisms.
- Their diet usually consists largely of sessile and slow-moving animals, such as sponges, ectoprocts, tunicates, snails, worms, and small crustaceans.

Predation



Stalker predator



Ectoprocts and Bryozoans

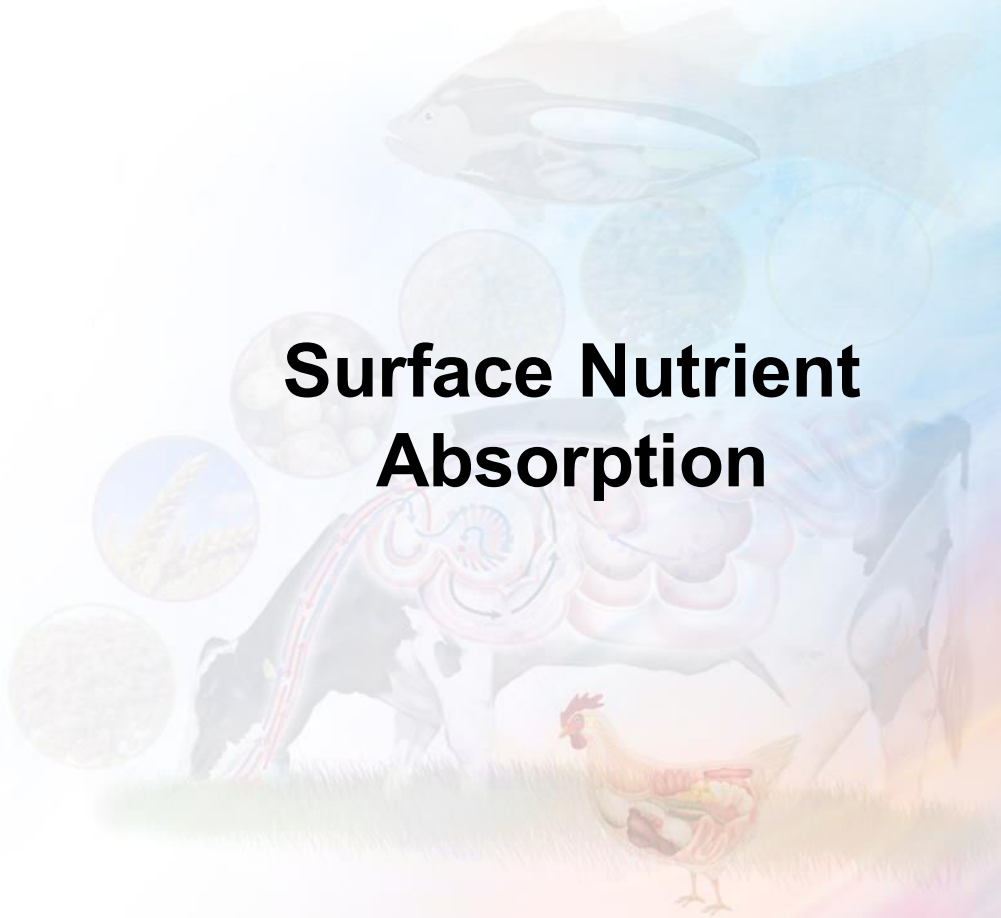
Animal Form & Function II

Module 19



Animal Form & Function II

Surface Nutrient Absorption



Surface Nutrient Absorption

- Some highly specialized animals have dispensed entirely with all mechanisms for prey capture, ingestion of food particles, and digestive processes.
- Instead they directly absorb nutrients from the external medium across their body surfaces.

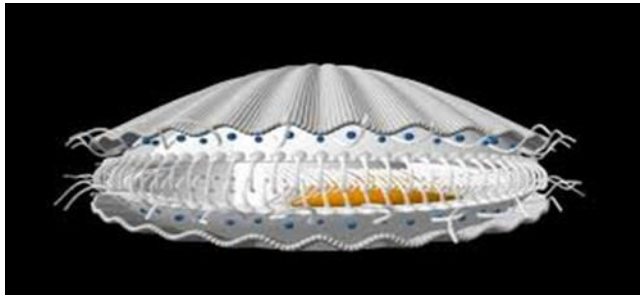
Surface Nutrient Absorption

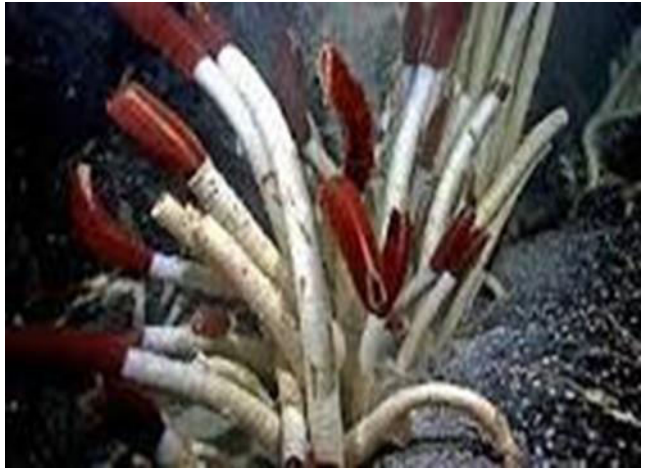
- This medium may be:
- Nutrient-rich seawater
- Fluid in other animals' digestive tracts,
- Body fluids of other animals.
- Examples:
- some free-living protozoa, such as Chilomonas, absorb all of their nutrients across their body surface.

Surface Nutrient Absorption

- **Endoparasites:**
- a) protozoa,
- b) cestode worms
- c) gastropods
- d) crustaceans
- All these parasires lack mouths and digestive systems.

Surface Nutrient Absorption





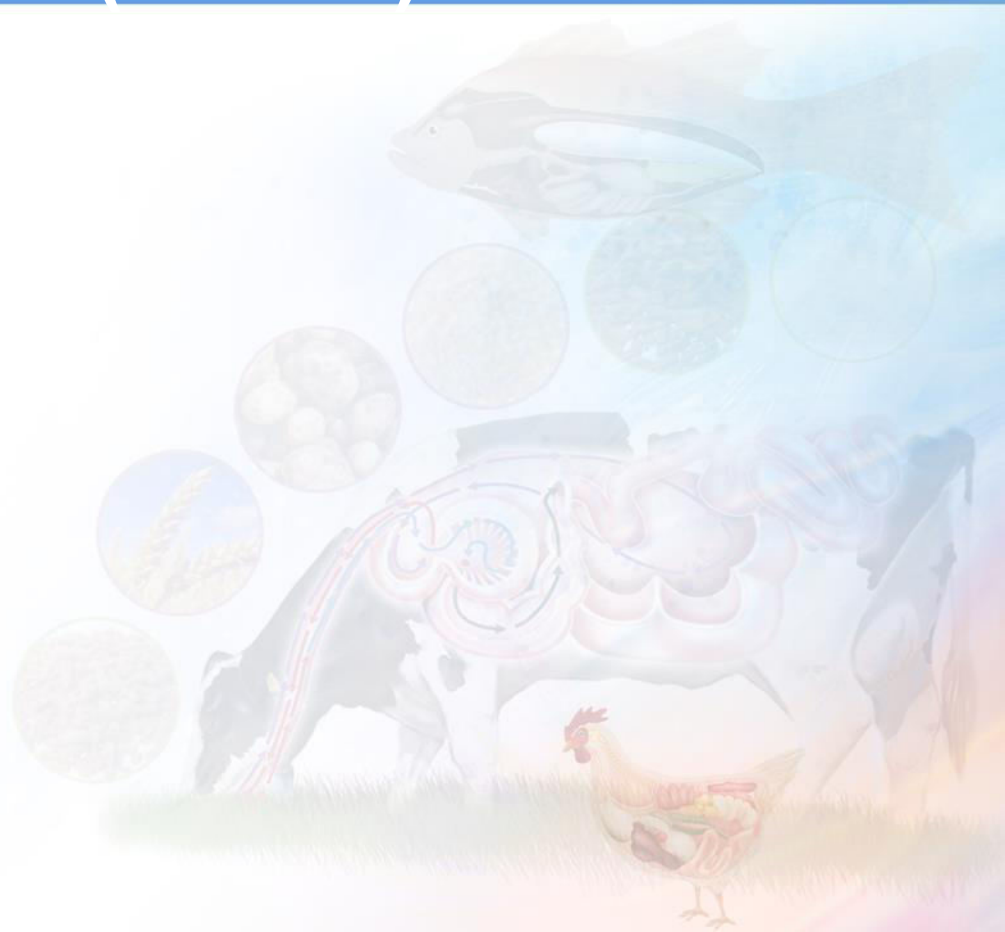
- A few nonparasitic multicellular animals also lack a mouth and digestive system and absorb nutrients across their body surface.
- Examples include the gutless bivalves and pogonophoran worms.

Prognophorans

Surface Nutrient Absorption

- Interestingly, many pogonophoran worms absorb some nutrients from seawater across their body surface.
- They also supplement their nutrition with organic carbon that symbiotic bacteria fix within the pogonophoran's tissues (Tube dwelling marine animals)

Surface Nutrient Absorption



Animal Form & Function II

Module 20



Animal Form & Function II

Fluid Feeders



Fluid Feeders

- **Fluid Feeders** are the animals which feed on biological fluids of animals and plants.
- These fluids are a rich source of nutrients.
- Fluid feeding is the characteristic of some parasites, such as the intestinal nematodes that bite and rasp off host tissue or suck blood.

Fluid Feeders

- **External parasites** (ectoparasites), such as leeches, ticks, mites, lampreys, and certain crustaceans, use a wide variety of mouthparts to feed on body fluids.
- For example, the sea lamprey has a funnel structure surrounding its mouth.

Fluid Feeders



Aphids

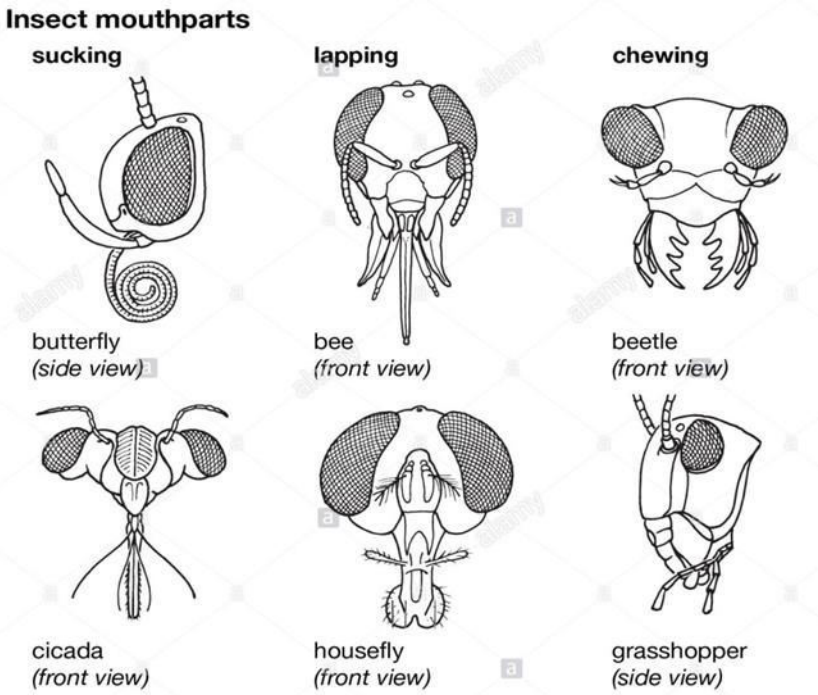


- Lampray

Fluid Feeders

- The funnel is lined with over 200 rasping teeth and a rasp like tongue.
- The lamprey uses the funnel like a suction cup to grip its fish host, and then with its tongue, rasps a hole in the fish's body wall.
- The lamprey then sucks blood and body fluids from the wound.

Fluid Feeders



- Insects have the most highly developed sucking structures for fluid feeding.
- For example, butterflies, moths, and aphids have tube-like mouthparts that enable them to suck up plant fluids.
-
- Blood-sucking mosquitoes have complex mouthparts with piercing stylets.

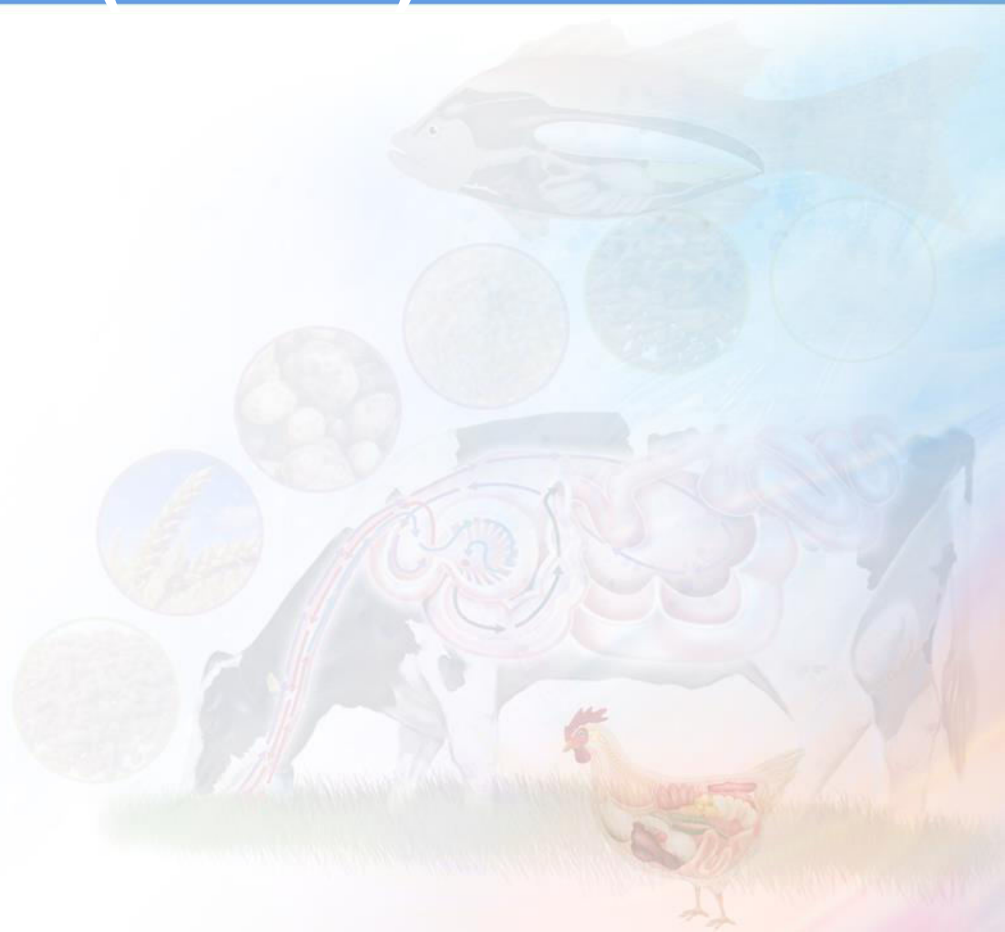
Fluid Feeders



Humming bird

- Some nectar-feeding birds have long bills and tongues.
- In fact, the bill is often specialized for particular types of flowers.
- Other nectar-feeding birds have short bills; they make a hole in the base of a flower and use their tongue to obtain nectar through the hole.

Fluid Feeders



Fluid Feeders



Desmodus

- The only mammals that feed exclusively on blood are the vampire bats.
- Such as *Desmodus*, of tropical South and Central America.
- These bats attack birds, cattle, and horses, using knife sharp front teeth to pierce the surface blood vessels, and then lap at the oozing wound.

Fluid Feeders



Nectar- feeding bat

- Nectar-feeding bats have a long tongue to extract the nectar from flowering plants

.

- The nectar-feeding honey possum has a long, brush-tipped tongue and reduced dentition.

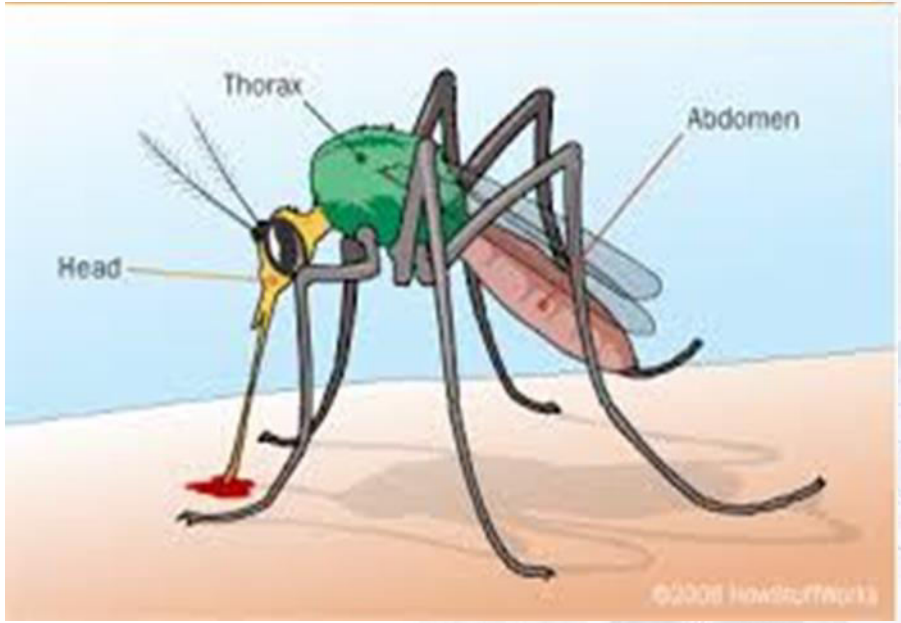
Fluid Feeders



Honey Possum

- The nectar-feeding honey possum has a long, brush-tipped tongue and reduced dentition.

Fluid Feeders



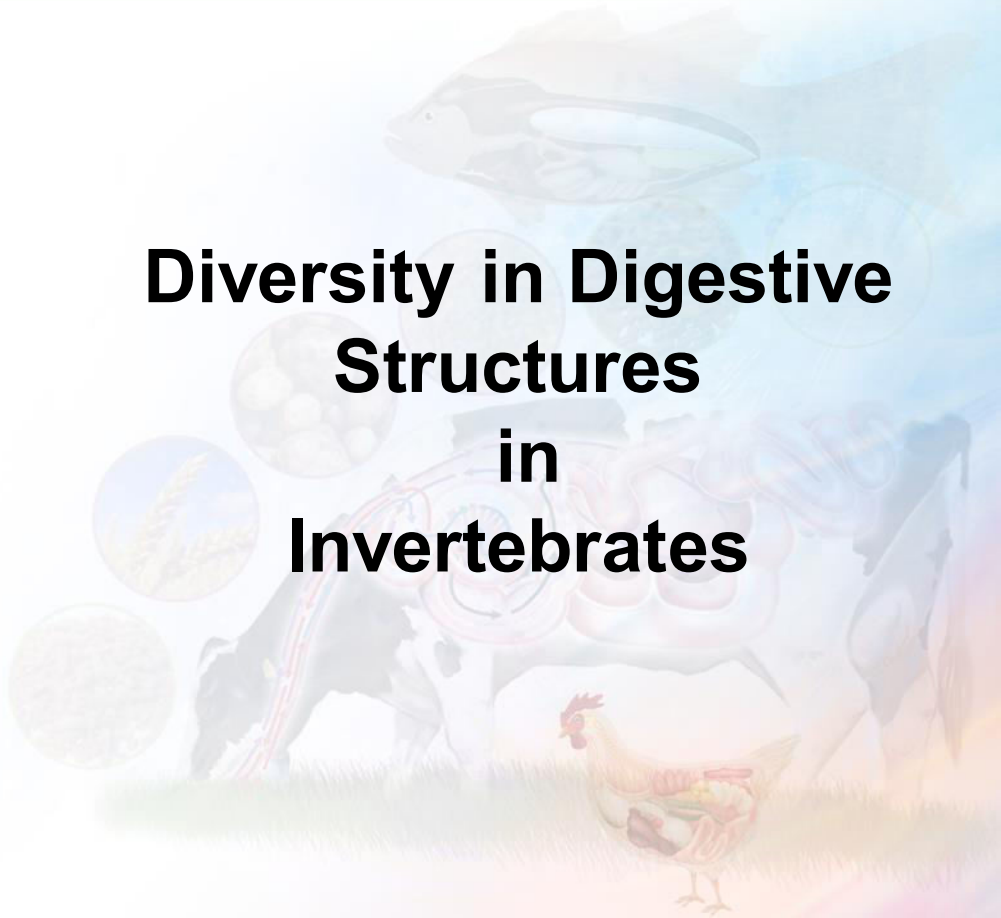
Animal Form & Function II

Module 21



Animal Form & Function II

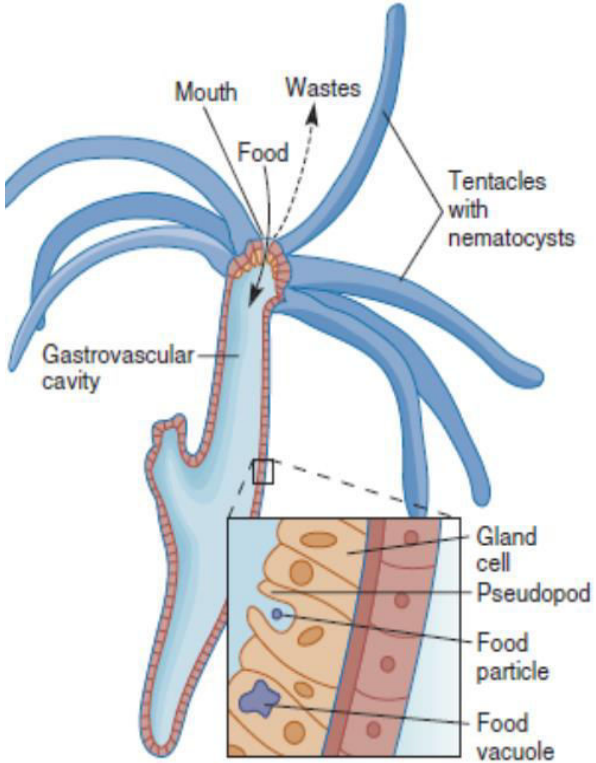
Diversity in Digestive Structures in Invertebrates



Diversity in Digestive Structures in Invertebrates

- In primitive, multicellular animals, such as cnidarians, the gut is a blind (closed) sac called a gastrovascular cavity.
- It has only one opening that is both entrance and exit thus, it is an incomplete digestive tract.

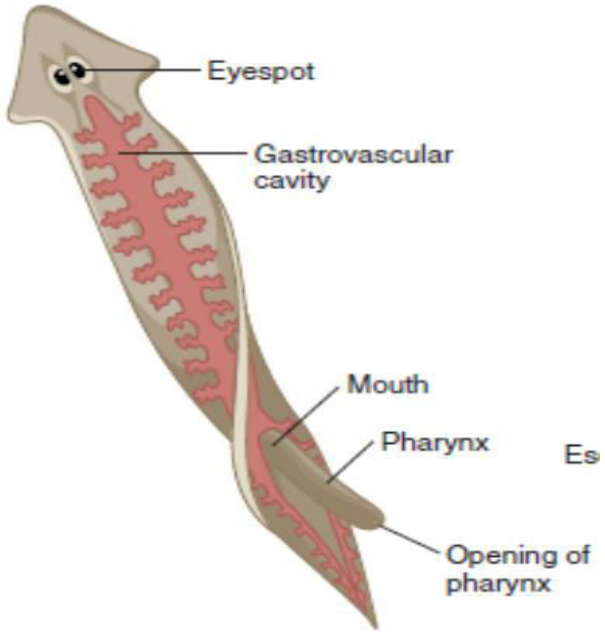
Diversity in Digestive Structures in Invertebrates



a) Cnidarian *(Hydra)*

- **Some specialized cells in the cavity secrete digestive enzymes that begin the process of extracellular digestion.**
- **Other phagocytic cells that line the cavity engulf food material and continue intracellular digestion inside food vacuoles.**

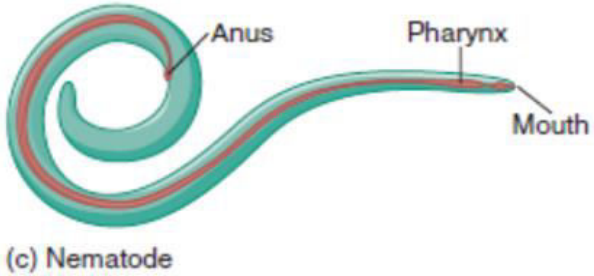
Diversity in Digestive Structures in Invertebrates



(b) Planarian

- **In Platyhelminth (planarian) Gastric cavity branches extensively having Single opening.**
- **When a planarian feeds, it sticks its muscular pharynx out of its mouth and sucks in food.**

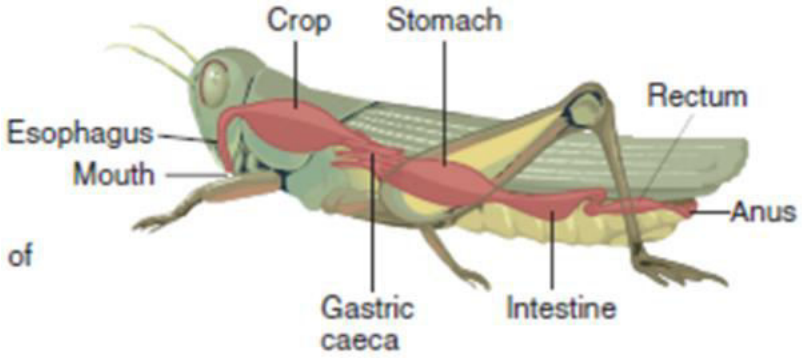
Diversity in Digestive Structures in Invertebrates)



(c) Nematode

- **In Aschelminths** the digestive tract is complete, that permits the one-way flow of ingested food without mixing it with previously ingested food or waste.
- A nematode (Ascaris) has a complete digestive tract with a mouth, pharynx, and anus.

Diversity in Digestive Structures in Invertebrates)



(d) Insect

- In insects (grasshopper) the complete digestive tract has an expanded region called a crop that functions as a food storage organ.
- These are evolutionary changes.

Animal Form & Function II

Module 22



Animal Form & Function II

Digestion in Protozoa

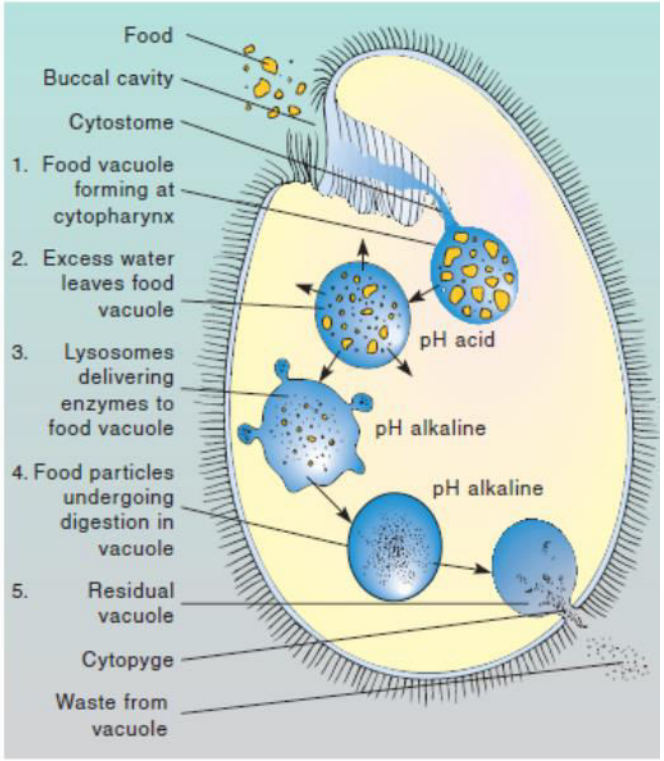


Digestion in Protozoa

Digestion in Protozoa

- Ciliated protozoa are good examples of protists that utilize heterotrophic nutrition.
- Ciliary action directs food from the environment into the buccal cavity and cytostome.

Digestion in Protozoa



- **The residual vacuole then excretes its waste products via the cytopyge.**

Digestion in Protozoa

- The cytostome opens into the cytopharynx, which enlarges as food enters and pinches off a food-containing vacuole.
- The detached food vacuole then moves through the cytoplasm.

Digestion in Protozoa

- During this movement, excess water is removed from the vacuole, the contents are acidified and then made alkaline and a lysosome adds digestive enzymes.
- The food particles are then digested within the vacuole and the nutrients absorbed into the cytoplasm.

Digestion in Protozoa

- **Intracellular Digestion in a Ciliated Protozoan.**
  Cilia direct food toward the cytostome (“mouth”).
- **The food enters the cytopharynx, where a food vacuole forms and detaches from the cytopharynx.**

Digestion in Protozoa

- The detached vacuole undergoes acidic and alkaline digestion, and the waste vacuole moves to the cytopyge (“anus”) for excretion.

Animal Form & Function II

Module 23

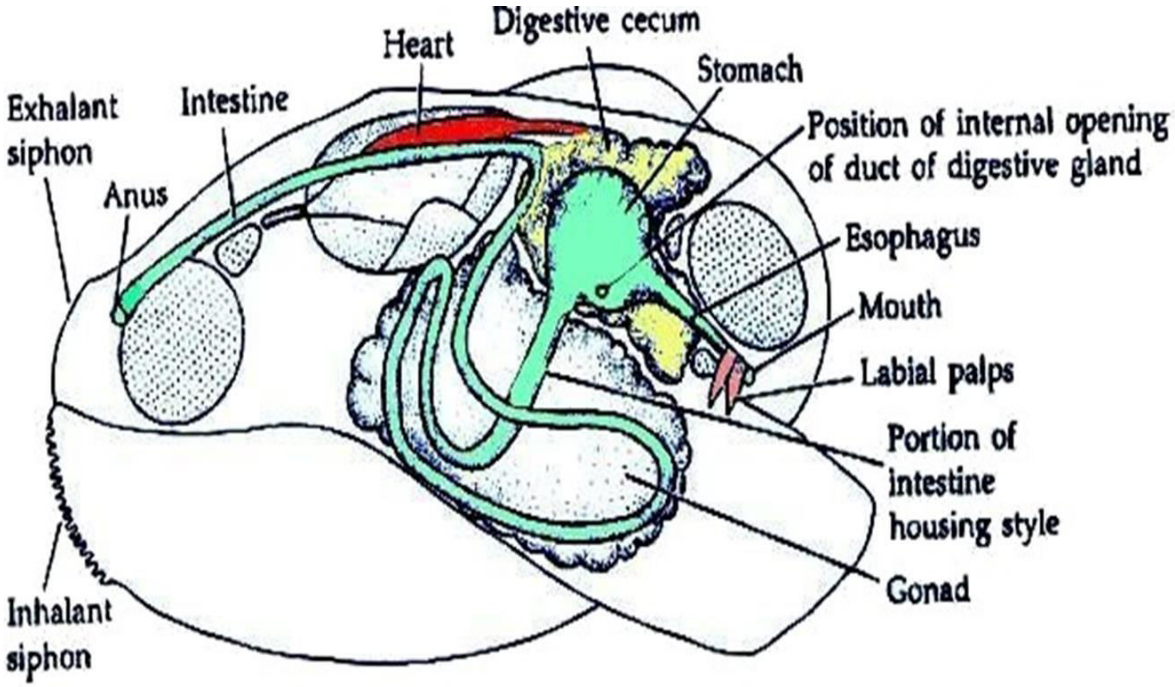
Animal Form & Function II

Digestion in Mollusca

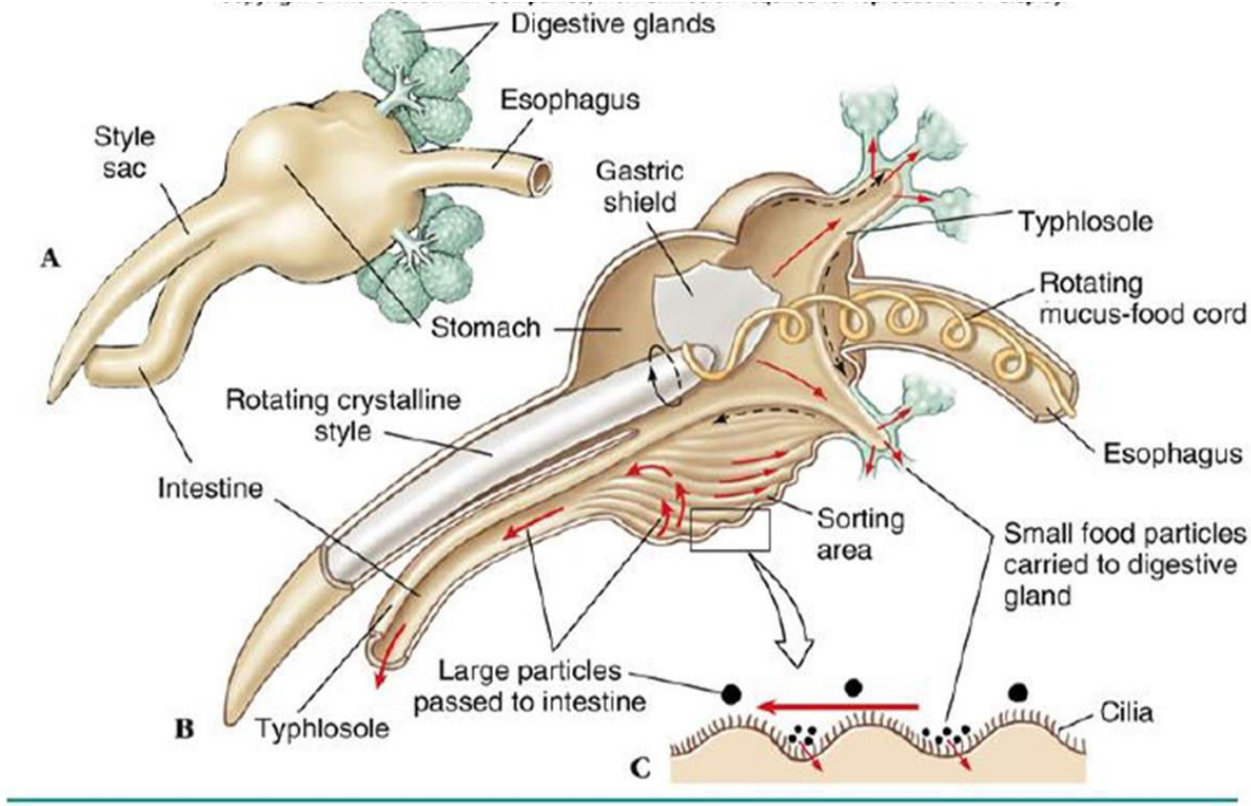
Digestion in Mollusca

- Many bivalve feed or ingest suspended small food particles.
- The digestive tract has a short esophagus opening into a stomach, midgut, hindgut, and rectum.
- The stomach contains a crystalline style, gastric shield, and diverticulated region.

Digestion in Mollusca



Digestion in Mollusca



Digestion in Mollusca

- These diverticulae are blind-ending sacs that increase the surface area for absorption and intracellular digestion.

Digestion in Mollusca

- The midgut, hindgut, and rectum function in extracellular digestion and absorption
- Digestion is a coordination of three main steps:
  (1) feeding,
  (2) extracellular digestion,
  (3) intracellular digestion.
- The resting phase is preparative for extracellular digestion.

Digestion in Mollusca

- (a) Extracellular digestion begins before food ingestion fragmentation spherules in the stomach.
- (b) As food enters the stomach, the rotating style and the enzymes released by the gastric shield mechanically and enzymatically break it down.

Digestion in Mollusca

- (c) The small food particles then move into the digestive diverticulae for intracellular digestion.
- (d) A progressive passage of food particles from the stomach to the digestive diverticulae follows cessation of feeding.

Digestion in Mollusca

- **e) During this resting phase, the stomach empties and the style reforms, while intracellular digestion in the diverticulae is completed, and fragmentation spherules begin to form again. The movement of fragmentation spherules starts the next feeding cycle.**

Animal Form & Function II

Module 24



Animal Form & Function II



Digestion in Insects

Digestion in Insects

- The grasshopper is a representative insect with a complete digestive tract.
- During feeding, the mandibles and maxillae first break up (masticate) the food, which is then taken into mouth and passed to the crop via the esophagus.

Digestion in Insects

- During mastication, the salivary glands add saliva to the food to lubricate it for passage through the digestive tract.
- These are the evolutionary changes in the digestive system.

Digestion in Insects

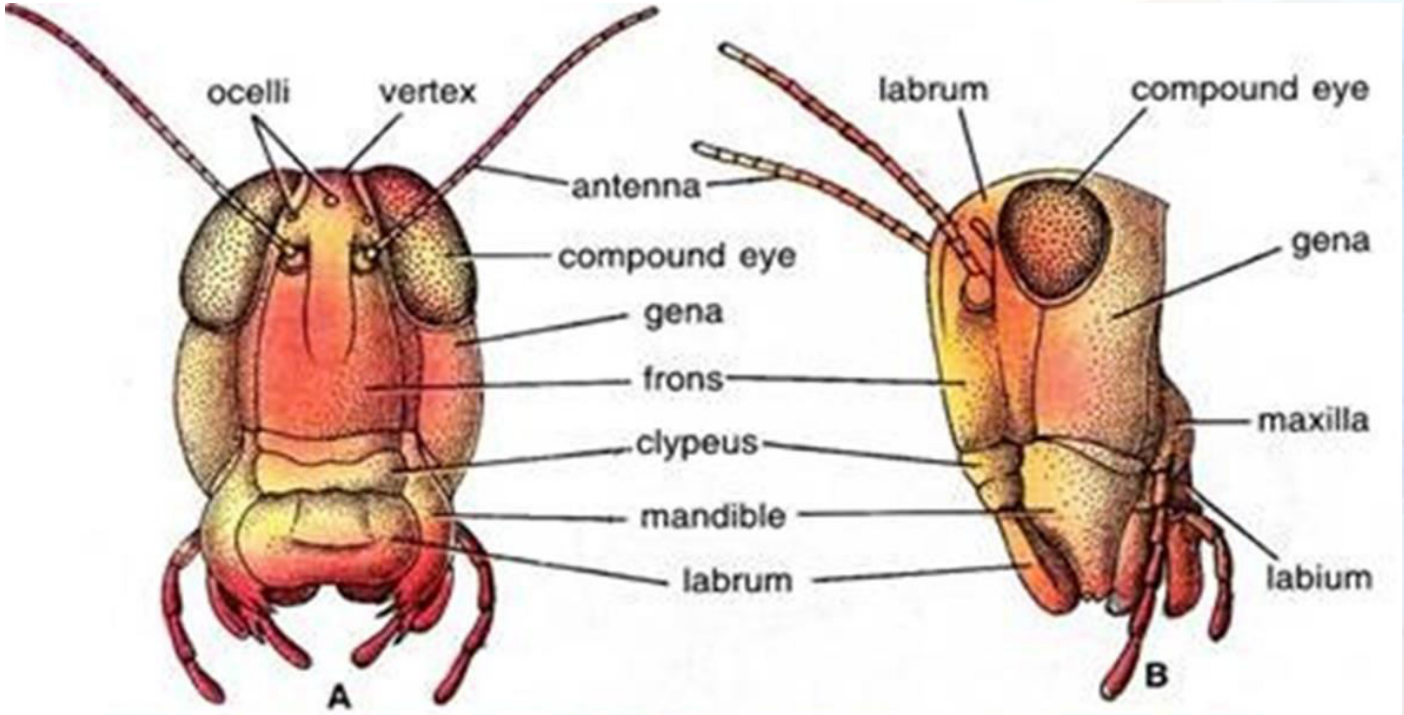
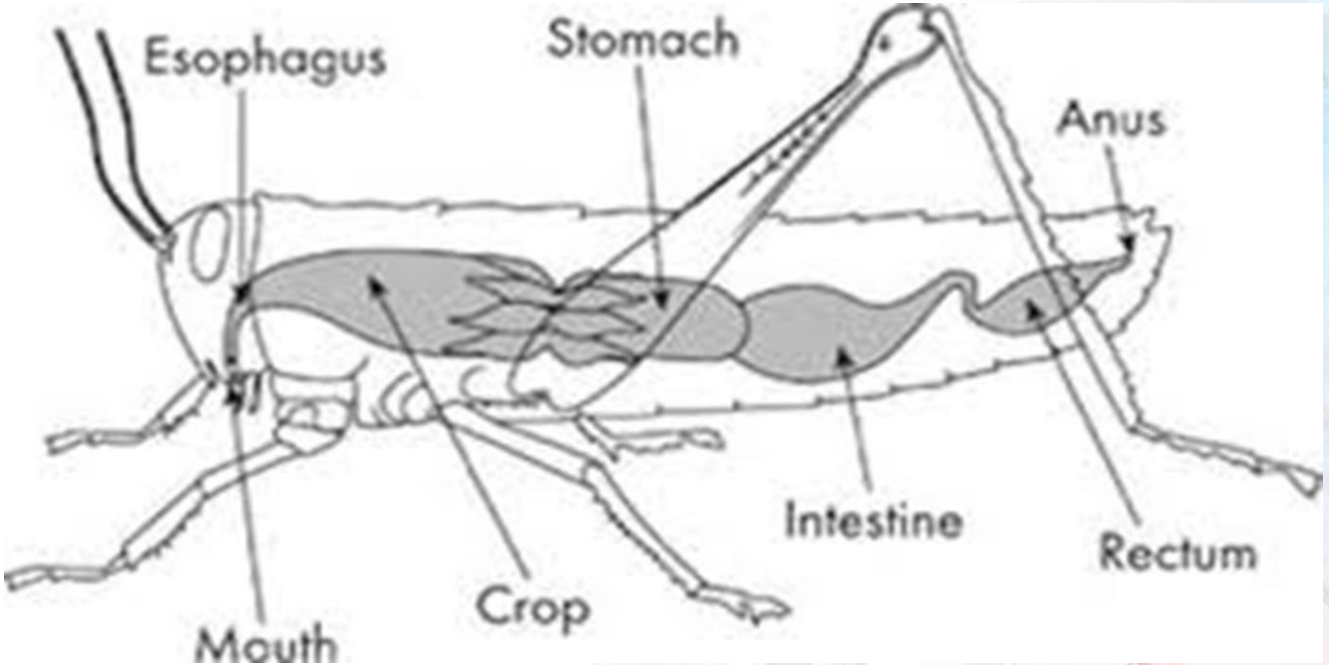


Fig. 74.2. Grasshopper. Head. A—Dorsal view; B—Lateral view.

Digestion in Insects



Digestion in Insects

Effort By: AARISH ALI

- Saliva also contains the enzyme amylase, which begins the enzymatic digestion of carbohydrates.
- This digestion continues during food storage in the crop.
- The midgut secretes other enzymes (carbohydrases, lipases, proteases) that enter the crop.

Digestion in Insects

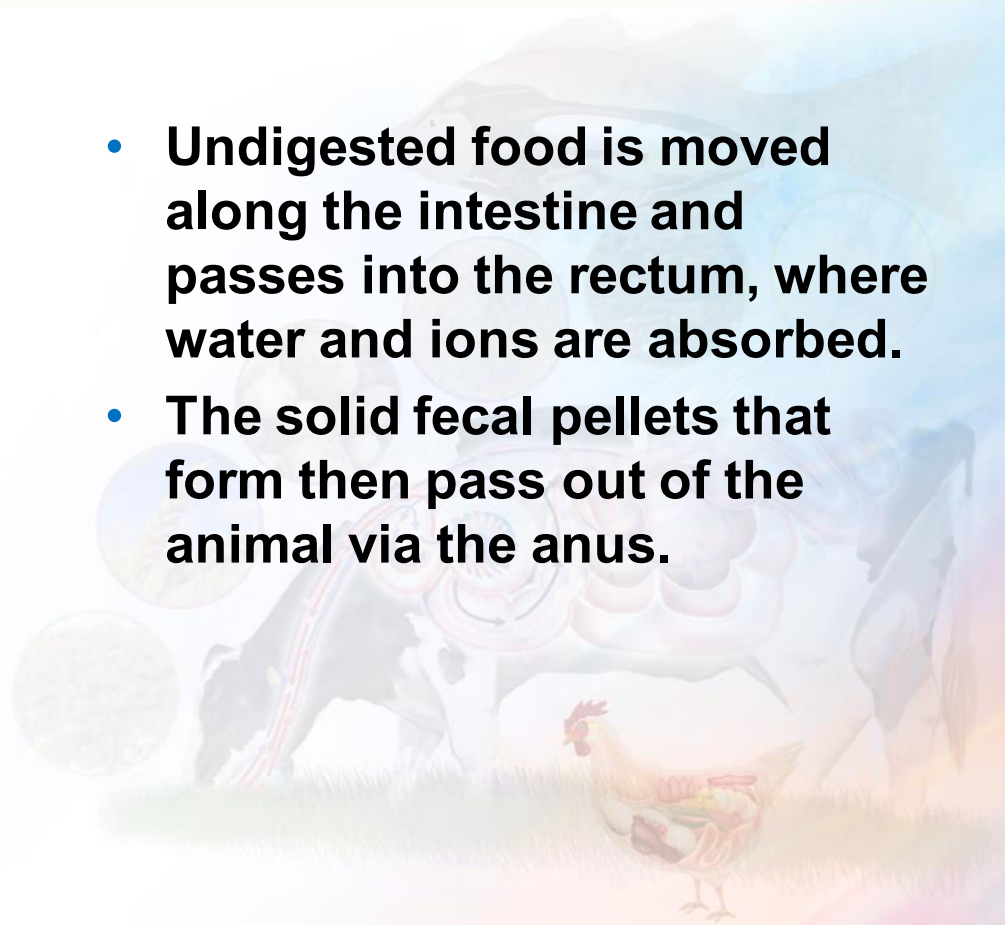
- Food passes slowly from the crop to the stomach, where it is mechanically reduced and the nutrient particles sorted.
- The glands in the walls of stomach and hepatic caeca secrete a few enzymes which bring about digestion.

Digestion in Insects

- The slightly alkaline or acidic secretions of midgut contain maltase, lipase, lactase, protease, trypsin and erepsin.
- The absorption of food material takes place in the midgut.
- Most nutrient absorption then occurs in the intestine.

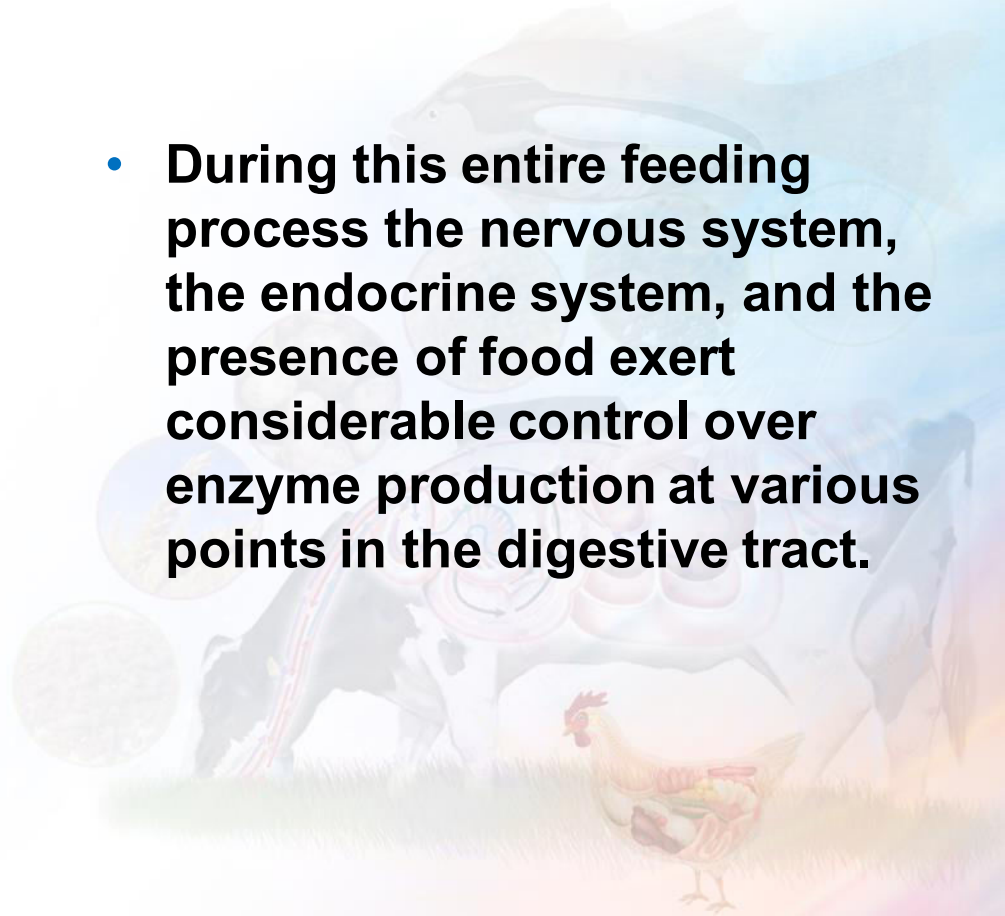
Digestion in Insects

- Undigested food is moved along the intestine and passes into the rectum, where water and ions are absorbed.
- The solid fecal pellets that form then pass out of the animal via the anus.



Digestion in Insects

- During this entire feeding process the nervous system, the endocrine system, and the presence of food exert considerable control over enzyme production at various points in the digestive tract.



Animal Form & Function II

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Animal Form & Function II

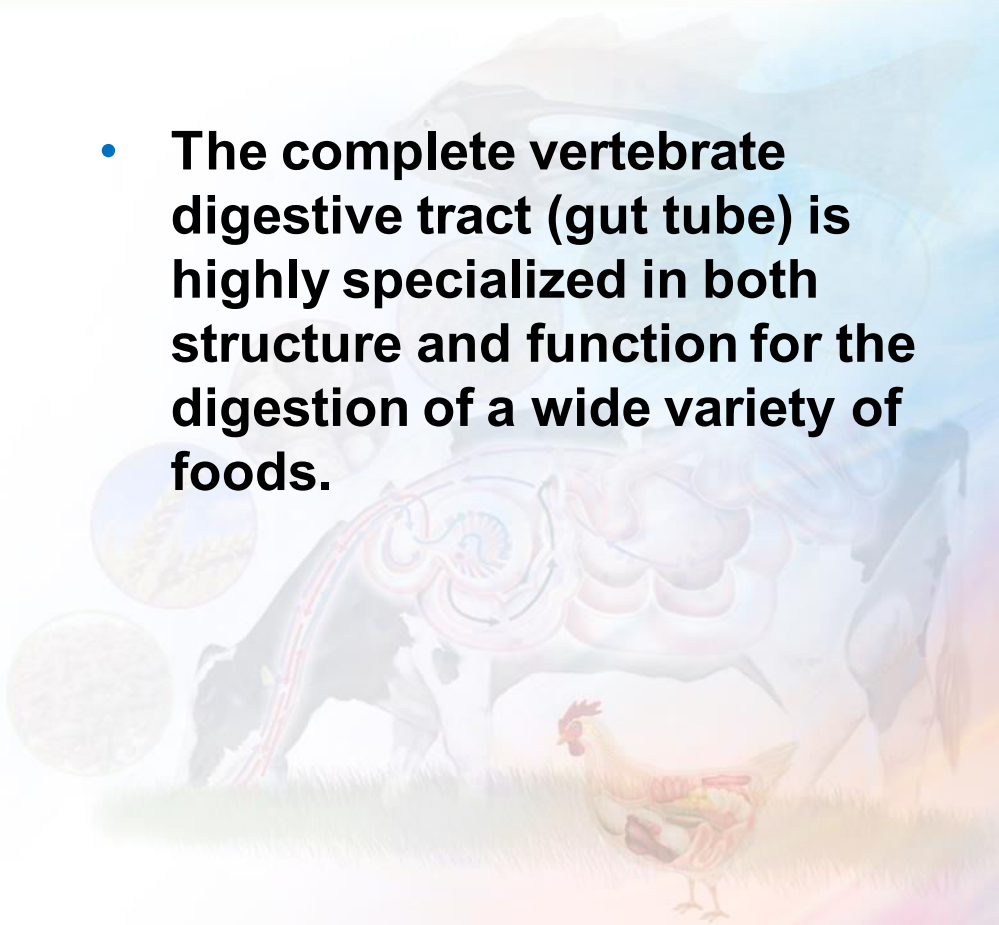
Diversity in Digestive Structures

(Vertebrates)



Diversity in Digestive Structures (Vertebrates)

- The complete vertebrate digestive tract (gut tube) is highly specialized in both structure and function for the digestion of a wide variety of foods.



Diversity in Digestive Structures (Vertebrates)

The basic structures of the gut tube include:

- Buccal cavity
- Pharynx,
- Esophagus
- Stomach,
- Small intestine
- Large intestine,
- Rectum, and Anus/Cloaca

Diversity in Digestive Structures (Vertebrates)

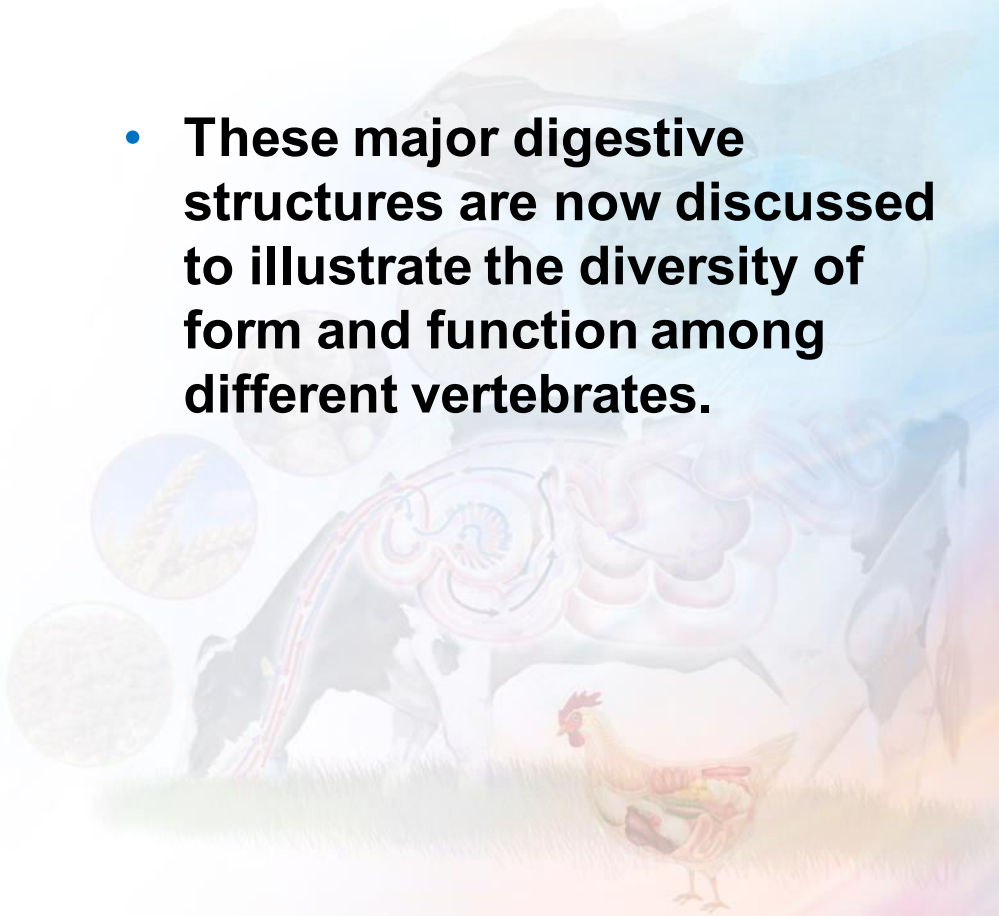
- In addition, three important glandular systems associated with the digestive tract are:
  i) The salivary glands.
  ii) The liver, gallbladder, bile duct.
  iii) The pancreas and pancreatic duct.

Diversity in Digestive Structures (Vertebrates)

- Most vertebrates spend the majority of their time acquiring food.
- The oral cavity (mouth), teeth, intestines, and other major digestive structures usually reflect the way an animal gathers food, the type of food it eats, and the way it digests that food.

Diversity in Digestive Structures (Vertebrates)

- These major digestive structures are now discussed to illustrate the diversity of form and function among different vertebrates.



Animal Form & Function II

Module 26



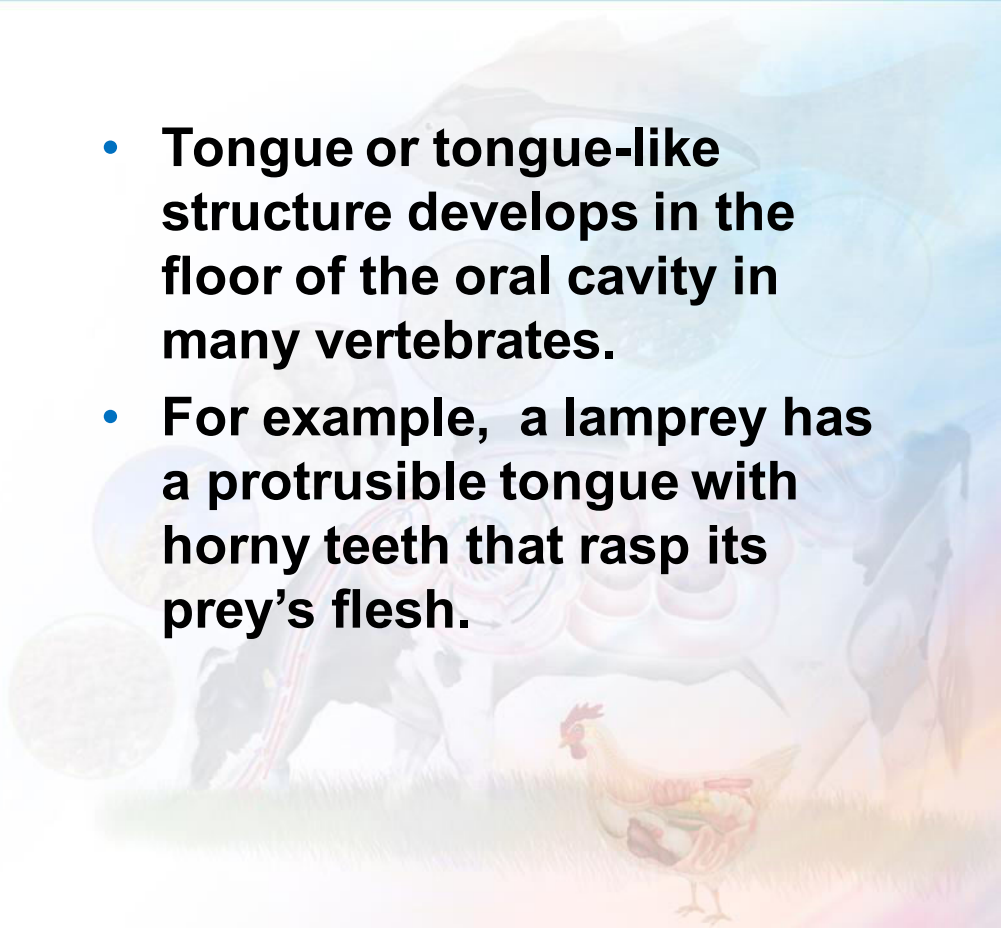
Animal Form & Function II

Tongues in Vertebrates



Tongues in Vertebrates

- Tongue or tongue-like structure develops in the floor of the oral cavity in many vertebrates.
- For example, a lamprey has a protrusible tongue with horny teeth that rasp its prey's flesh.

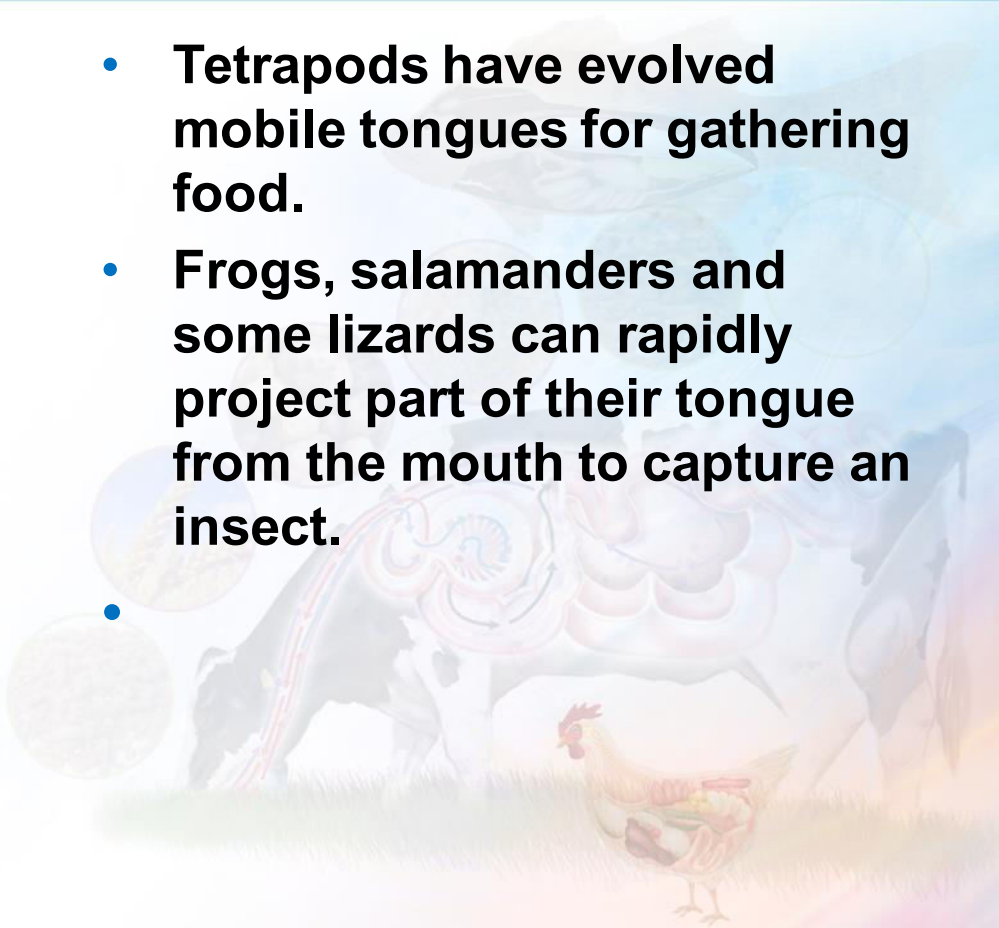


Tongues in Vertebrates

- The tongue of a fish is formed from a fold in the floor of the mouth. In some species of bony fishes the tongue has teeth which help to hold prey items.
- However, this type of tongue is not muscular.
-

Tongues in Vertebrates

- Tetrapods have evolved mobile tongues for gathering food.
- Frogs, salamanders and some lizards can rapidly project part of their tongue from the mouth to capture an insect.
-



Tongues in Vertebrates

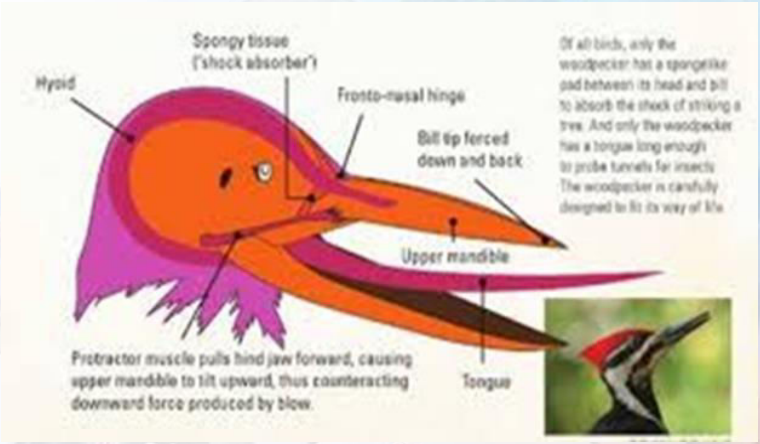


Frog



Lizard

Tongues in Vertebrates

- **A woodpecker has a long, spiny tongue for gathering insects and grubs ants.**
- 

- **Woodpecker**

Tongues in Vertebrates





- Spiny papillae on the tongues of cats and other carnivores help these animals rasp flesh from a bone.
- Pangolins have long, sticky tongues, which are often longer than their body.
- They need to have long to reach far inside ant nests.

Animal Form & Function II

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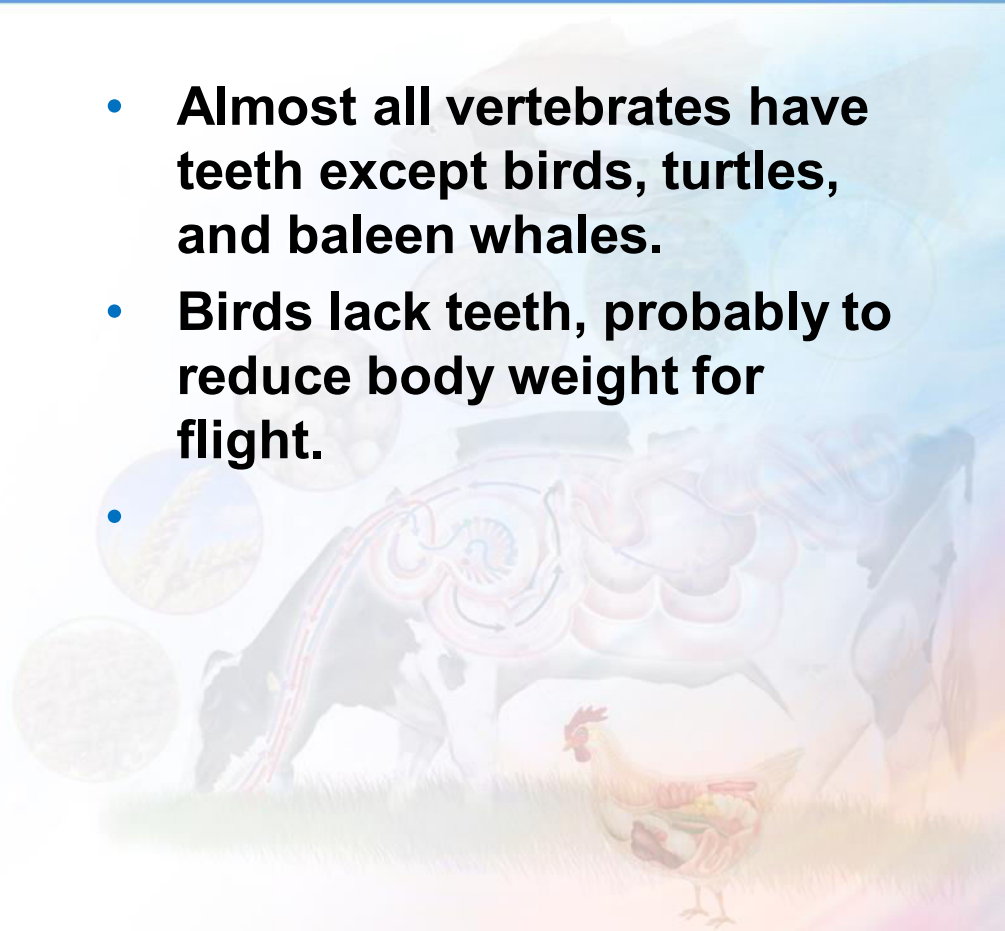
Animal Form & Function II



Teeth in Vertebrates

Teeth in Vertebrates

- Almost all vertebrates have teeth except birds, turtles, and baleen whales.
- Birds lack teeth, probably to reduce body weight for flight.
-



Teeth in Vertebrates

- The teeth of snakes slope backward to aid in the retention of prey while swallowing and the canine teeth of wolves are specialized for ripping food.



Teeth in Vertebrates





- **Teeth are specialized, depending on whether an animal feeds on plants or animals, and on how it obtains its food**

Teeth in Vertebrates



Beaver

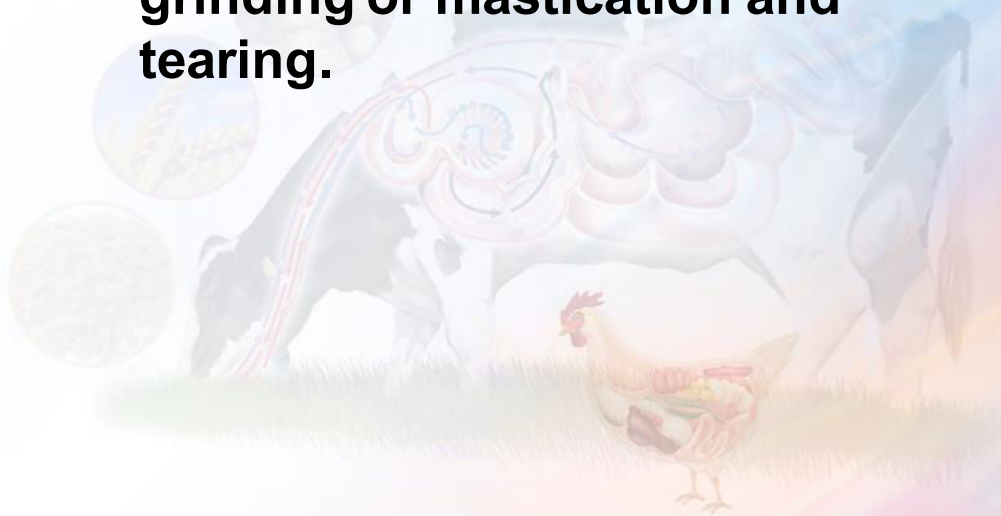


Elephant

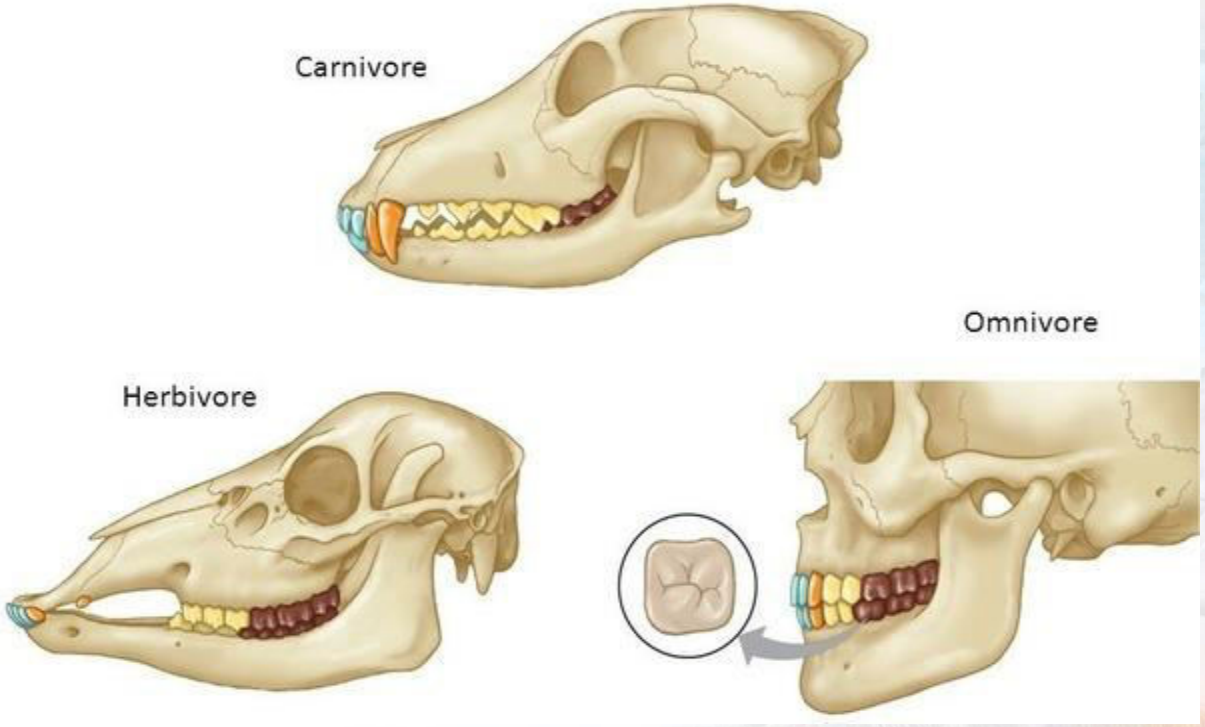
- Herbivores, such as deer, have predominantly grinding teeth.
- The front teeth of a beaver are used for chiseling trees and branches.
- The elephant has two of its upper, front teeth specialized as weapons and for moving objects.

Teeth in Vertebrates

- Because humans, pigs, bears, raccoons, and a few other mammals are omnivores, they have teeth that can perform a number of tasks—cutting grinding or mastication and tearing.



Teeth in Vertebrates



Animal Form & Function II

Module 28

Salivary Glands in Vertebrates

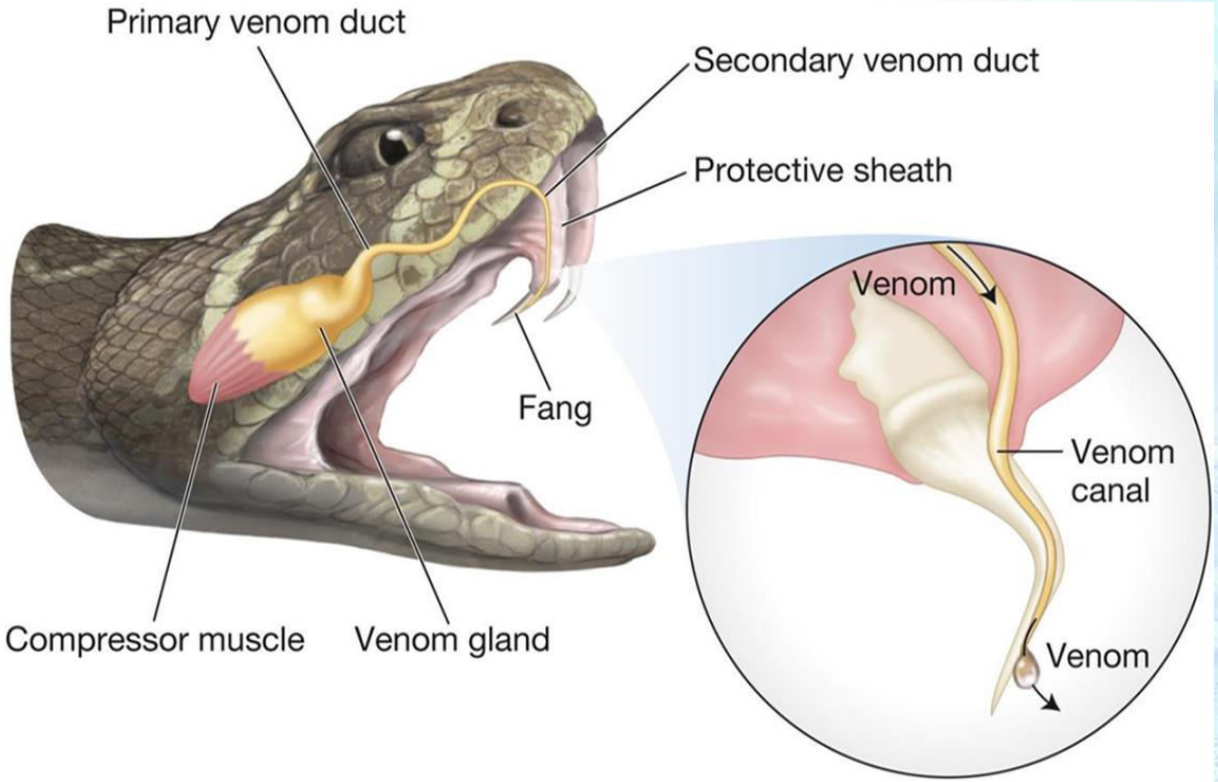
Salivary Glands in Vertebrates

- Most fishes lack salivary glands in the head region.
- Lampreys are an exception because they have a pair of glands that secrete an anticoagulant needed to keep their prey's blood flowing as they feed.

Salivary Glands in Vertebrates

- Modified salivary glands of some snakes produce venom that is injected through fangs to immobilize prey.
- Because the secretion of oral digestive enzymes is not an important function in amphibians or reptiles, salivary glands are absent.

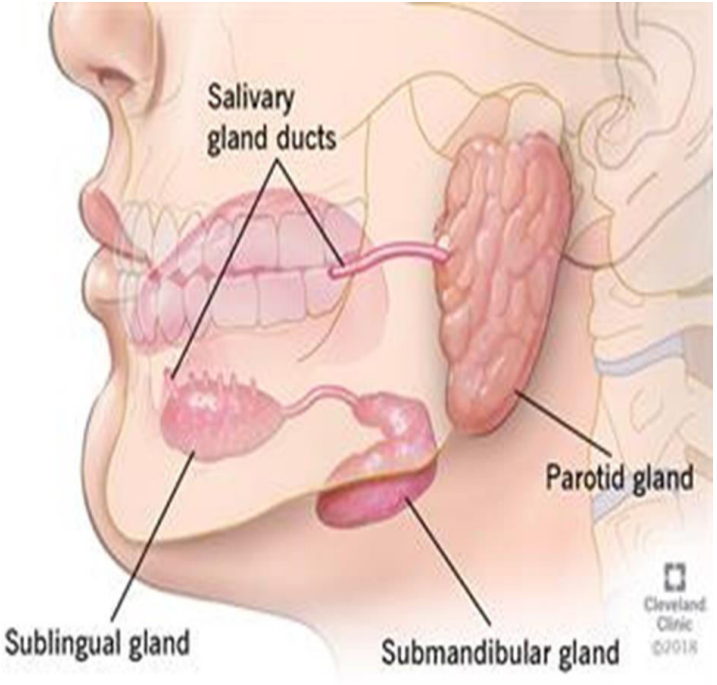
Salivary Glands in Vertebrates

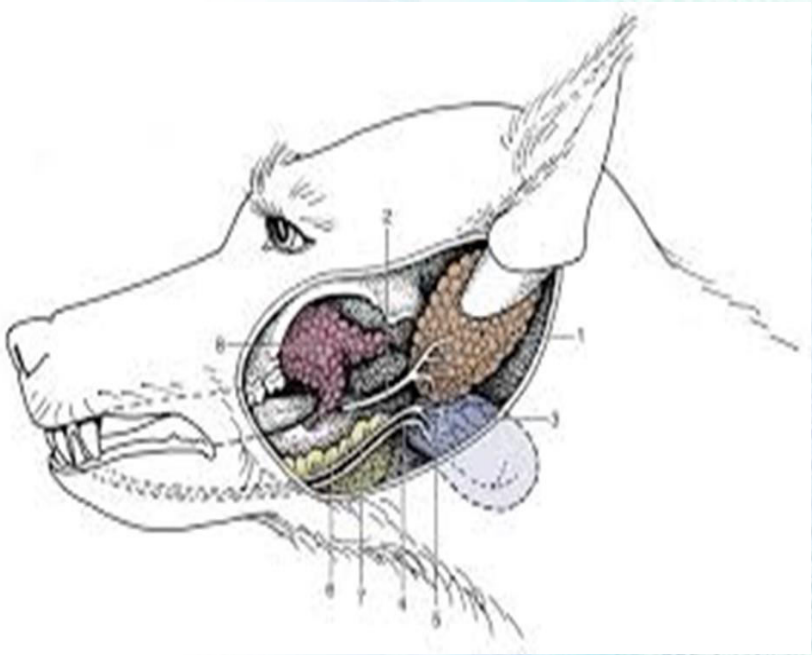


Salivary Glands in Vertebrates

- Most birds lack salivary glands, while all mammals have them.

Salivary Glands in Vertebrates





Salivary Glands in Vertebrates



Animal Form & Function II

Module 29

Animal Form & Function II

Esophagi In Vertebrates

Esophagi in Vertebrates

- The esophagus is a muscular tube that connects the throat (pharynx) with the stomach.
- The esophagus is lined by moist tissue called mucosa. Esophagus runs behind the windpipe (trachea).
- In fishes and amphibians, the Esophagus is short.

Esophagi in Vertebrates

- In amniotes the esophagus in much longer due to their longer necks.
- In birds particularly grain and seed-eaters, the esophagi have a crop that develops from the caudal portion of the esophagus.

Esophagi in Vertebrates

- A bird's crop is an expandable “muscular pouch near the gullet or throat.” It is used to store excess food for later digestion.
- Storing food in the crop ensures an almost continuous supply of food to the stomach and intestine for digestion.

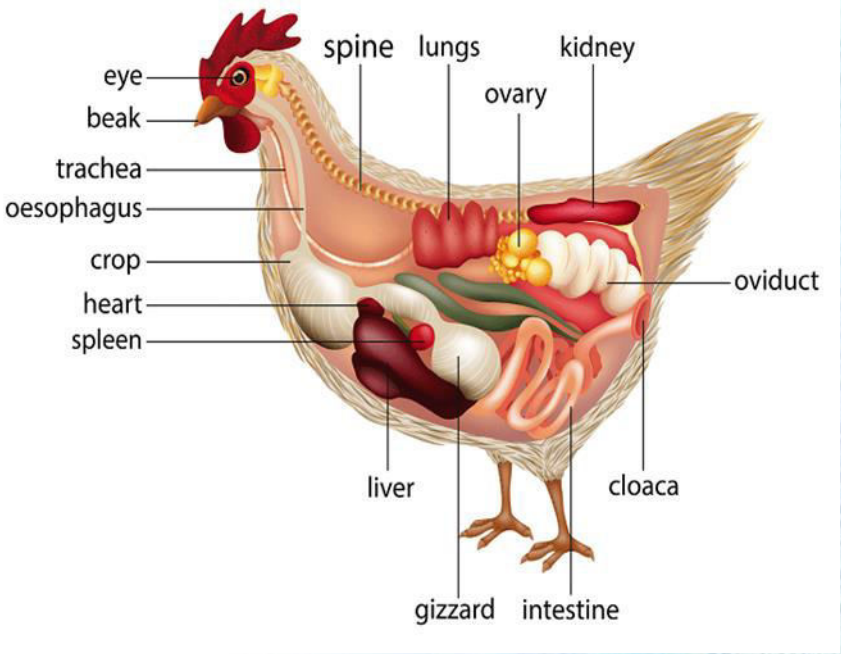
Esophagi in Vertebrates

- This structure allows these birds to reduce the frequency of feeding and still maintain a high metabolic rate.
- Birds that have more visible crops include vultures, hawks, falcons, eagles, gulls, and many types of quail.

Esophagi in Vertebrates



Crop in Vultures



Esophagus and Crop in Hen

Animal Form & Function II

Module 30

Animal Form & Function II

Stomach in Vertebrates

Stomach in Vertebrates

Stomach

- The stomach is a muscular organ located on the left side of the upper abdomen.
- The stomach receives food from the esophagus.
- The stomach secretes acid and enzymes that digest food.

Stomach in Vertebrates

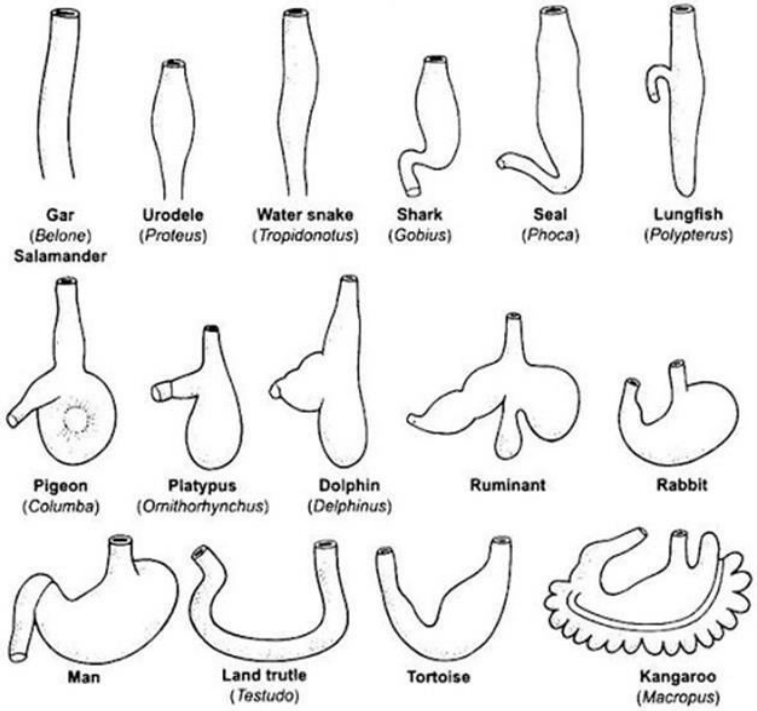


Fig. 43.10. Different shapes of vertebrate stomachs including human.

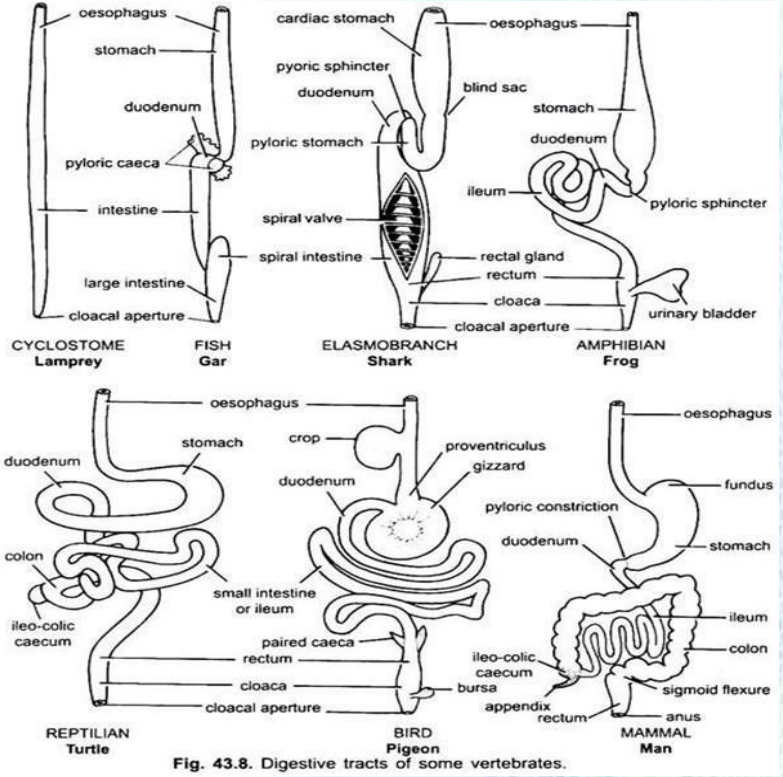


Fig. 43.8. Digestive tracts of some vertebrates.

Stomach in Vertebrates

Functions of Stomach:

- Temporary storage for food, which passes from the esophagus to the stomach where it is held for 2 hours or longer.
- Mixing and breakdown of food by contraction and relaxation of the muscle layers in the stomach.
- Digestion of food.

Animal Form & Function II

Module 31

Animal Form & Function II

Gizzard in Vertebrates

Gizzard in Vertebrates

Gizzard

- The gizzard, is an organ found in the digestive tract of some animals.
- Gizzard is also referred to as the ventriculus, gastric mill, and gigerium.
- Gizzard has a thick muscular wall .

Gizzard in Vertebrates

- Some fishes, some reptiles such as crocodile, and all birds have a gizzard for grinding up food.
- The bird's gizzard develops from the posterior part of the stomach called the ventriculus.

Gizzard in Vertebrates

- Grain-eating birds swallow pebbles (grit) which are often retained in the gizzard to facilitate the grinding process grains.

Gizzard in Vertebrates



END

Animal Form & Function II

Module 32

Animal Form & Function II

Stomach in Ruminants

Stomach in Ruminants

- Ruminant mammals are the animals that “chew their cud,” such as cows, sheep, and deer—show some of the most unusual modifications of the stomach.
- This method of digestion has evolved in animals that need to eat large amounts of food relatively quickly, but can chew the food at a more comfortable or safer location.

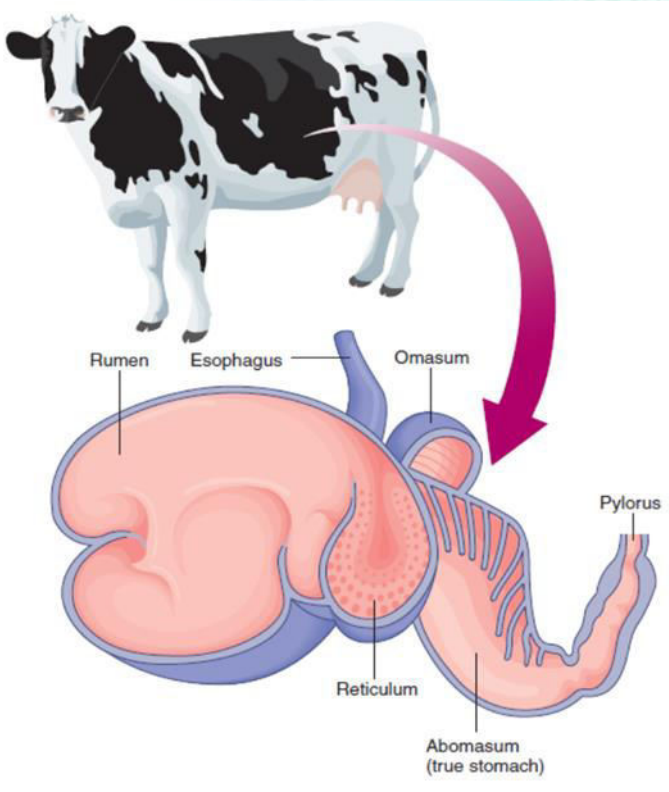
Stomach in Ruminants

- The ruminant stomach provides an opportunity for large numbers of microorganisms to digest the cellulose walls of grass and other vegetation.
- Cellulose contains a large amount of energy; however, animals generally lack the ability to produce the enzyme cellulase for digesting cellulose and obtaining its energy.

Stomach in Ruminants

- **Because gut microorganisms can produce cellulase, they have made the herbivorous lifestyle more effective.**

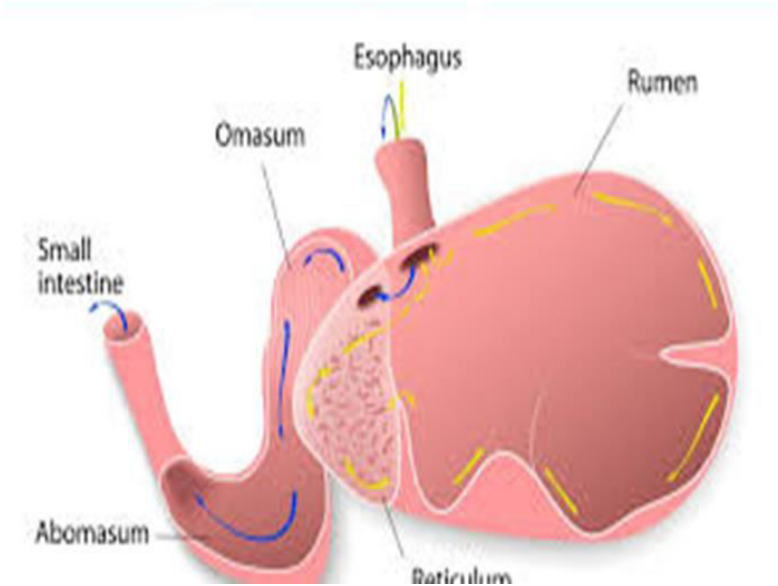
Stomach in Ruminants

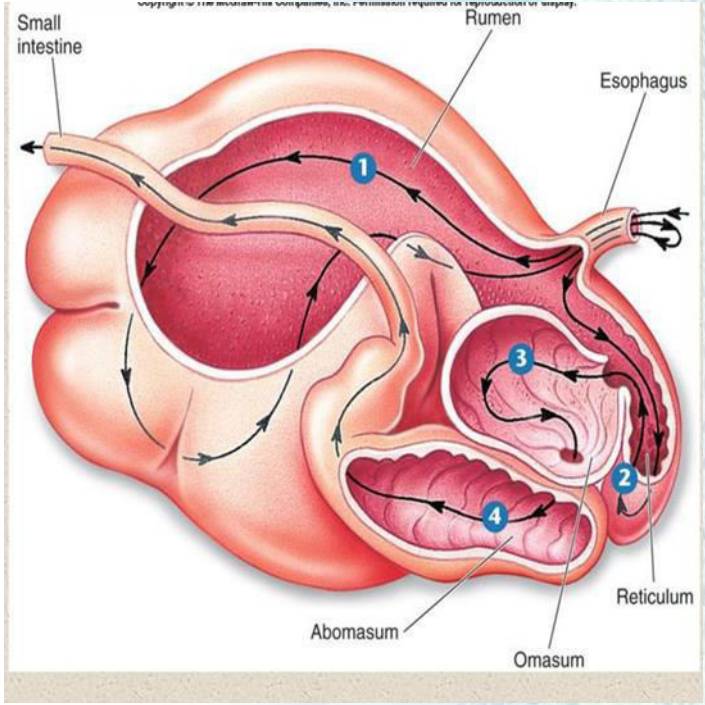


Stomach in Ruminants

- In ruminants, the upper portion of the stomach expands to form a large pouch, the rumen, and a smaller reticulum.
- The lower portion of the stomach consists of a small antechamber, the omasum, with a “true” stomach, or abomasum, behind it.

Stomach in Ruminants





Animal Form & Function II

Module 33

Animal Form & Function II

Caece in Vertebrates

Caece in Vertebrates

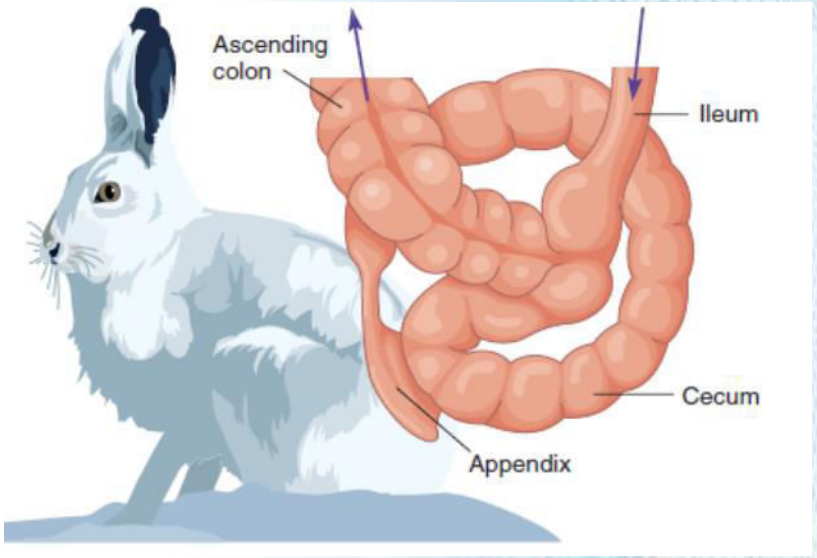
Caece

- The caecum is a short, pouch-like part of the large intestine between the ascending colon and vermiform appendix.
- In ruminants microorganisms attack the food before gastric digestion. but In typical non ruminant herbivores microbial action on cellulose occurs after digestion.

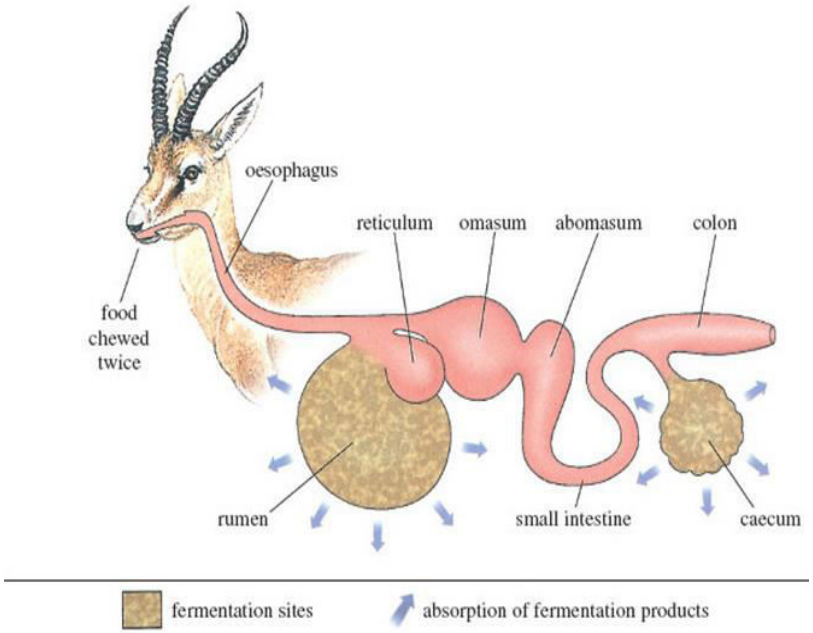
Caece in Vertebrates

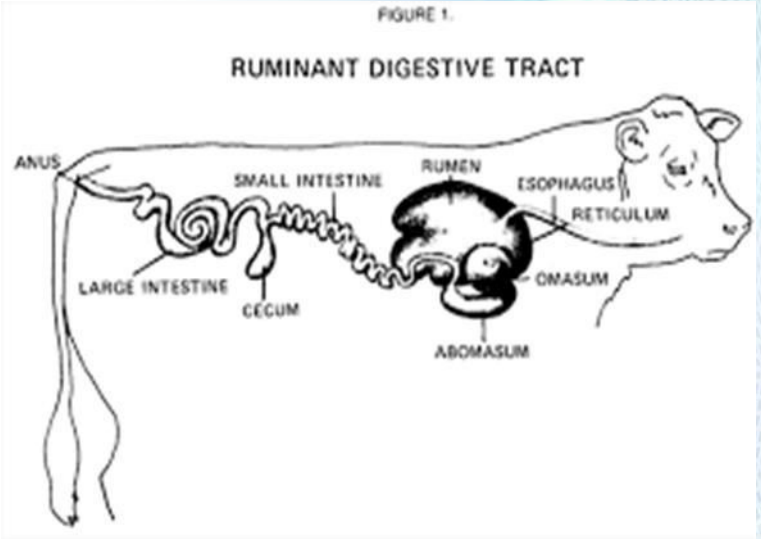
- Rabbits, horses, and rats digest cellulose by maintaining a population of microorganisms in their unusually large caecum
- Adding to this efficiency, a few non-ruminant herbivores, such as mice and rabbits.

Caece in Vertebrates



Caece in Vertebrates





Animal Form & Function II

Module 34

Animal Form & Function II

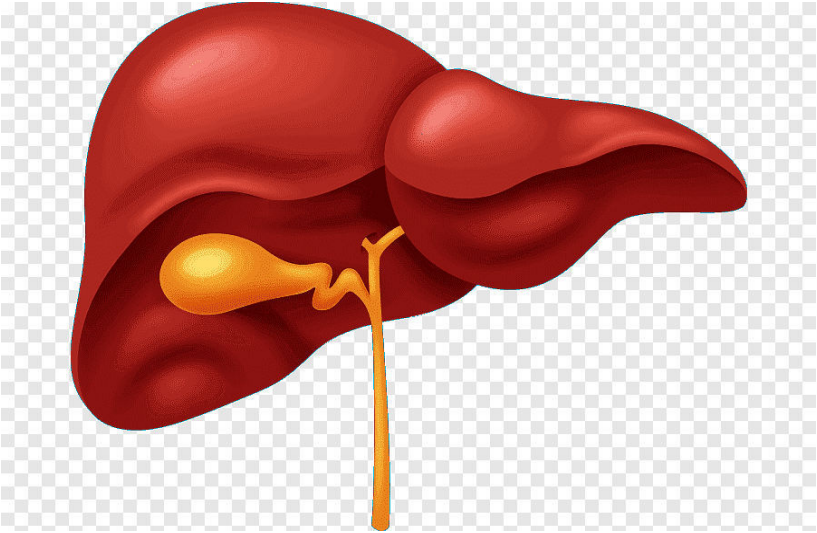
Livers in Vertebrates

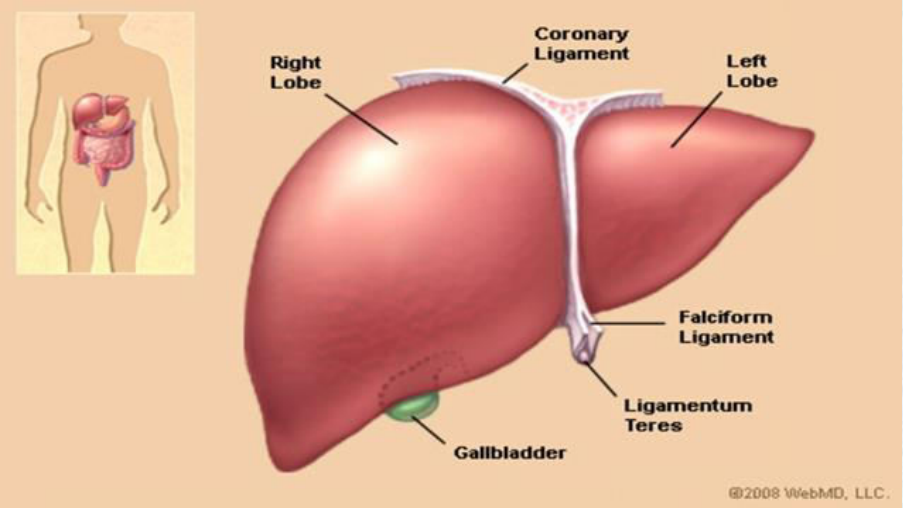
Livers in Vertebrates

The Liver

- In humans, the liver is an organ located in the right upper quadrant of the abdomen, below the diaphragm.
- Liver is found only in vertebrates.

Livers in Vertebrates





Livers in Vertebrates

Functions of Liver

- The liver manufactures bile.
- Bile is a fluid containing bile salts and bile pigments.
- Bile salts play an important role in the digestion of fats, although they are not digestive enzymes.

Livers in Vertebrates

- They emulsify dietary fat, breaking it into small globules (emulsification) on the surface of which the fat-digesting enzyme lipase can function.
- RBCs are phagocytized to form Bile Pigments in the spleen, liver and red bone marrow.

Livers in Vertebrates

- Phagocytosis cleaves the hemoglobin molecule, releasing iron, and the remainder of the molecule is converted into pigments that enter the circulation.
- These pigments are subsequently extracted from the circulation in the liver and excreted in the bile as bilirubin (“red bile”) and biliverdin (“green bile”).

Livers in Vertebrates

- Liver detoxifies various metabolites.
- Liver synthesizes proteins and produces bio chemicals necessary for digestion and growth.

Livers in Vertebrates

- Destruction of RBCs and formation of bile pigments.
- Protein metabolism
- Synthesis of plasma proteins
- Enzyme synthesis.
- Conversion of ammonia to urea.

Livers in Vertebrates

Carbohydrate metabolism

- Fat metabolism
- Removal or excretion of drugs

Livers in Vertebrates

- **Gallbladder** is a small hollow organ where bile is stored and concentrated before it is released into the small intestine.
- The structure and position of the gallbladder can vary significantly among animal species.

Livers in Vertebrates

- Because of the importance of bile in fat digestion, the gallbladder is relatively large in carnivores and vertebrates.
- Gallbladder is much reduced or absent in bloodsuckers, such as the lamprey, and in animals that feed primarily on plant food (e.g., some teleosts, many birds, and rats).

Animal Form & Function II

Module 35

Animal Form & Function II

Pancreata in Vertebrates

Pancreata in Vertebrates

- The pancreas is a gland organ.
- Pancreas produces enzymes, or digestive juices, and are released into the small intestine.
- There, it continues breaking down food that has left the stomach.

Pancreata in Vertebrates

- The pancreas also produces the hormones:
- i) Insulin and
- ii) Glucagon,
- These hormones are secreted into the bloodstream, where they regulates the body's glucose or sugar level.

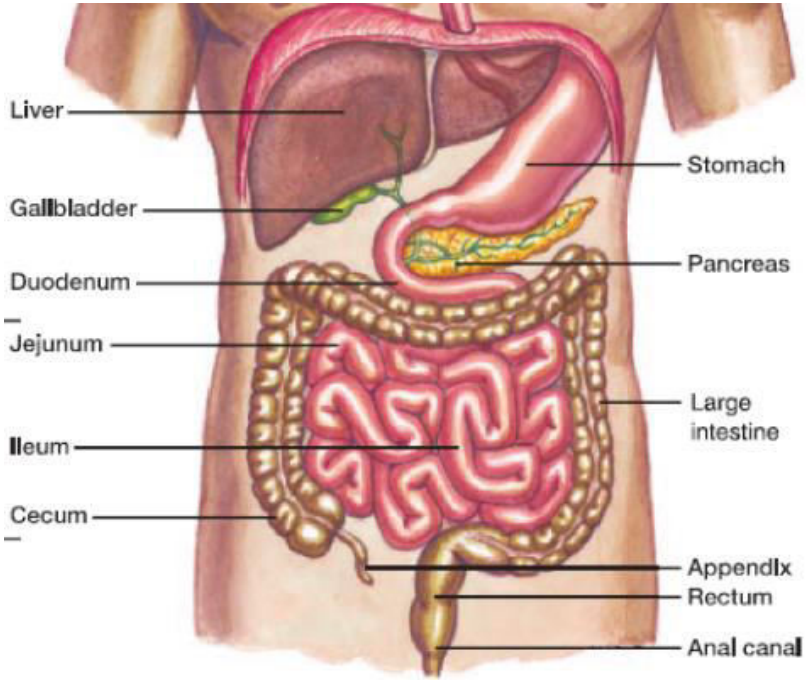
Pancreata in Vertebrates

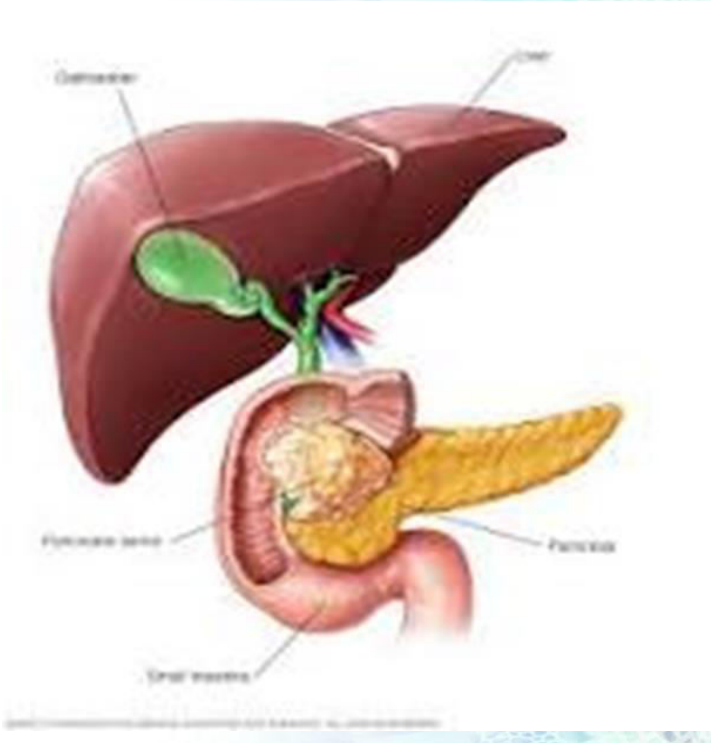
- Every vertebrate has a pancreas, however, in lampreys and lungfishes it is embedded in the wall of the intestine and is not a visible organ.
- Both endocrine and exocrine tissues are present, but the cell composition varies.

Pancreata in Vertebrates

- Pancreatic fluid containing many enzymes empties into the small intestine via the pancreatic duct.

Pancreata in Vertebrates





Animal Form & Function II

Module 36

Animal Form & Function II

Intestines in Vertebrates

Intestine in Vertebrates

- The intestines are a long, continuous tube running from the stomach to the anus.
- Most absorption of nutrients and water takes place in the intestines.
- The intestines include the small intestine, large intestine, and rectum.

Intestine in Vertebrates

- The small intestine (small bowel) is about 20 feet long and about an inch in diameter.
- The configuration and divisions of the small and large intestines vary greatly among vertebrates.
- Intestines are closely related to the animal's type of food, body size, and levels of activity.

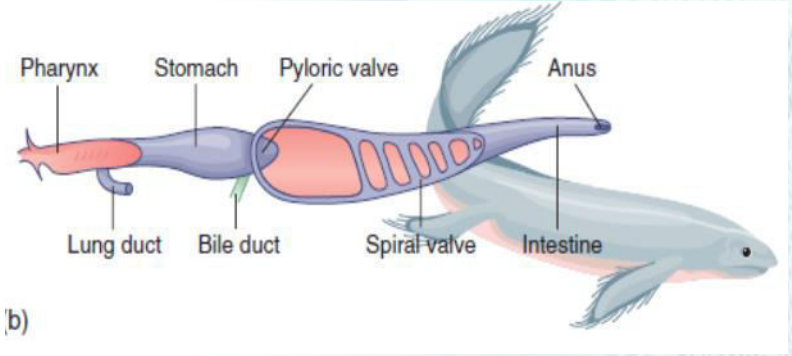
Intestine in Vertebrates

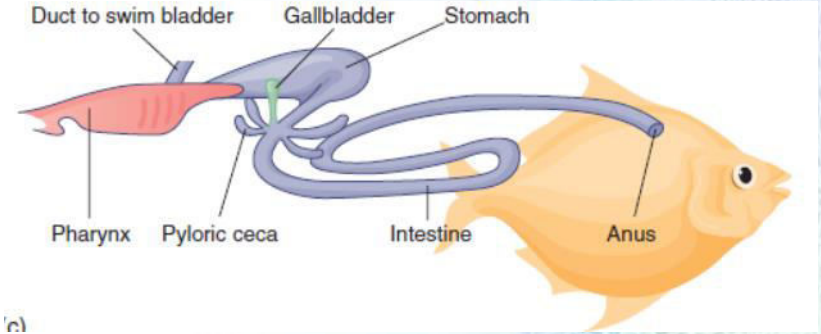
- For example, cyclostomes, chondrichthian fishes, and primitive bony fishes have short, nearly straight intestines that extend from the stomach to the anus.
- In more advanced bony fishes, the intestine increases in length and begins to coil.
- The intestines are moderately long in most amphibians and reptiles.

Intestine in Vertebrates

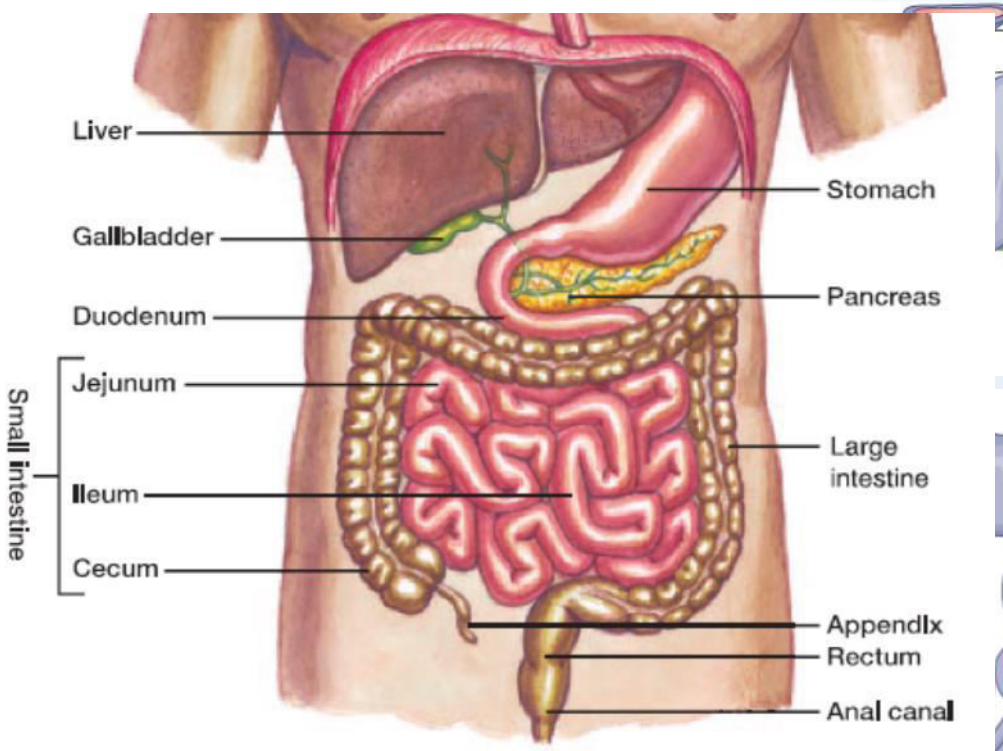
- In birds and mammals, the intestines are longer and have more surface area than those of other tetrapods.
- Birds typically have two ceca, and mammals have a single cecum at the beginning of the large intestine.
- The large intestine is much longer in mammals than in birds, and it empties into the cloaca in most vertebrates.

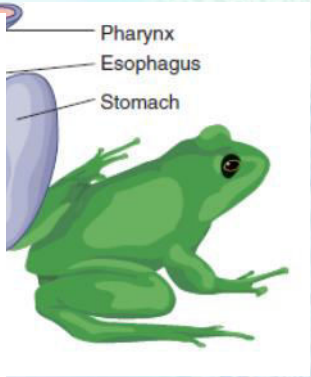
Intestine in Vertebrates

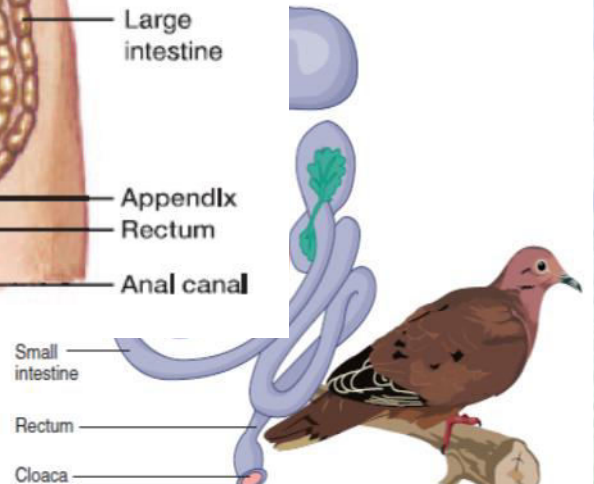




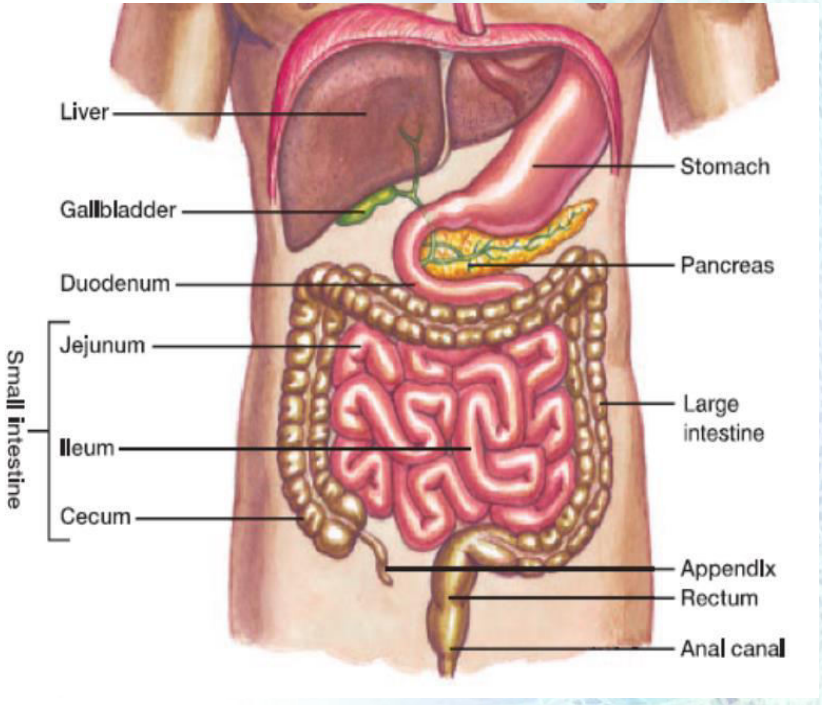
Intestine in Vertebrates







Intestine in Vertebrates



Animal Form & Function II

Module 37

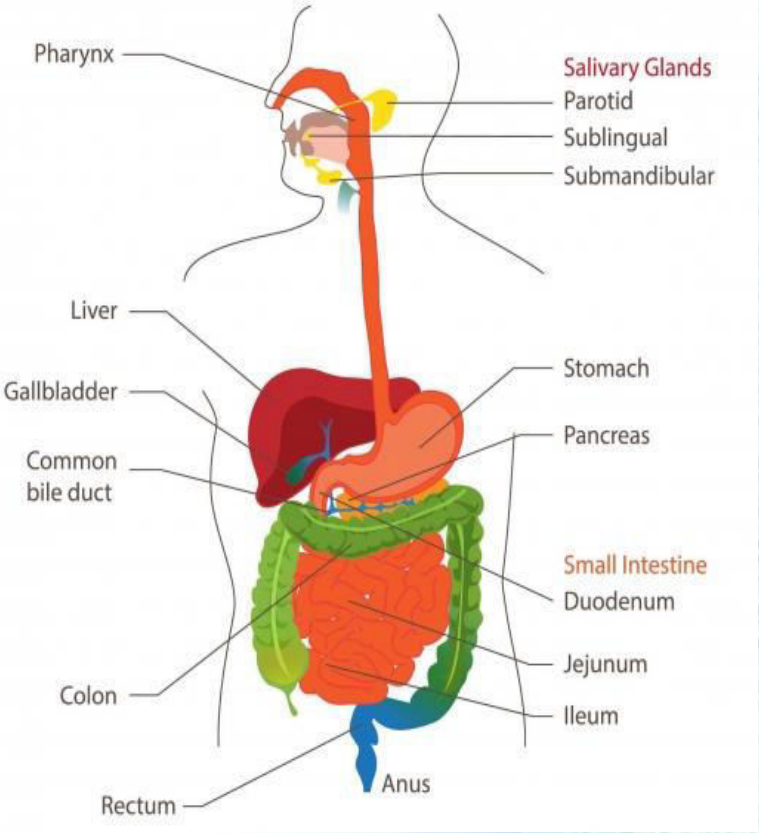
Mammalian Digestive System

Mammalian Digestive System

The Mammalian Digestive System

- The digestive system of an omnivore has the mechanical and chemical ability to process many kinds of foods.
- Humans, pigs, bears, raccoons, and a few other mammals are omnivores.

Mammalian Digestive System



Mammalian Digestive System

Major Organs and Parts of the Human Digestive System.

- Food passes from the mouth through the pharynx and esophagus to the stomach.
- From the stomach, it passes to the small intestine, where nutrients are broken down and absorbed into the circulatory and lymphatic systems.

Mammalian Digestive System

- Nutrients then move to the large intestine, where water is reabsorbed, and feces form.
- Feces exit the body via the anal canal.

Mammalian Digestive System

- In mammals the process of digestion and absorption of nutrients includes:
- 1. Ingestion—eating.
- 2. Peristalsis—the involuntary, sequential muscular contractions that move ingested nutrients along the digestive tract.
- 3. Segmentation—mixing the contents in the digestive tract.

Mammalian Digestive System

- **4. Secretion—the release of hormones, enzymes, and specific ions and chemicals that take part in digestion.**
- **5. Digestion—the conversion of large nutrient particles or molecules into small particles or molecules.**

Mammalian Digestive System

- **6. Absorption—the passage of usable nutrient molecules from the small intestine into the bloodstream and lymphatic system for the final passage to body cells.**
- **7. Defecation—the elimination from the body of undigested and unabsorbed material as waste.**

Animal Form & Function II

Module 38

Animal Form & Function II

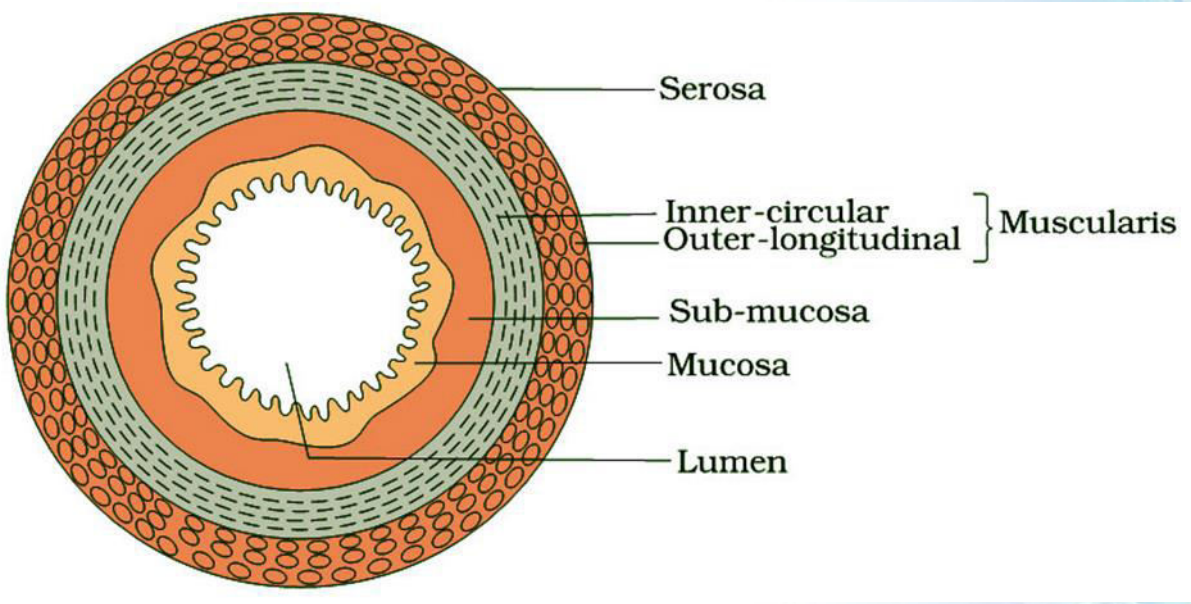
Layers of Mammalian Gastro Intestinal (GI)Tract

Layers of Mammalian GI Tract

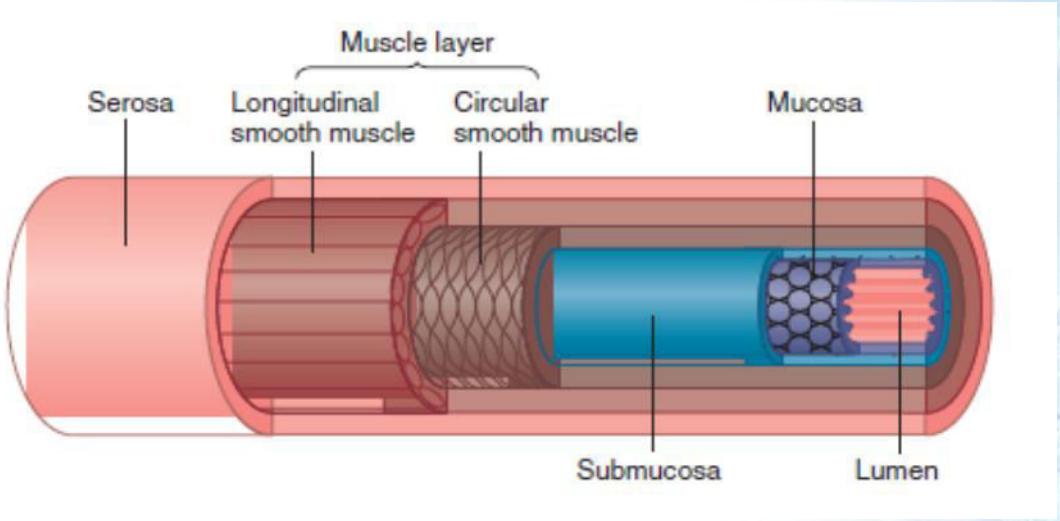
Layers of GI Tract

- The GI tract is composed of four layers.
- Each layer has different tissues and functions.
- From the inside out they are called: mucosa, submucosa, muscularis, and serosa.

Layers of Mammalian GI Tract



Layers of Mammalian GI Tract



Layers of Mammalian GI Tract

- **The mucosa** is the innermost layer, and functions in absorption and secretion.
- It is composed of epithelium cells and a thin connective tissue.
- The mucosa contains specialized goblet cells that secrete sticky mucus throughout the GI tract.

Layers of Mammalian GI Tract

- On the mucosa layer, small finger-like projections called villi and microvilli help to increase surface area for nutrient absorption.
- the submucosa is the layer of connective tissue that supports the mucosa, as well as joins the mucosa to the bulk of underlying smooth muscle.

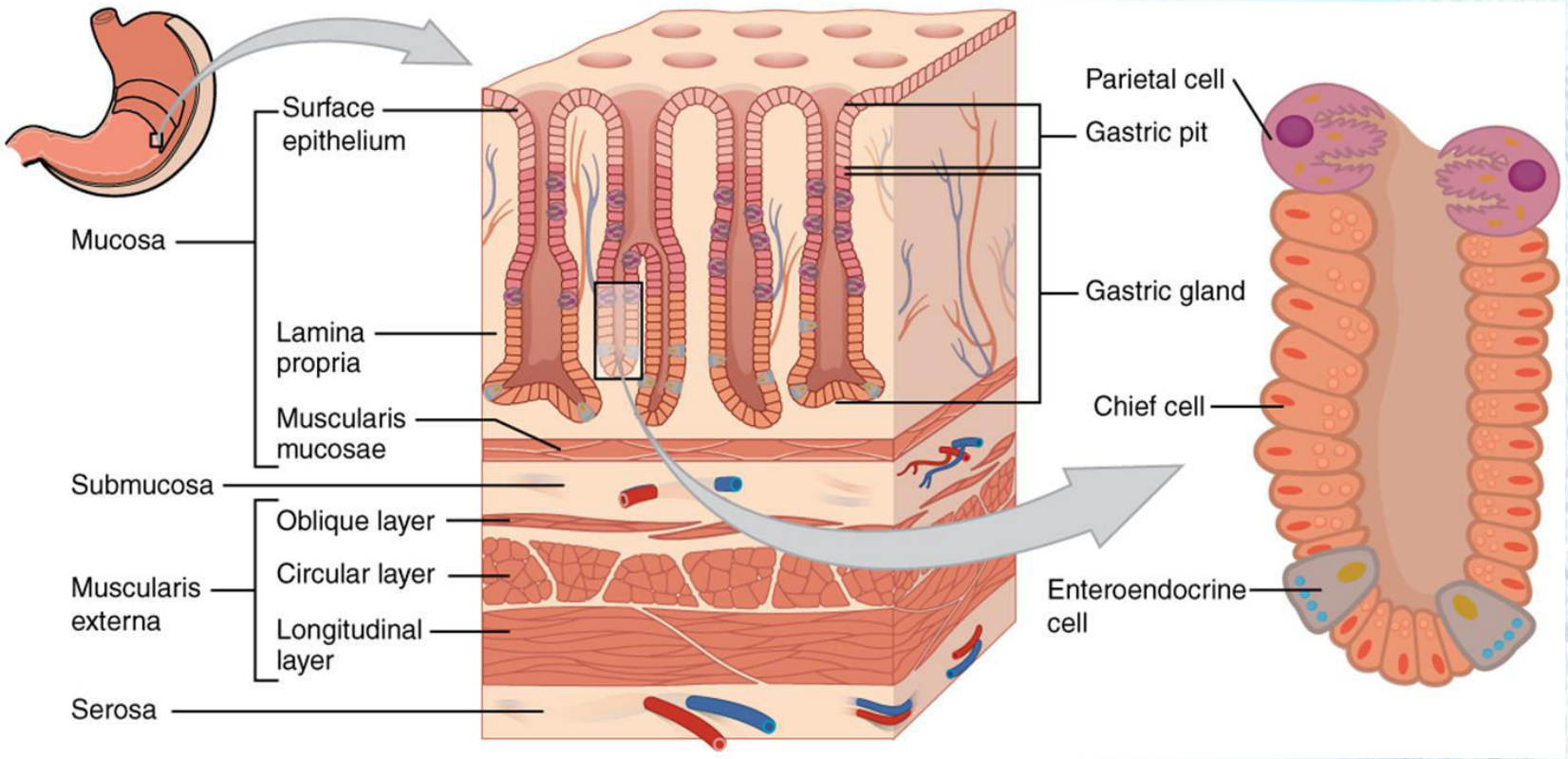
Layers of Mammalian GI Tract

- **The muscularis, or muscularis externa, consists of an inner circular muscular layer and a longitudinal outer muscular layer.**
- **The coordinated contractions of these layers is called peristalsis, which propels the food through the GI tract.**

Layers of Mammalian GI Tract

- **Serosa.** This structure consists of connective tissue covered by a simple squamous epithelium

Layers of Mammalian GI Tract



Animal Form & Function II

Module 39

Animal Form & Function II

Human Oral Cavity

Human Oral Cavity

- A pair of lips protects the oral cavity(mouth).
- The lips are highly vascularized, skeletal muscle tissue with an abundance of sensory nerve endings.
- Lips help retain food as it is being chewed and play a role in phonation (the modification of sound).

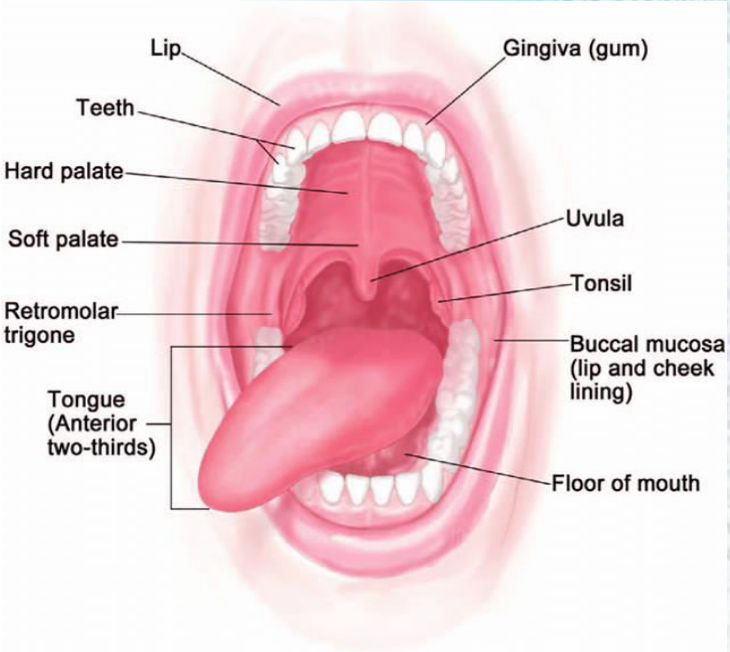
Human Oral Cavity

- **The oral cavity** is bounded by the roof, the hard and soft palate the floor of the mouth under the tongue, the side cheeks and the small area behind the wisdom teeth.
- The oral cavity contains the tongue, gums and teeth.

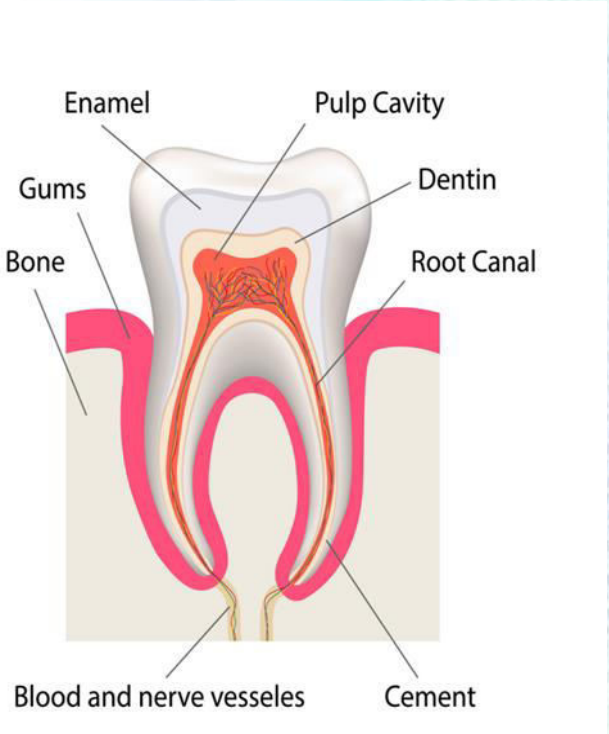
Human Oral Cavity

- Mammals can mechanically process a wide range of foods because of their teeth.
- Teeth are covered with enamel (the hardest material in the body).
- In mammals there are four types of teeth:
- Incisors, Canine, Premolars and Molars.

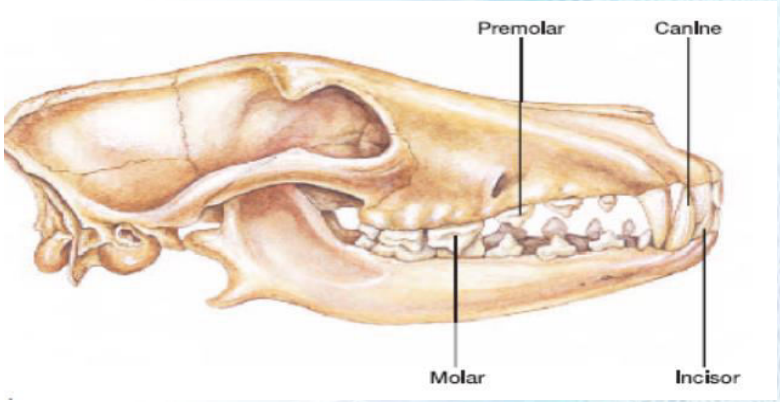
Human Oral Cavity



Human Oral Cavity



Human Oral Cavity



Human Oral Cavity

- Saliva moistens food, binds it with mucins (glycoproteins), and forms the ingested food into a moist mass called a bolus.
- Saliva also contains bicarbonate ions (HCO_3), which buffer chemicals in the mouth, the enzyme lysozyme, which kill microorganisms.

Human Oral Cavity

- It also contributes an enzyme (amylase) necessary for the initiation of carbohydrate digestion.

Animal Form & Function II

Module 40

Animal Form & Function II

Digestion in Oral Cavity

Digestion in Oral Cavity

Following are the main functions of the oral cavity:

- Selection of food
- Grinding or mastication of food
- Lubrication and
- Digestion

Selection of Food:

Food is selected by the senses of smell, sight and taste.

Digestion in Oral Cavity

Grinding of food:

- After selection, food is ground by means of molar teeth into smaller pieces.
- *This is useful because:*
- Esophagus allows only smaller pieces to pass through
- Smaller pieces have much more surface for the enzymes to enter.

Digestion in Oral Cavity

Lubrication and Digestion:

- These are the chief functions of the oral cavity accomplished by saliva.
- Saliva is secreted by three pairs of salivary glands:
- Sublingual, sub maxillary and parotid.

Digestion in Oral Cavity

Saliva produced by these glands contain:

- Water and mucus - serves to moisten and lubricate food.
- Sodium bicarbonate which buffer chemicals in mouth,.
- An enzyme, lysozyme, and thiocyanate ions (SCN) that kill microorganisms.

Digestion in Oral Cavity

- Carbohydrate digesting enzyme –amylase or ptyalin for the initiation of carbohydrate digestion.
- Starch and Glycogen-- Maltose.
- Fresh saliva is alkaline with pH nearly 8, quickly loses CO_2 and gets to pH 6.

Digestion in Oral Cavity

Animal Form & Function II

Module 41

Animal Form & Function II

Swallowing of Food

Swallowing of Food

Swallowing of Food:

- Masticated, softened, partly digested and slimy food mass is rolled into bolus.
- Bolus is then pushed back of the mouth by the action tongue and muscles of the pharynx.

Swallowing of Food

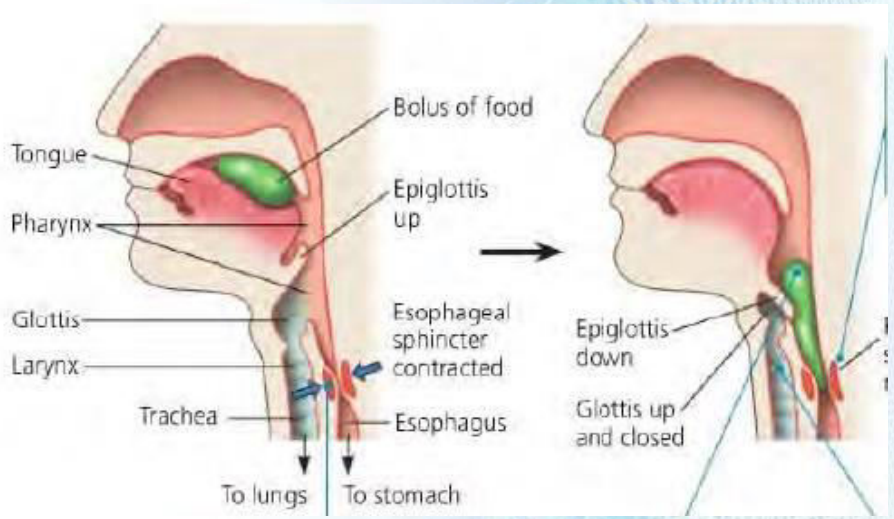
Events of Swallowing

- The tongue moves upward and backward forcing the bolus to the back of the mouth cavity.
- The backward movement of the tongue pushes the soft palate up and closes the nasal opening at the back.
- At the same time the tongue forces the epiglottis into more or less horizontal position thus closing the opening of the wind pipe.

Swallowing of Food

- The or bolus is then diverted safely down the esophagus.
- The beginning of swallowing action is voluntary but once the food reached the back of the mouth, swallowing becomes automatic.
- The food is then forced down the esophagus by peristalsis.

Swallowing of Food



Swallowing of Food

Animal Form & Function II

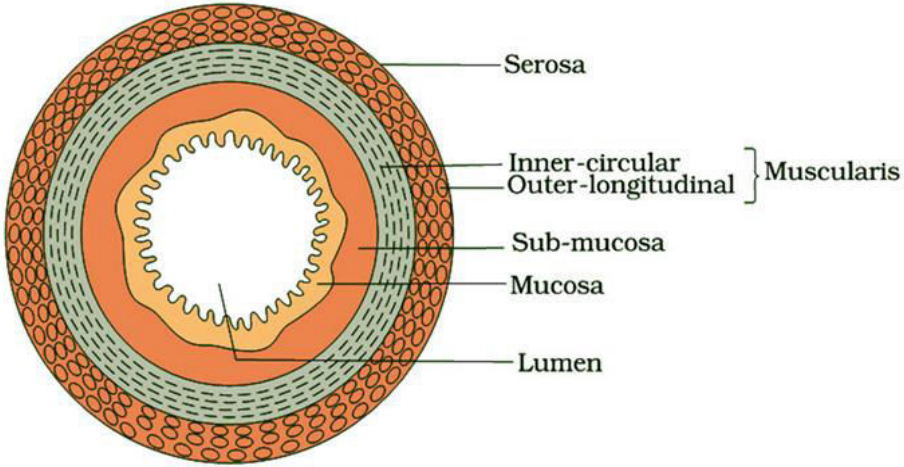
Module 42

Gastrointestinal Motility

Gastrointestinal Motility

- As with any organ, the function of the gastrointestinal tract is determined by the type of tissues it contains.
- Most of the mammalian gastrointestinal tract has the same anatomical structure along its entire length.

Gastrointestinal Motility



- From the outside inward is a thin layer of connective tissue called the serosa.
- The serosa forms a moist epithelial sheet called the peritoneum.
- This peritoneum lines the entire abdominal cavity and covers all internal organs.

Gastrointestinal Motility

- Next are the longitudinal smooth-muscle layer and circular smooth-muscle layer.
- Underneath this muscle layer is the submucosa.
- The submucosa contains connective tissue, blood, and lymphatic vessels.

Gastrointestinal Motility

- The mucosa faces the central opening, which is called lumen.
- The coordinated contractions of the muscle layers of the gastrointestinal tract mix the food material with various secretions and move the food from the oral cavity to the rectum.

Gastrointestinal Motility

- The two types of movement involved are peristalsis and segmentation.

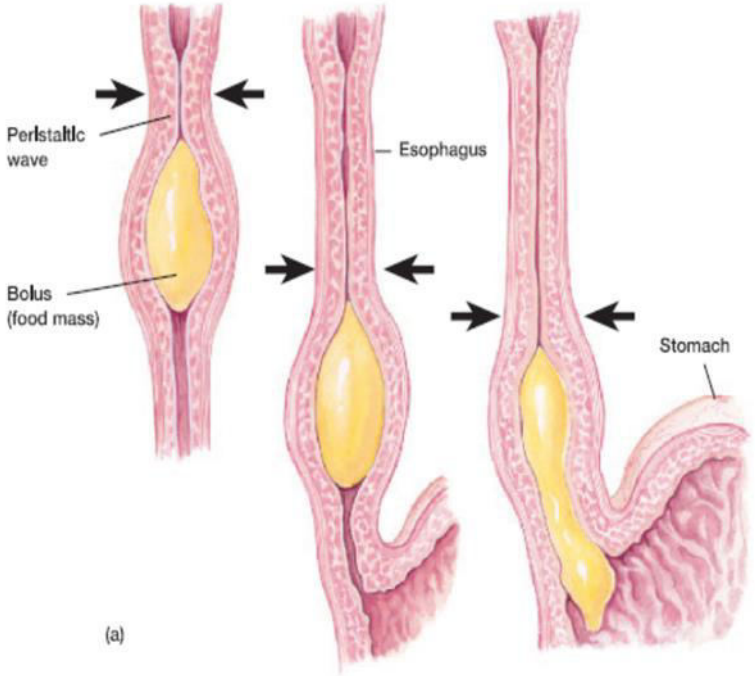
Animal Form & Function II

Module 43

Animal Form & Function II

Peristalsis and Segmentation

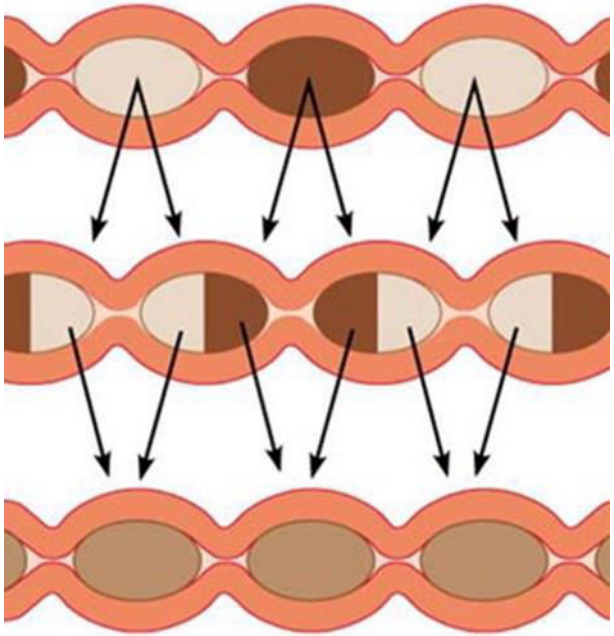
Peristalsis and Segmentation



Peristalsis:

- The characteristic involuntary movements of the digestive tract by which food moves along the cavity of the canal.
- It comprises relaxation of circular smooth muscles, then their contraction behind the chewed material to keep it from moving backward, then longitudinal contraction to push it forward.

Peristalsis and Segmentation



Segmentation

- In segmentation, simultaneous muscular contractions of many sections of the intestine help mix nutrients with digestive secretions.
- This is coordinated contractions of the muscle layers of the GI tract.

Peristalsis and Segmentation

Antiperistalsis

- The reverse peristalsis is called Antiperistalsis, which occurs occasionally.
- These movements cause the passage of food from the intestine back into the stomach or even into the mouth.
- This is also called vomiting.

Peristalsis and Segmentation

- During peristalsis the food advances through the GI tract when the rings of circular smooth muscle contract behind it and relax in front of it.
- The small and large intestines also have rings of smooth muscles that repeatedly contract and relax.

Peristalsis and Segmentation

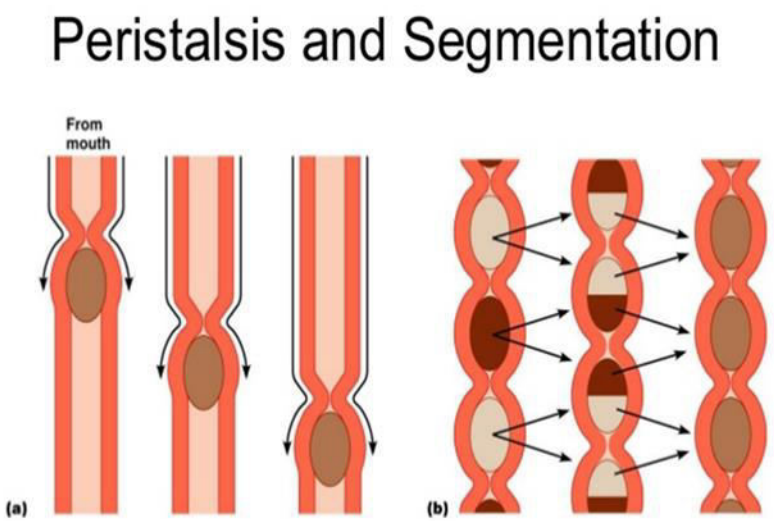
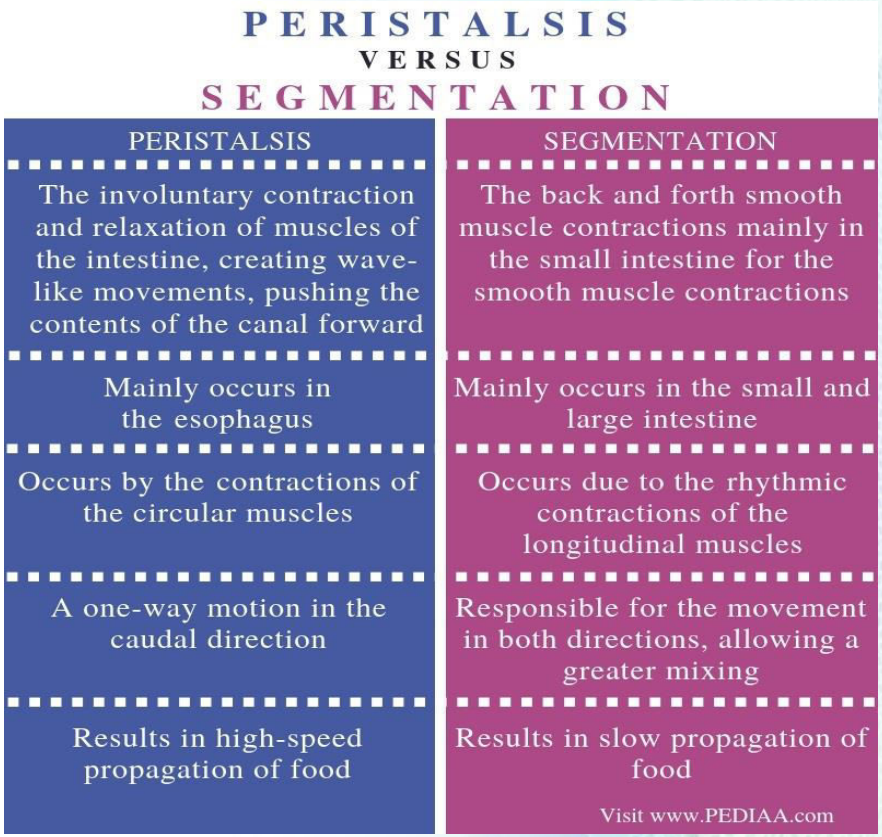


Figure 23.3

- This create an oscillating back-and-forth movement in the same place, called segmentation.
- This movement mixes the food with digestive secretions and increases the efficiency of absorption.

Peristalsis and Segmentation



Animal Form & Function II

Module 44

Control of Gastrointestinal Tract Motility

Control of GI Tract Motility

- Control of gastrointestinal activity is based on the volume and composition of food in the lumen of the gut.
- Ingested food distends the gut and stimulates mechanical receptors in the gut wall.
- In addition, digestion of carbs, lipids, and proteins stimulates various chemical receptors in the gut wall.

Control of GI Tract Motility

Nervous Control

- Signals from these mechanical and chemical stimuli travel through nerve plexuses in the gut wall to control the muscular contraction that leads to peristalsis and segmentation.
- This also controls the secretion of various substances (e.g., mucus, enzymes) into the gut lumen.

Control of GI Tract Motility

- In addition to this local control, long-distance nerve pathways connect the receptors and effectors with the CNS.
- Either or both of these pathways function to maintain homeostasis in the gut.

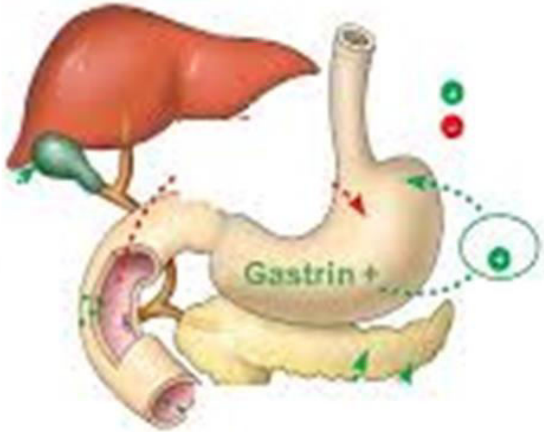
Control of GI Tract Motility

Endocrine Control

- The endocrine cells of the gastrointestinal tract also produce hormones that help regulate secretion, digestion, and absorption. (Gastrin, Secretin)

Control of GI Tract Motility

Gastrin



- Enters the bloodstream →→→→
- goes back to the stomach affecting the **parietal cells**

Animal Form & Function II

Module 45

Animal Form & Function II

Pharynx in Vertebrates

Pharynx in Vertebrates

Pharynx

- It is the common passageway for both the digestive and respiratory tracts.
- How both air and swallowed foods and liquids pass from the mouth into the respiratory and digestive tract respectively?
- It is the epiglottis that controls the traffic.

Pharynx in Vertebrates

- Initiation of the swallowing reflex can be voluntary, but most of the time it is involuntary.
- When swallowing begins, sequential, involuntary contractions of smooth muscles in the walls of the esophagus propel the bolus or liquid to the stomach.

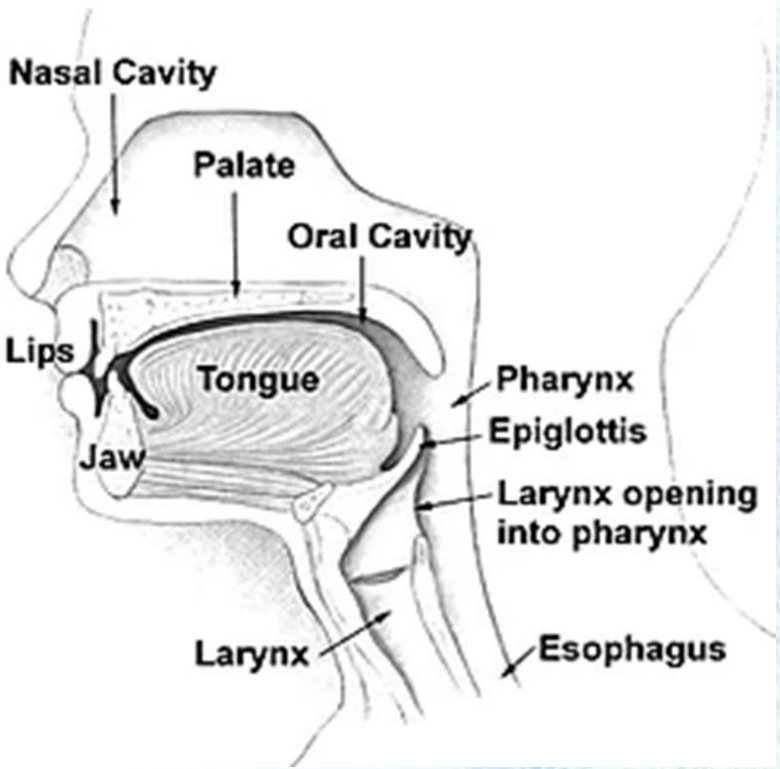
Pharynx in Vertebrates

- Neither the pharynx nor the esophagus contribute to digestion.

The epiglottis

- The epiglottis temporarily seals off the opening (glottis) to the trachea so that swallowed food does not enter the trachea.

Pharynx in Vertebrates



Animal Form & Function II

Module 46

Stomach in Vertebrates
Part I

Stomach in Vertebrates Part I

Structure of Stomach

- The mammalian stomach is a muscular, distensible sac.
- The stomach is made up of an inner mucous membrane containing thousands of gastric glands.
- These gastric glands are composed of three types of cells:

Stomach in Vertebrates Part I

- **Parietal cells,** secrete a solution containing HCl.
- **Chief cells** secrete pepsinogen, the precursor of the enzyme pepsin.
- Both of the cells are in the pits of the gastric glands.

Stomach in Vertebrates Part I

- **Mucous cells** secrete mucus that coats the surface of the stomach and protects it from the HCl and digestive enzymes.
- The esophagus and mouth have a much thinner mucous-cell layer.

Stomach in Vertebrates Part I

Functions of Stomach

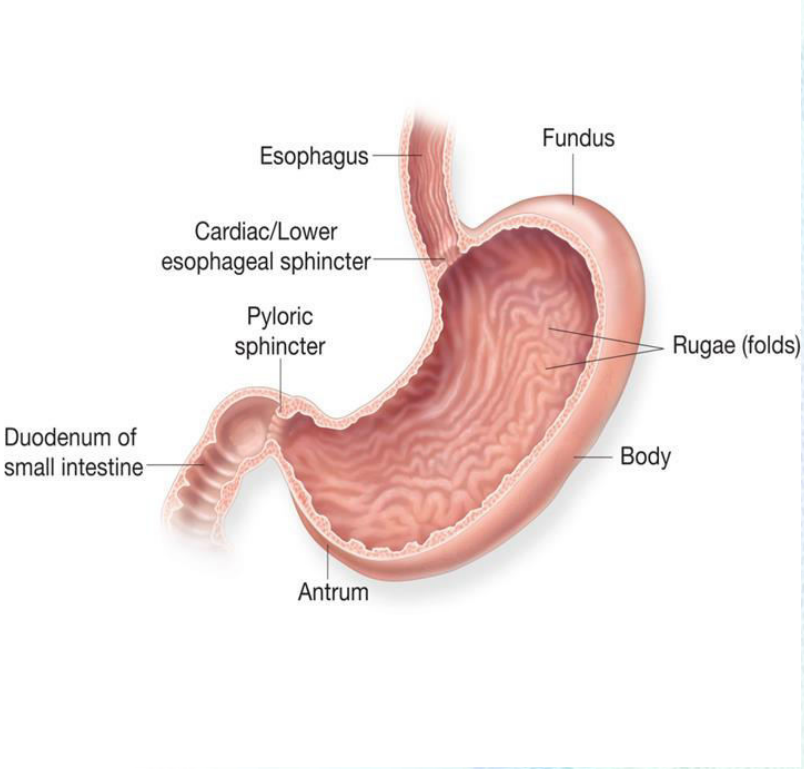
The mammalian stomach has three main functions.

- **1. stores and mixes the food bolus received from the esophagus.**
- **2. Secretes substances (enzymes, mucus, and hydrochloric acid (HCl) that start the digestion of proteins.**

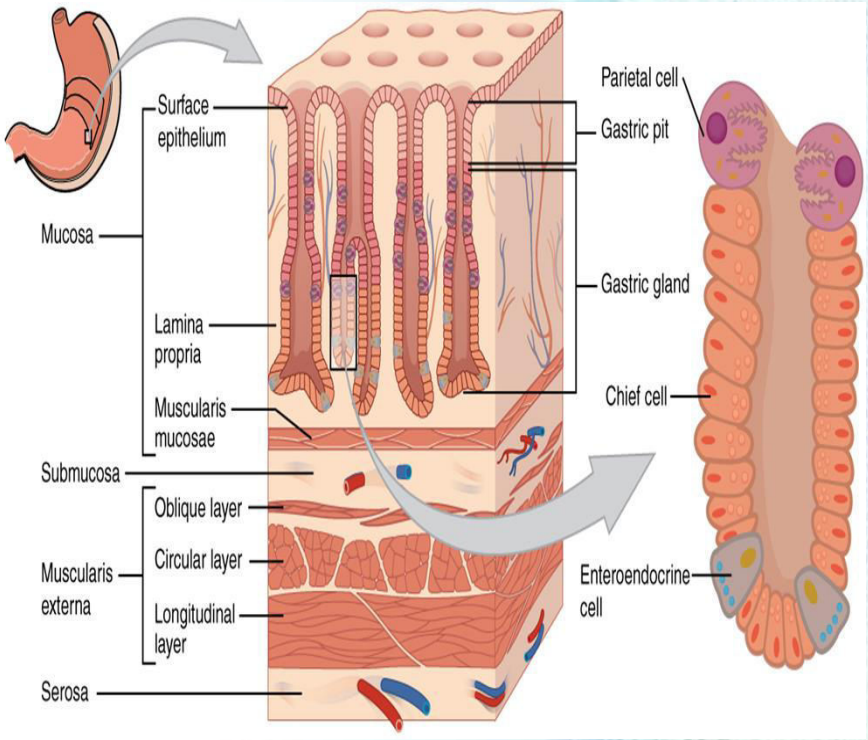
Stomach in Vertebrates Part I

- 3. helps control the rate at which food moves into the small intestine via the pyloric sphincter.
- Endocrine cells in stomach mucosa release the hormone, gastrin, which travels to target cells in the gastric glands, further stimulating them to release gastric juice.

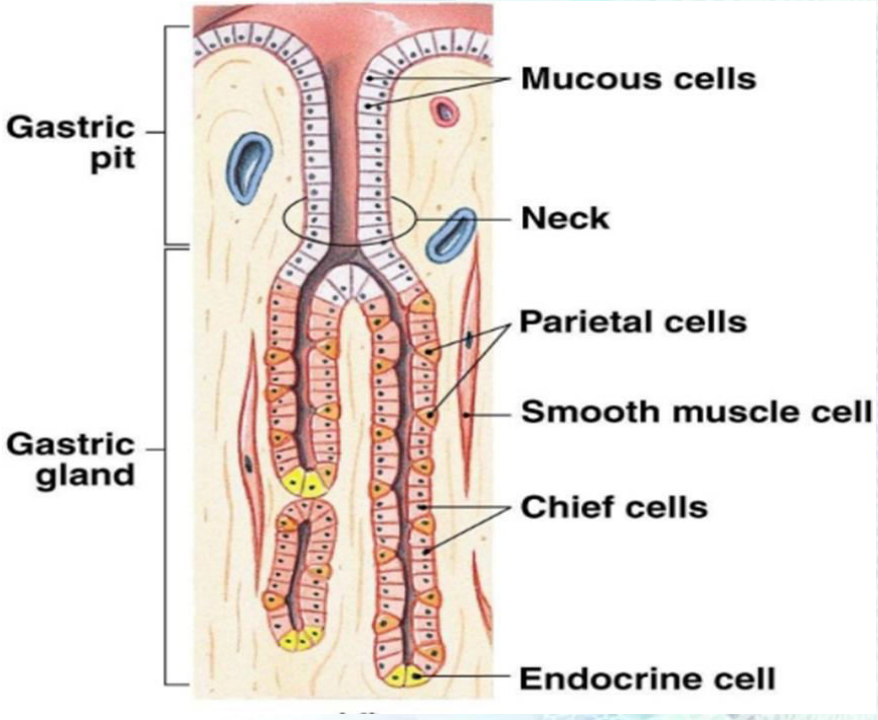
Stomach in Vertebrates Part I



Stomach in Vertebrates Part I



Stomach in Vertebrates Part I



Animal Form & Function II

Module 47

Animal Form & Function II

Stomach
Part II

Stomach Part II

- When the bolus of food enters the stomach, it distends the walls of the stomach.
- This distention, as well as the act of eating, causes the gastric pits to secrete HCl and pepsinogen.

Stomach Part II

- The H+ ions cause pepsinogen to be converted into the active enzyme pepsin.
- As pepsin, mucus, and HCl mix with and begin to break down proteins, smooth mucosal muscles contract and vigorously churn and mix the food bolus.

Stomach Part II

- About three to four hours after a meal, the stomach contents have been sufficiently mixed and forms semiliquid mass called chyme.
- The pyloric sphincter regulates the release of the chyme into the small intestine.
-

Stomach Part II

- When the stomach is empty, peristaltic waves cease, however, after about 10 hours of fasting, new waves may occur in the upper region of the stomach.
- These waves can cause “hunger pangs” as sensory nerve fibers carry impulses to the brain.

Stomach Part II

Animal Form & Function II

Module 48

Small Intestine
(Parts and Structure)

Small Intestine (Parts and Structure)

Small Intestine

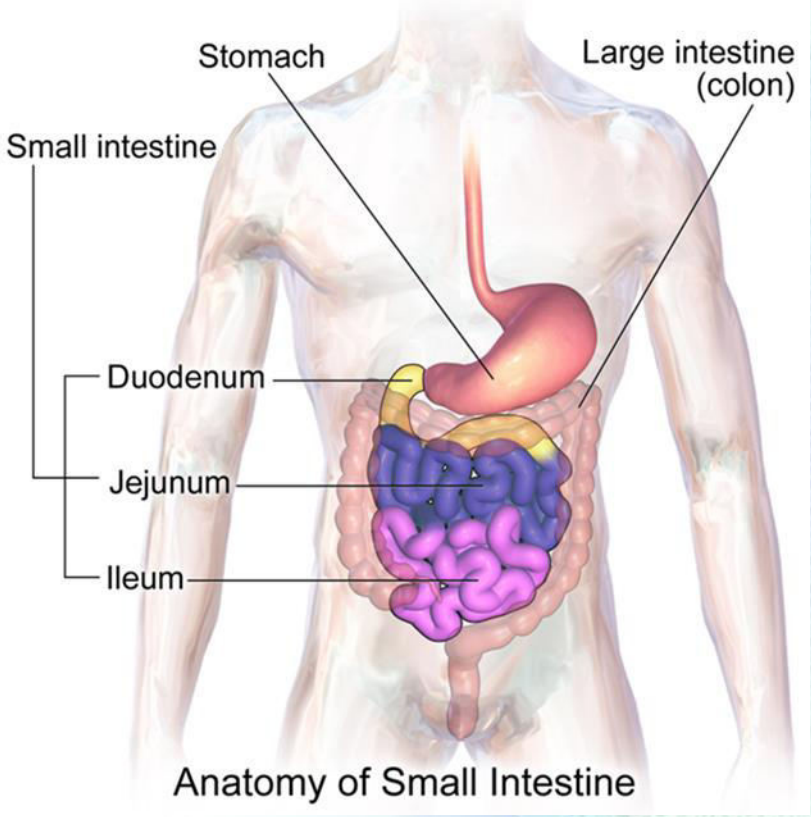
- The human small intestine is about 4 cm in diameter and 7 to 8 m in length.
- Length of human small intestine is intermediate between the small intestines of typical carnivores and herbivores of similar size.
- It reflects the human's omnivorous eating habits.

Small Intestine (Parts and Structure)

The small intestine has three parts:

- 1. Duodenum- It is about 6 m in length and functions primarily in digestion.
- 2. Jejunum and
- 3. Ileum .
- Both function in nutrient absorption.

Small Intestine (Parts and Structure)



Small Intestine (Parts and Structure)

- There is no clear border between the jejunum and the ileum.
- Histologically, the jejunum differs from the rest of the small intestine by the absence of Brunner's glands (which are present in the duodenum and Peyer's patches (which are present in the ileum .

Small Intestine (Parts and Structure)

- The Brunner's glands produce alkaline mucus, which neutralizes acid from the stomach as it enters the duodenum.
- Peyer's patches or nodules of lymphoid follicles in the mucus membrane that lines small intestine, the Ileum.

Small Intestine (Parts and Structure)

- The wall of the jejunum contains the following features that increase its surface area:
- Circular folds;
- Villi,
- Microvilli.
- These features are also found in the ileum, though with slight differences.

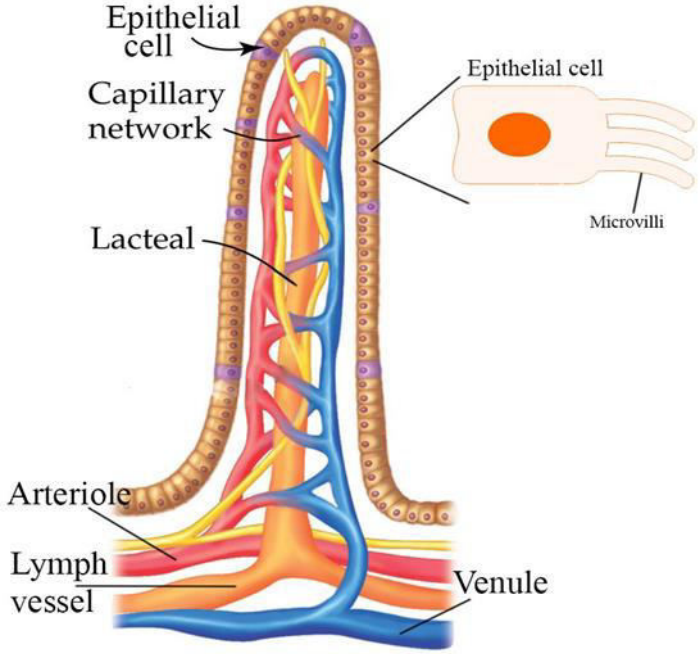
Small Intestine (Parts and Structure)

- The small intestine has many circular folds and minute projections called Villi on the inner gut surface.
- This is to increase the surface for absorption.
- Simple columnar epithelial cells, each bearing numerous microvilli, cover the villi .

Small Intestine (Parts and Structure)

- These minute projections are so dense that the inner wall of the human small intestine has a total surface area of approximately 300 m^2—the size of a tennis court.

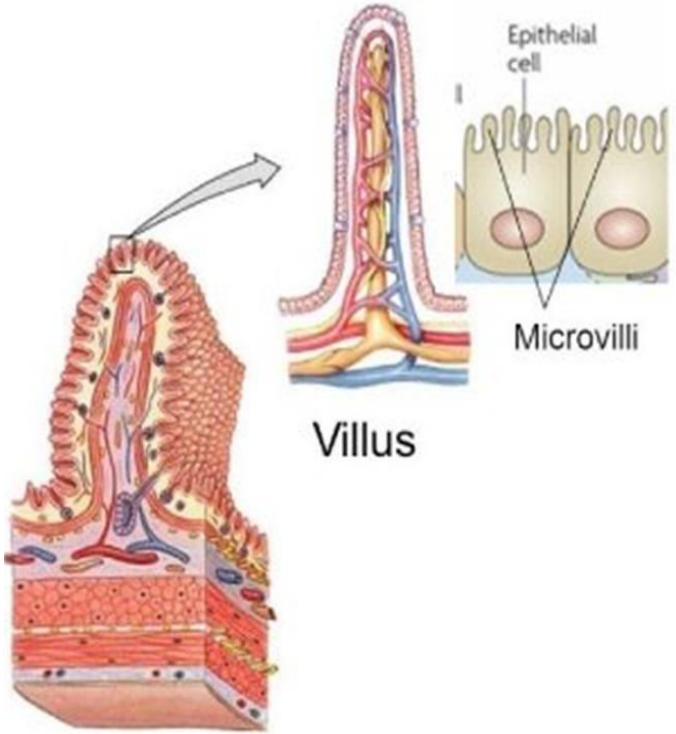
Small Intestine (Parts and Structure)



Structure of Villus

- **A single villus contains a central capillary network and a lymphatic lacteal, both of which transport nutrients absorbed from the lumen of the intestine.**
- **The plasma membrane of the simple columnar epithelial cells covering the villi fold into microvilli, further increase the surface area.**

Small Intestine (Parts and Structure)



Animal Form & Function II

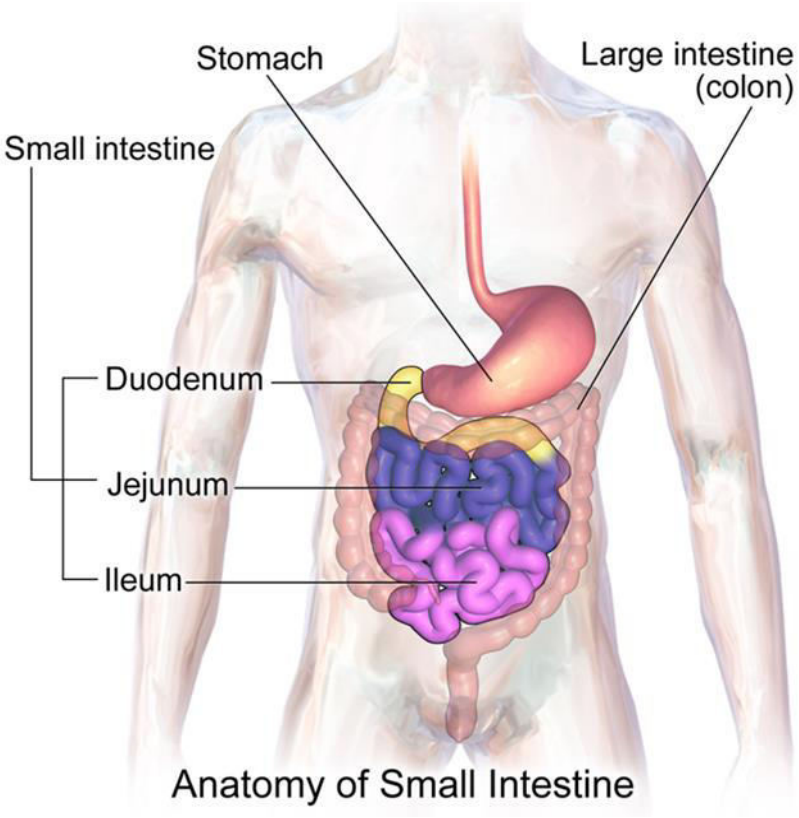
Module 49

Animal Form & Function II

Digestion in Small Intestine

Effort By: AARISH ALI

Digestion in Small Intestine



Anatomy of Small Intestine

Digestion in Small Intestine

Duodenum:

- **When chyme passes from the stomach into duodenum, its acidity stimulates the release of three secretions:**
- ***1. Digestive enzymes***
- ***2. Pancreatic enzymes***
- ***3. Bile***

Digestion in Small Intestine

Intestinal Enzymes:

- ***Enterokinase*** **Converts inactive trypsinogen into active trypsin.**
- ***Lactase*** **Digests lactose into glucose and galactose.**
- ***Maltase*** **Digests maltose into glucose.**
- ***Peptidase*** **Digests polypeptides into amino acids.**
- ***Sucrase*** **digests sucrose into glucose and fructose.**

Digestion in Small Intestine

Pancreatic Enzymes

- ***Pancreatic Amylase*- Digests starch into maltose**
- ***Chymotrypsin*- Digests proteins into peptides and amino acids**
- ***Lipase*- Digests lipids into fatty acids and glycerol (requires bile salts).**
- ***Nuclease*- Digests nucleic acids into mononucleotides.**

Digestion in Small Intestine

- *Trypsin*- Digests proteins into peptides and amino acids.

- **Liver**
- Secretes Bile which contain no enzyme.
- It emulsifies lipids; activates lipase.

Digestion in Small Intestine

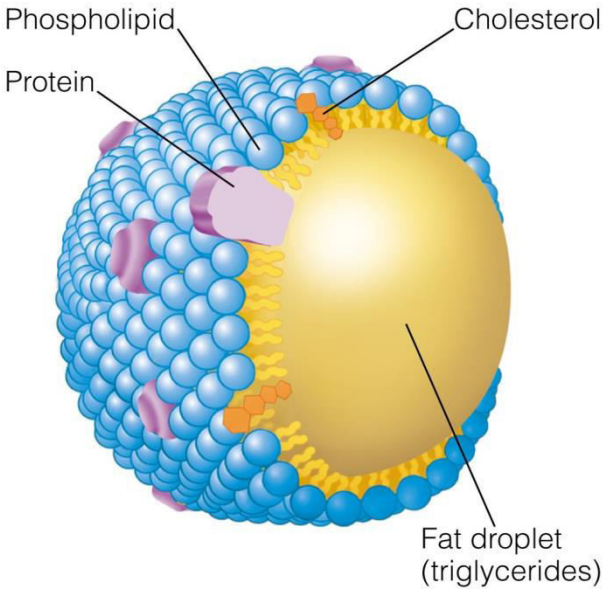
- In the duodenum, digestion of carbohydrates and proteins is completed, and most lipids are digested.
- The jejunum and ileum absorb the end products of digestion (amino acids, simple sugars, fatty acids, glycerol, nucleotides, water).

Digestion in Small Intestine

Absorption of Foods:

- Much of this absorption involves active transport and the sodium dependent ATPase pump.
- Sugars and amino acids are absorbed into the capillaries of the villi.

Digestion in Small Intestine



- **Free fatty acids enter the epithelial cells of the villi and recombine with glycerol to form triglycerides.**
- **The triglycerides are coated with proteins to form small droplets called chylomicrons, which enter the lacteals of villi.**

Animal Form & Function II

Module 50

Role of Pancreas in Digestion

Role of Pancreas in Digestion

- **The pancreas** is an organ that lies just ventral to the stomach and has both endocrine and exocrine functions.
- Exocrine cells secrete digestive enzymes into the pancreatic duct, which merges with the hepatic duct to form a common bile duct that enters the duodenum.

Role of Pancreas in Digestion

- The pancreas also secretes bicarbonate (HCO_3) ions that help neutralize the acidic food residue coming from the stomach.
- Bicarbonate raises the pH from 2 to 7 for optimal digestion.
- Without such neutralization, pancreatic enzymes could not function.

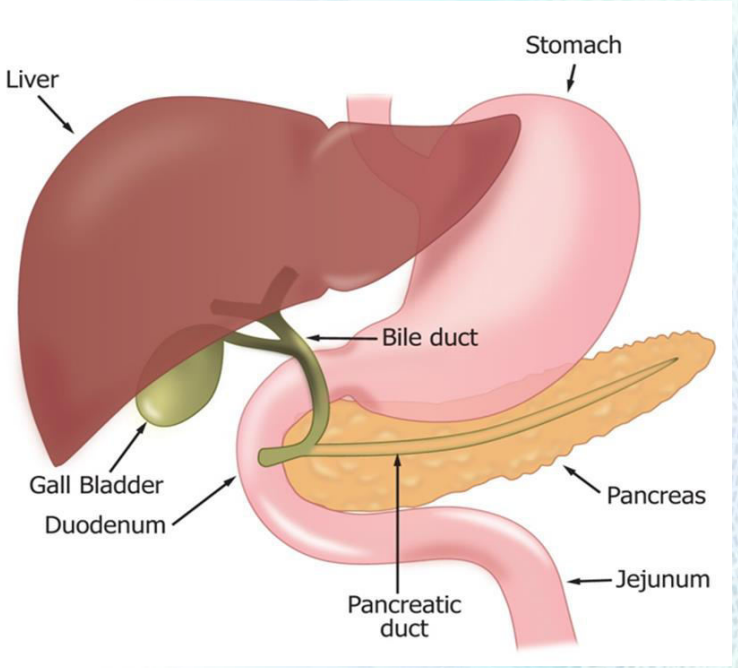
Role of Pancreas in Digestion

- **Amylase:** Digests starch into maltose
- **Chymotrypsin:** Digests proteins into peptides and amino acids
- **Pancreatic Lipase:** Digests lipids (triglycerides) into smaller, absorbable glycerol and free fatty acids.

Role of Pancreas in Digestion

- **Nuclease:** Digests nucleic acids into mononucleotides,
- **Trypsin:** Digests proteins into amino acids.

Role of Pancreas in Digestion



Animal Form & Function II

Module 51

Animal Form & Function II

Role of Liver in Digestion

Role of Liver in Digestion

- **The liver is the largest organ in the mammalian body, is just under the diaphragm.**
- **It is composed of millions of specialized cells called hepatocytes.**
- **They also manufacture the blood proteins, prothrombin and albumin.**

Role of Liver in Digestion

- **The liver is the largest organ in the mammalian body, is just under the diaphragm.**
- **It is composed of millions of specialized cells called hepatocytes.**
- **They also manufacture the blood proteins, prothrombin and albumin.**

Role of Liver in Digestion

- Major metabolic functions of the liver include:
- 1. Removal of amino acids from organic compounds.
- 2. Urea formation from proteins and conversion of excess amino acids into urea to decrease body levels of ammonia.
- 3. Synthesis of nonessential amino acids.

Role of Liver in Digestion

- 4. Conversion of galactose and fructose to glucose.
- 5. Oxidation of fatty acids.
- 6. Manufacture of plasma protein, the albumin.
- 7. Formation of fetal RBCs.
- 8. Destruction of worn-out RBCs.

Role of Liver in Digestion

- 9. Synthesis of the blood-clotting agents prothrombin and fibrinogen from amino acids.
- 10. Formation of lipoproteins, cholesterol, and phospholipids (essential cell membrane components).
- 11. Conversion of carbohydrates and proteins into fat.

Role of Liver in Digestion

- 12. Modification of waste products, toxic drugs, and poisons (detoxification).
- 13. Synthesis of vitamin A from carotene, and with the kidneys, participation in the activation of vitamin D.
- 14. Maintenance of a stable body temperature.

Role of Liver in Digestion

- 15. liver is the major heat producer in a mammal's body.
- 16. Manufacture of bile salts, which are used for the emulsification of fats in the small intestine.

Role of Liver

Main storage center.

- The liver stores glucose in the form of glycogen, and with the help of insulin and enzymes, converts glycogen back into glucose as the body needs it.
- The liver also stores fat-soluble vitamins (A, D, E, and K), and minerals, such as iron, from the diet.

Role of Liver

- The liver can also store fats and amino acids, and convert them into usable glucose as required.

Animal Form & Function II

Module 52

Animal Form & Function II

Gallbladder and Bile

Gallbladder and Bile

Role of Gall Bladder

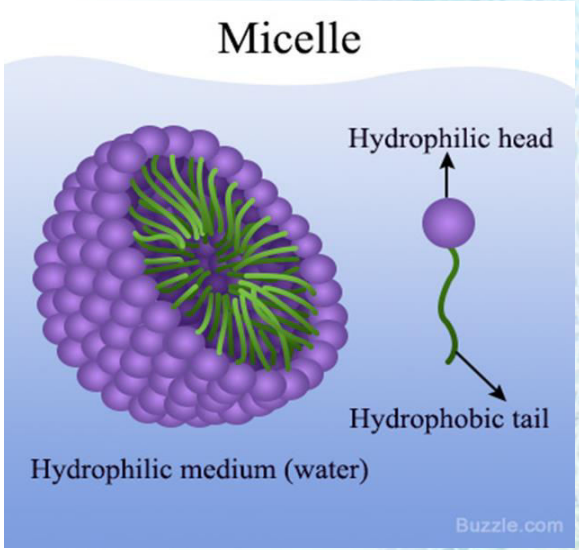
- The gallbladder is a small bag near the liver.
- The gallbladder stores the greenish fluid, the bile, that the liver cells continuously produce.
- Bile is very alkaline and contains pigments e.g. biliverdin (green) and bilirubin (Red)

Gallbladder and Bile

- Bile also contain salts that act as detergents to emulsify fats and aid in fat digestion and absorption.
- Bile salts also combine with the end products of fat digestion to form micelles.
- Micelles are lipid aggregates (fatty acids and glycerol) with a surface coat of bile salts.

Gallbladder and Bile

- **Because they are so small, they can cross the microvilli of the intestinal epithelium.**



Gallbladder and Bile

- Micelles are lipid molecules that arrange themselves in a spherical form in aqueous solutions.
- The formation of a micelle is a response to the amphipathic nature of fatty acids, meaning that they contain both hydrophilic regions (polar head groups) as well as hydrophobic regions (the long hydrophobic chain).

Gallbladder and Bile

Animal Form & Function II

Module 53

Digestion in Jejunum and Ileum

Digestion in Jejunum and Ileum

- **The jejunum is the second portion of the small intestine extending from duodenum to ileum.**
- **Jejunum makes up about two-fifths of the small intestine.**
- **It is about 2.4 m in length.**
- **Lower three fifth of the small intestine from jejunum to ileocolic sphincter is the ileum.**

Digestion in Jejunum and Ileum

- The main function of the jejunum is absorption of important nutrients-sugars, fatty acids, and amino acids.
- Nutrients absorbed by the jejunum enter the bloodstream, where they can then be distributed to the organs of the body.

Digestion in Jejunum and Ileum

- The food which escapes undigested from the duodenum is completely digested in jejunum and ileum by a group of following enzymes.
- These enzymes, their substrates and final products are:

Digestion in Jejunum and Ileum

i) Polypeptides -- *Amino peptidase* Dipeptides

ii) Dipeptides --- *Erypsin*--- Amino acids.

iii) Fats---*Lipase*---- Fatty acids + Glycerol

iv) Maltose-- *Maltase*– Glucose

v) Lactose – *Lactase*– Glucose and Galctose.

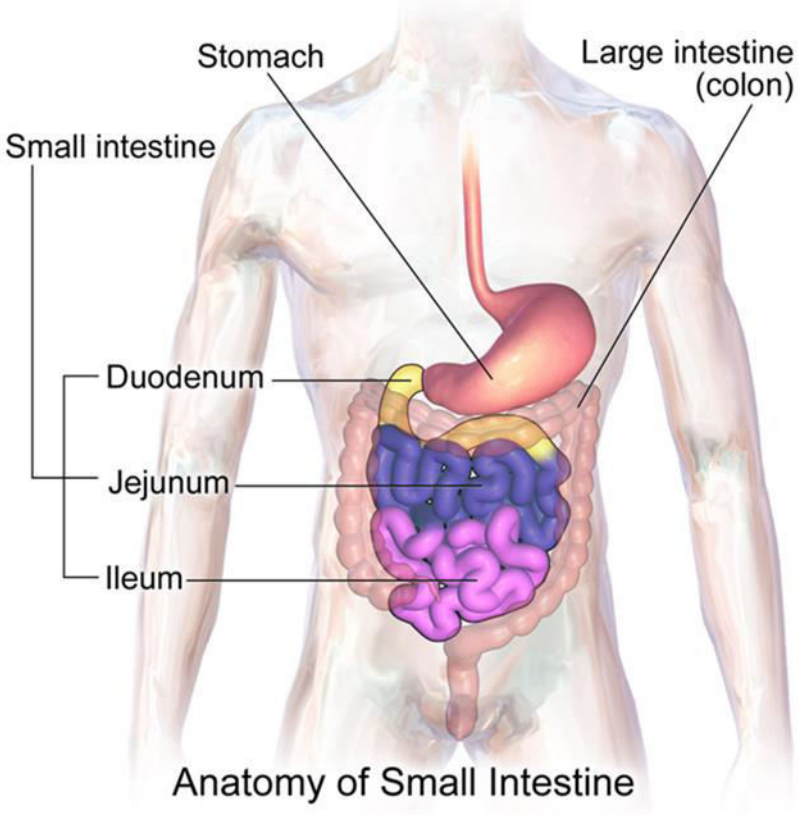
Digestion in Jejunum and Ileum

- The enzymes which complete the digestive process are NOT secreted into the lumen of the intestine.
- In fact these final processes take place in the membranes of the microvilli of the epithelial mucosa because the enzymes are located within the cell membranes.

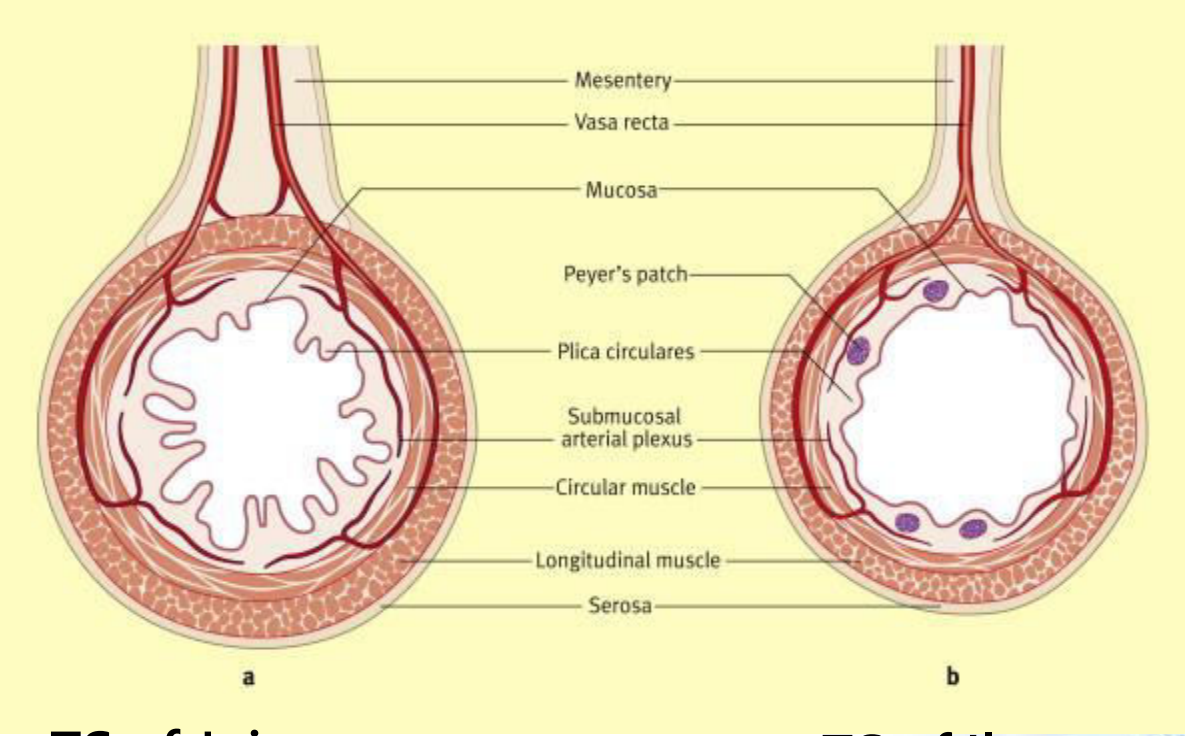
Digestion in Jejunum and Ileum

- The end products of digestion are then liberated into the lumen of intestine on their way to blood stream.
- The function of the ileum is mainly to absorb vitamin B12, bile salts, and any products of digestion that were not absorbed by the jejunum.

Digestion in Jejunum and Ileum



Digestion in Jejunum and Ileum



TS of Jejunum

TS of Ileum

Animal Form & Function II

Module 54

Animal Form & Function II

Structure and Function of Villi

Structure and Function of Villi

Villi

- Located in the circular folds measuring 0.5-1mm in length, finger-like projections known as villi extend into the intestinal lumen available for nutrient absorption. Each villus contains a:

Structure and Function of Villi

- Capillary bed – comprising an arteriole and a venule.
- Lymphatic capillary – central lacteal.
- The venules allow glucose and amino acids to be absorbed directly into the bloodstream, while products from the breakdown of lipids (fatty acids and glycerol) are absorbed into the lymphatic system via the lacteals.

Structure and Function of Villi

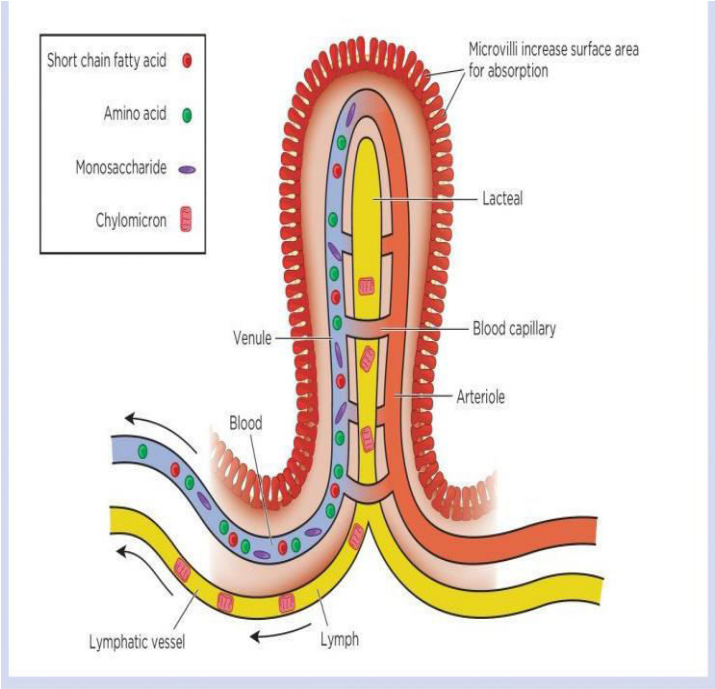
Microvilli

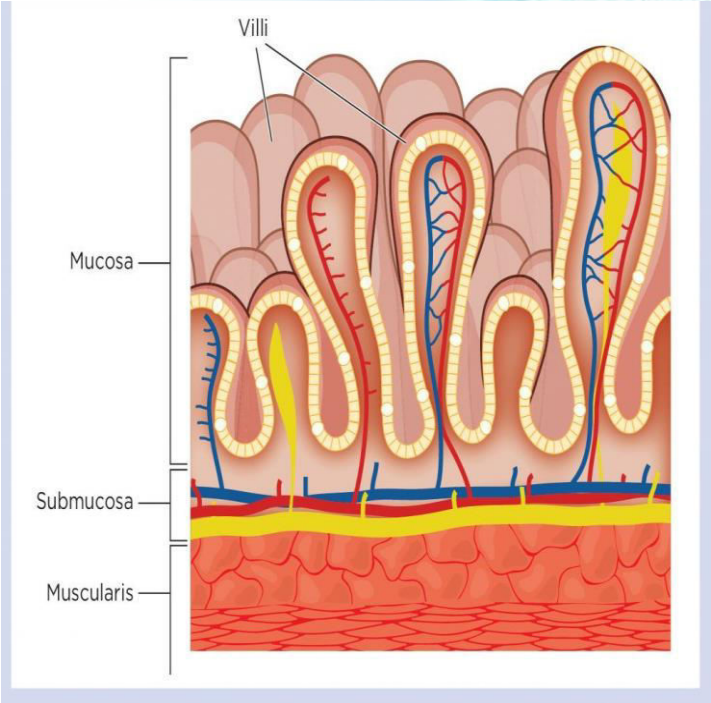
- The mucosal epithelial cells have thin, hair-like extensions about 1µm (0.001mm) in length, jutting out into the intestinal lumen.
- These tiny projections are known as microvilli and there are approximately 200 million of them per $1mm^2$.

Structure and Function of Villi

- Microscopically, they appear as a mass of bristles and are, therefore, termed the brush border.
- Fixed to the surface of the microvilli are a series of enzymes that finish chemical digestion.

Structure and Function of Villi





Animal Form & Function II

Module 55

Animal Form & Function II

Absorption of Digested Food

Absorption of Digested Food

- Nearly all absorption of the products of digestion take place in the ileum through Villi.
- Each villus is richly supplied with blood capillaries and a vessel called lacteal of lymphatic system with a covering of epithelial cells having microvilli.

Absorption of Digested Food

- The total area of absorption becomes extensively large due to in-foldings, villi and microvilli.
- Monosaccharides and amino acids are absorbed either by diffusion or active transport into blood capillaries through microvilli.

Absorption of Digested Food

- Some of the fatty acids and glycerol are also absorbed into the blood stream.
- However, a large proportion of fatty acids and glycerol enter the epithelial cells of villi where they recombine into fats which then enter the lacteals.

Absorption of Digested Food

- Protein present in lymph vessels combine with fat molecules to form lipo-protein droplets.
- These lipoprotein molecules subsequently hydrolyzed by blood plasma enzyme back to fatty acids and glycerol.

Absorption of Digested Food

- These molecules are then absorbed by the body cells where they are used in respiration or stored as fats in liver, muscles or under the skin

Animal Form & Function II

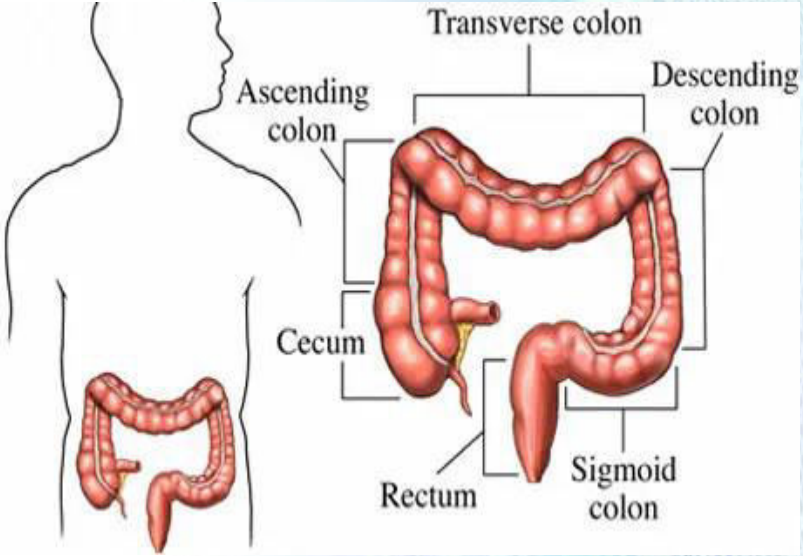
Module 56

Large Intestine

Large Intestine

- Large Intestine is composed of caecum, colon and rectum.
- Caecum is blind sac that projects from the ileum and colon.
- From the caecum there arises a finger-like process called appendix.

Large Intestine



Large Intestine

Appendicitis:

- The appendix sometimes get inflamed due to entrapping and the putrefaction of food causing appendicitis.
- Appendix has to removed surgically in many instances.

Large Intestine

- The material that passes from the small intestine to the large intestine contains:
- Water
- Dissolved salts
- Undigested material
- Water and salts are absorbed into blood while undigested material is rejected as feces.

Large Intestine

- Large intestine also harbors a large population of useful bacteria that synthesize some vitamins especially Vitamin K which absorbed in blood.

Large Intestine

Animal Form & Function II

Module 57

Animal Form & Function II

Composition of Fecal Matter

Composition of Fecal Matter

Composition of Fecal Matter:

- Fecal matter is composed of:
- Undigested material
- Bacteria
- Plant fibers
- Sloughed off mucosal cells
- Mucus
- Bile pigments
- Metabolic wastes
- Water

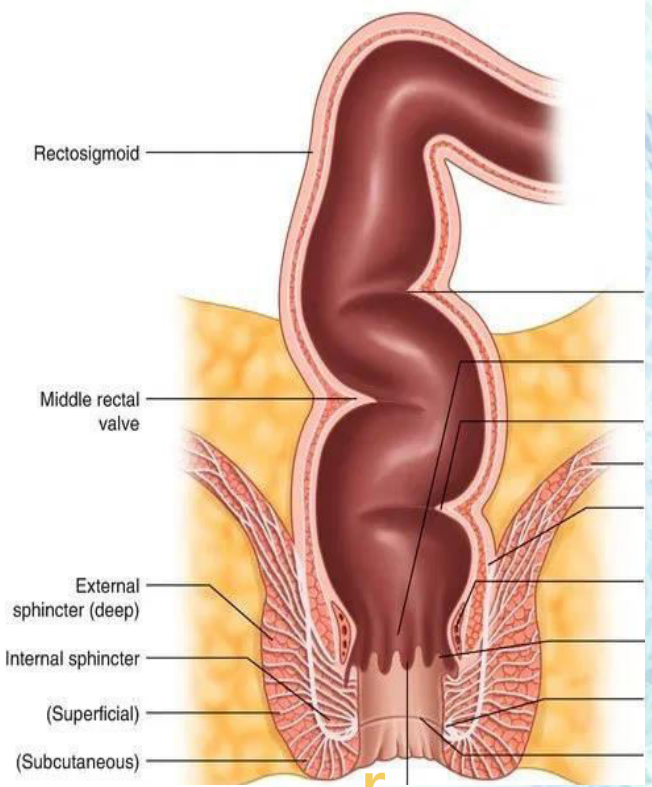
Composition of Fecal Matter

- Normally human feces are semisolid, with a mucus coating.
- In the medical literature, the term "stool" is more commonly used than "feces".
- Human feces together with human urine are collectively referred to as human waste or human excreta.

Composition of Fecal Matter

- Defecation is the final act of digestion, by which organisms eliminate solid, semisolid, or liquid waste material from the digestive tract via the anus.

Composition of Fecal Matter



Rectum

Rectal sinus

Animal Form & Function II

Module 58

Some Common Diseases Related to Nutrition

Some Common Diseases Related to Nutrition

Following are some common diseases related to nutrition and digestive system:

- Dental Caries
- Dyspepsia
- Food poisoning
- Anorexia Nervosa
- Obesity
- Piles or Hemorrhoids

Some Common Diseases Related to Nutrition



Dental Caries:

- Also known as Tooth decay or cavities.
- This is the break down of teeth due to the acids made by bacteria.
- Caries is also associated with poor cleaning of the mouth.
- Receding gums resulting in exposure of the roots of the teeth is also dental caries.

Some Common Diseases Related to Nutrition

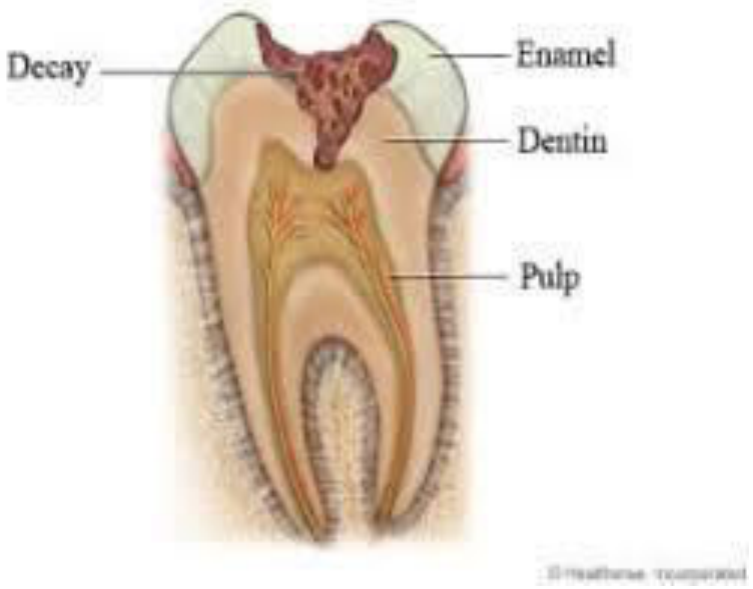
Major causes of tooth decay:

- Sugary, sticky foods and beverages.
- The more the sugar consumed, the more the acid, gets produced leading to decay.
- Sugar combines with plaque to weaken the enamel leaving you vulnerable to tooth decay.

Some Common Diseases Related to Nutrition

- Cavities are permanently damaged areas in the hard surface of teeth that develop into tiny openings or holes.
- Cavities start small and gradually become bigger when they're left untreated.
- Tatar, calculus, root canal.

Some Common Diseases Related to Nutrition





Animal Form & Function II

Module 59

Animal Form & Function II

Dyspepsia or Indigestion

Dyspepsia or Indigestion

Dyspepsia:

- It is characterized by incomplete of imperfect digestion.
- It is not a disease but sometimes symptoms of other disorders or diseases.

Symptoms:

- Abdominal discomfort
- Flatulance, Nausea, Vomiting

Dyspepsia or Indigestion

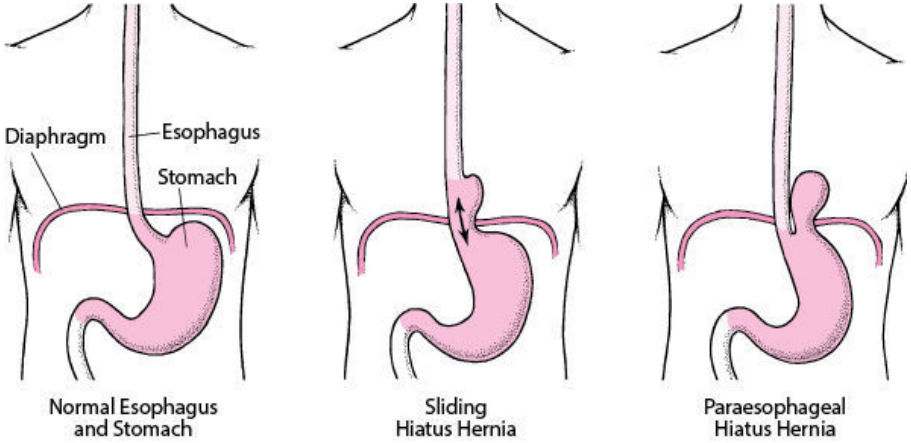
Causes:

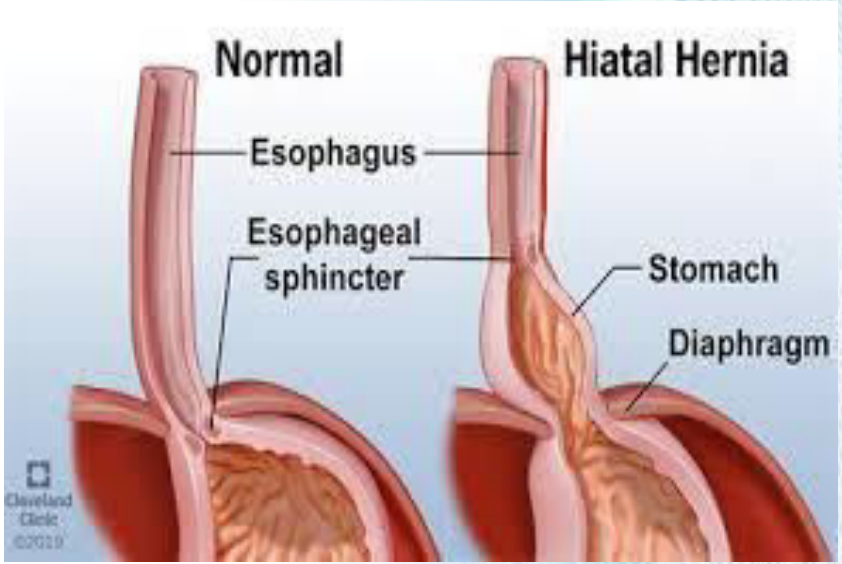
- Excessive acidity in stomach
- Faulty function of stomach and intestine.
- Insufficient quantity of bile secretions.
- Eating too much and too rapidly.
- Eating fatty, greasy, or spicy foods.

Dyspepsia or Indigestion

- Drinking too much caffeine or alcohol.
- Consuming too much chocolate or soda.
- Emotional trauma.
- Gallstones.
- Gastritis, or inflammation of the stomach.
- Hiatus hernia.

Dyspepsia or Indigestion





Animal Form & Function II

Module 60

Animal Form & Function II

Food Poisoning

Food Poisoning

Illness from indigestion of food containing toxic substances is called food poisoning.

Symptoms:

- **Diarrhea**
- **Vomiting**
- **Abdominal pain**
- **Occurs between 12-24 hours after eating contaminated food.**

Food Poisoning

Causes:

- Eating improperly cooked (especially meat or eggs
- Drinking unpasteurized milk contaminated with *Salmonella* and *Campylobacte*r bacteria).

Preventions:

- The liquid that escapes during defrosting frozen meat contain *Salmonella.*

Food Poisoning

- **The dishes and utensils while thawing must not be allowed to come in contact with any other food meat.**

Animal Form & Function II

Module 60

Animal Form & Function II

Anorexia Nervosa

Anorexia Nervosa

Anorexia Nervosa:

- This term is employed to the loss of appetite due to the fear of becoming obese.
- Such a feeling is common in human females between the age of 12 to 21 years just on the onset of puberty.
- Fear is not diminished even when weight is dropped to dangerous level.

Anorexia Nervosa



Anorexia Nervosa

Symptoms:

- Loss of appetite.
- Body weight drops to dangerous level.
- Young girls feel unable to cope with the changes of puberty and emerging sexuality psychologically.
- Anorexia girls have a strong feeling of becoming obese.

Anorexia Nervosa

Treatment:

- Psychiatric therapy.
- Patients are fed intramuscularly or intravenously.
- Recovery is slow.

Animal Form & Function II

Module 62

Gastric Ulcer

Gastric Ulcer

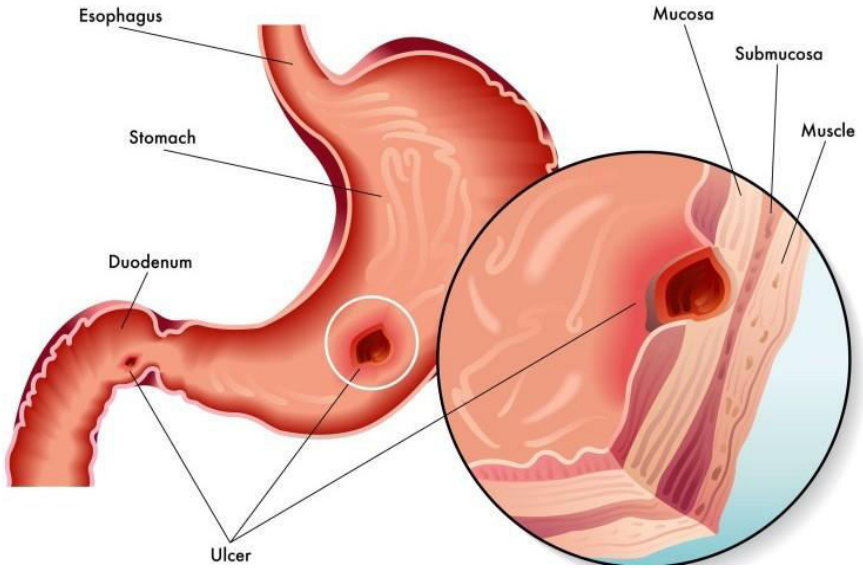
Gastric ulcer:

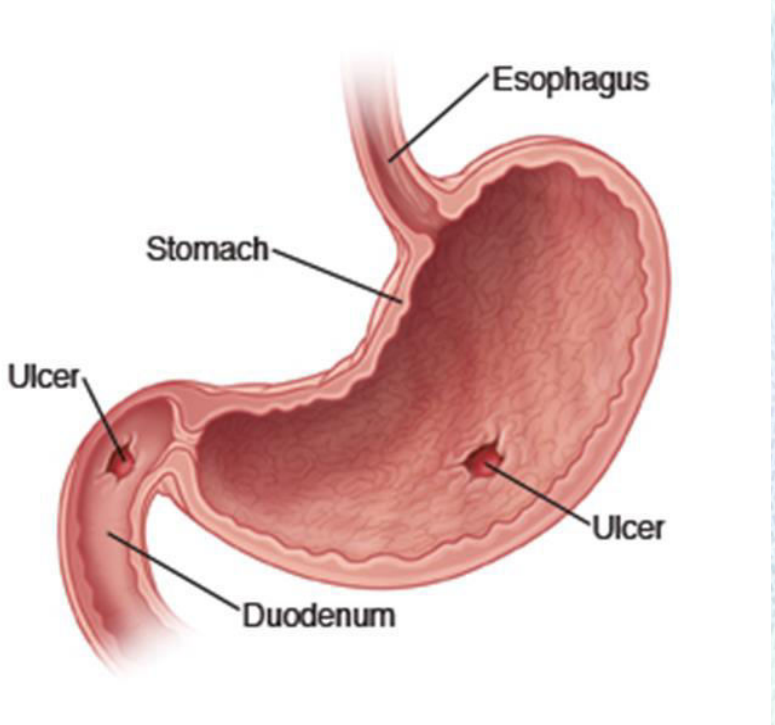
- Digestive tract is coated with a mucus layer that normally protects against acid.
- But if the amount of acid is increased or the amount of mucus is decreased, an ulcer develop
- Sore in digestive tract particularly in stomach and duodenum.

Gastric Ulcer

- This is formed when digestive enzymes begin to eat away the walls of GIT due to the break down of mucus layer lining the stomach.
- A hole in the lining of the stomach corroded by the acidic digestive juices which are secreted by the stomach cells.

Gastric Ulcer





Gastric Ulcer

Causes:

- Use of spicy food.
- *Helicobacter pylori* bacteria
- Alcoholic beverages.
- Regular use of certain pain relievers
- Untreated stress.

Gastric Ulcer

Symptoms:

- Point pain in affected area.
- Burning sensation.
- Nausea or vomiting
- Vomiting or vomiting blood — which may appear black.
- Dark blood in stools,
- Feeling faint
- Unexplained weight loss.
- Fever and chills.

Gastric Ulcer

Treatment:

- Avoid smoking,
- Avoid eating spicy food.
- Stop using alcohol.
- Avoid coffee, tea and stress.
- Surgery in extreme cases.

Animal Form & Function II

Module 63

Animal Form & Function II

Gallstones

Gallstones

Gallstones

- Gallstones are solid pebbles that form from bile cholesterol and bilirubin in the gallbladder.
- The gallbladder is a small pear-shaped saclike organ located under the liver.
- Liver in the present on the upper right part of the abdomen.

Gallstones



Gallstones

Symptoms:

- Gallstones, may be asymptomatic, even for years.
- A characteristic symptom of a gallstone attack is the presence of colicky pain in the upper-right side of the abdomen, often accompanied by nausea and vomiting.

Gallstones

- The pain steadily increases for approximately 30 minutes to several hours.
- Often, attacks occur after a particularly fatty meal and almost always happen at night, and after drinking.

Gallstones

- On the basis of composition of stones Gall stones can be classified as
  - a. Cholesterol stone,
  - b. Pigment stone.
  - c. Mixed stone

Cholesterol stones:

- Are composed of 70 to 80% cholesterol by weight.
- Between 35% and 90% of stones are cholesterol stones.

Gallstones

Pigment stones:

- **These stones are small, often appearing black, and usually numerous.**
- **They are composed of bilirubin and $CaPO_4$ salts that are found in bile.**
- **They contain less than 20% of cholesterol.**
- **Between 2% and 30% of stones are bilirubin stones.**

Gallstones

- Nutritional factors that may increase risk of gallstones include constipation; eating fewer meals per day; low intake of the nutrients folate, magnesium, calcium, and vitamin C; low fluid consumption;

Gallstones

Mixed stones:

- Mixed stones contain 20–80% cholesterol.
- Other common constituents are calcium carbonate, palmitate phosphate, bilirubin and other bile pigments.
- Between 4% and 20% of stones are mixed.

Gallstones

- Nutritional factors that may increase risk of gallstones include:
- Constipation,
- Eating fewer meals per day,
- Low intake of the nutrients folate, magnesium, calcium, and vitamin C,
- Low fluid consumption.

Animal Form & Function II

Module 64

Piles or Hemorrhoids

Piles or Hemorrhoids

Piles or Hemorrhoids:

- are the masses of dilated tortuous veins in the anorectic mucosa.

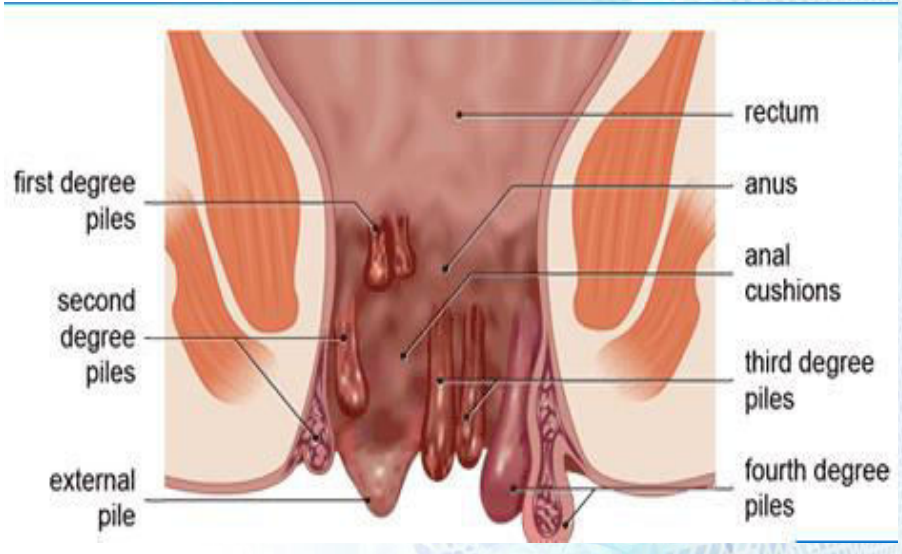
Symptoms:

- The masses may sometimes start bleeding during bowl movements.
- The urge to defecate is depressed and it becomes difficult to expel the feces.

Piles or Hemorrhoids

- **Bleeding may aggravate when the patient suffers from constipation.**

Piles or Hemorrhoids



Different degree piles

Piles or Hemorrhoids

Treatment:

- Improvement of hygiene
- Food must contain roughage and laxatives.
- The patient is advised not to sit on hard seats.
- Surgery in severe cases.

Animal Form & Function II

Module 65

Summary of the Chapter Part I

Summary of the Chapter Part I

- Nutrition describes all of those processes by which an animal takes in, digests, absorbs, stores, and uses food (nutrients) to meet its metabolic needs.
- Digestion is the mechanical and chemical breakdown of food into smaller particles that the individual cells of an animal can absorb.

Summary of the Chapter Part I

- Evolutionarily, animals have evolved the following nutritional types:
- insectivores,
- herbivores,
- carnivores, and
- omnivores.

Summary of the Chapter Part I

- The nutrients that a heterotroph ingests can be divided into macronutrients and micronutrients.
- Macronutrients are needed in large quantities and include the carbohydrates, lipids, and proteins.
- Micronutrients are needed in small quantities and include the vitamins and minerals.

Summary of the Chapter Part I

- Only a few protists and animals can absorb nutrients directly from their external environment.
- Most animals must work for their nutrients.
- Various specializations have evolved for food procurement
- (feeding) in animals.

Summary of the Chapter Part I

- Some examples include continuous versus discontinuous feeding, suspension feeding, deposit feeding, herbivory, predation, surface nutrient absorption, and fluid feeding.

Summary of the Chapter Part I

- The evolution and structure of the digestive system in various invertebrates and vertebrates reflect their eating habits, their rate of metabolism, and their body size.

Animal Form & Function II

Module 66

Summary of the Chapter

Part II

Summary of the Chapter Part II

- The digestive system of vertebrates is one-way, leading from the mouth (oral cavity), to the pharynx, esophagus, stomach, small intestine, large intestine, rectum, and anus.

Summary of the Chapter Part II

- The coordinated contractions of the muscle layer of the gastrointestinal tract mix food material with various secretions and move the food from the oral cavity to the rectum.
- The two types of movement involved are peristalsis and segmentation.

Summary of the Chapter Part II

- Most digestion occurs in the duodenal portion of the small intestine.
- The products of digestion are absorbed in the walls of the jejunum and ileum.
- In the process of digestion, bile that the liver secretes, makes fats soluble.

Summary of the Chapter Part II

- The liver has many diverse functions.
- It controls the fate of newly synthesized food molecules.
- Stores excess glucose as glycogen.
- Synthesizes many blood proteins, and converts nitrogenous and other wastes into a form that the kidneys can excrete.

Summary of the Chapter Part II

- The large intestine has little digestive or nutrient absorptive activity.
- It functions principally to absorb water, to compact the material left over from digestion, and to be a storehouse for microorganisms.

Summary of the Chapter Part II

Animal Form & Function II

Module 67

Animal Form & Function II

Homeostasis

Homeostasis

- **Homeostasis** refers to the ability of an organism to maintain the internal environment (physical, and chemical conditions) of the body within limits that allow it to survive.

Homeostasis

- This includes many variables, which are kept within certain pre-set limits.
- These variables are:
- Fluid balance.
- Body temperature
- pH of extra cellular fluid
- Concentrations of Na^{+}, K^{+} and Ca^{++} ions.
- Blood sugar level

Homeostasis

- Each of these variables is controlled by one or more regulators or homeostatic mechanisms, which together maintain life.
- Equilibrium is kept maintained by many regulatory mechanisms, specially nervous system and endocrine system.

Animal Form & Function II

Module 68

Concept of Thermoregulation

Concept of Thermoregulation

- **Thermoregulation** is the ability of an organism to keep its body temperature within certain limits, even when the surrounding temperature is very different.

Concept of Thermoregulation

- **A thermo-conforming organism, by contrast, simply adopts the surrounding temperature as its own body temperature, thus avoiding the need for internal thermoregulation.**

Concept of Thermoregulation

- The temperature of a living cell affects the rate of its metabolic processes.
- An animal can grow faster and respond to the environment more rapidly if its cells are kept warm.

Concept of Thermoregulation

- This ability to control the temperature of the body is called thermoregulation (“heat control”) and involves the nervous, endocrine, respiratory, and circulatory systems in higher animals.

Concept of Thermoregulation

- **In fact zoologists believe that the ability of some higher animals to maintain moderate body temperature is a major reason for their evolutionary success.**

END

Animal Form & Function II

Module 69

Animal Form and Function II

Effort By: AARISH ALI

Concept of Osmoregulation

Concept of Osmoregulation

Osmoregulation

- It is defined as the maintenance of constant osmotic pressure in the fluids of an organism by the control of water and salt concentrations.
- It means osmoregulation governs water and salt balance, and excretion of metabolic wastes.

Concept of Osmoregulation

- In fishes, reptiles, birds, and mammals, the kidneys are the primary osmoregulatory structures.
- To understand osmoregulation, lets us first study the response of the cell in various external environments i.e. in different concentrations when the control system is not operating.

Concept of Osmoregulation

- In hypertonic solution cell shrinks due to the loss of water.
- In hypotonic solution the cell swells up and may even burst.
- To prevent these situations the cell's control system operates and osmoregulate itself to keep water and salts in balance in both plants and animals.

Concept of Osmoregulation

Osmoregulation in Animals:

- Based on osmoregulation the animals are divided into two groups:

a. Osmoconformers

b. Osmoregulators

- In first group of animals the osmotic concentration of the body fluids of the animals equals that of the medium.

Concept of Osmoregulation

- In other words, the osmotic concentration of the environment changes, so does that of the animal's body fluids.
- The osmoregulators maintain their body fluids at a different osmotic concentration from that of its surrounding environment.

END

Animal Form & Function II

Module 70

Animal Form & Function II

Impact of Temperature on Animal Life

Impact of Temperature on Animal Life

- As we know that metabolism is sensitive to changes in internal temperature.
- The rate of cellular respiration increases with temperature up to a point.
- When the temperature rises above the temperature optima the rate declines as the enzymes begin to denature.

Impact of Temperature on Animal Life

- A digestive enzyme in a trout might function optimally at 10° C, whereas another enzyme in the human body catalyzes the same reaction functions best at 37° C.

Impact of Temperature on Animal Life

- Higher temperatures cause the proteins in nucleic acids to denature, and
- Lower temperatures may cause membranes to change from a fluid to a solid state, which can interfere with many cellular processes, such as active transport pumps.

Impact of Temperature on Animal Life

- It shows that enzyme evolution have frequently been enzymes with temperature optima that reflect an animal's habitat.
- Animals can guard against these damaging effects of temperature fluctuations by balancing heat gains and heat losses with their environment.

END

Animal Form & Function II

Module 71

Heat Gains and Losses in Animals

Heat Gains and Losses in Animals

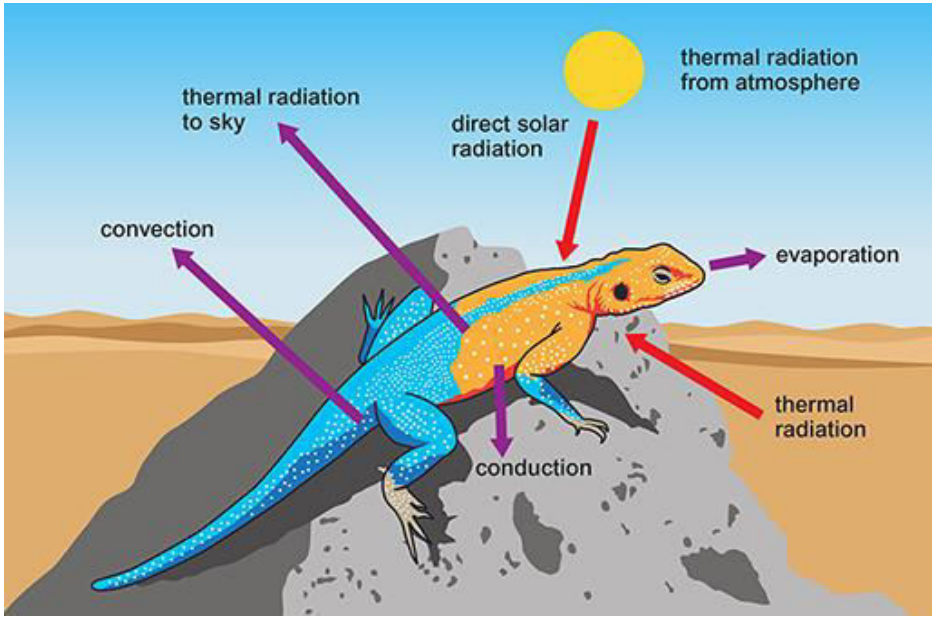
- Animals produce heat as a by-product of metabolism.
- The total body temperature is a result of an interaction of these factors and can be expressed as:
- Body temperature =

  heat produced metabolically

  + heat gained from the environment

  - heat lost to the environment

Heat Gains and Losses in Animals Part I



Animals use four physical processes to exchange heat with the environment:

1. **conduction,**
2. **convection,**
3. **radiation and**
4. **evaporation**

Heat Gains and Losses in Animals

- **Conduction** is the direct transfer of (heat) from the environment to the body and from the body to the environment of an animal
- This transfer is always from an area of higher temperature to one of lower temperature.
- For example, when we sit on the cold ground, we lose heat, and when we sit on warm sand, we gain heat.

Heat Gains and Losses in Animals

- **Convection** is the movement of air (or a liquid) over the surface of a body it contributes to heat loss if the air is cooler than the body or heat gain if the air is warmer than the body.
- On a cool day, our body loses heat by convection because our skin temperature is higher than the surrounding air temperature.

Heat Gains and Losses in Animals

- **Radiation** is the emission of electromagnetic waves that the sun, or an object, produce.
- An object may be another animal.
- Radiation can transfer heat between objects that are not in direct contact with each other, as happens when an animal suns itself.

Heat Gains and Losses in Animals

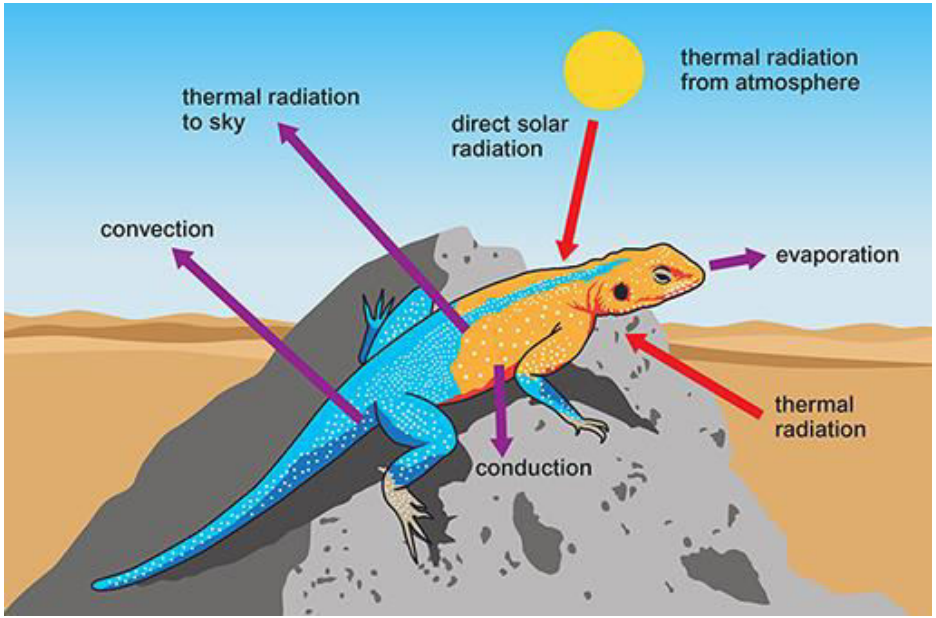
- **Evaporation** is the loss of heat from a surface of an animal.
- It is useful only to terrestrial animals.
- For example, humans and some other mammals (chimpanzees and horses) have sweat glands that actively move watery solutions through pores to the skin surface.

Heat Gains and Losses in Animals

- **When skin temperature is high, water at the surface absorbs enough thermal energy (heat) to break the hydrogen bonds holding the individual water molecules together, and they depart from the surface, carrying heat with them.**

END

Heat Gains and Losses in Animals Part I



Animals use four physical processes to exchange heat with the environment:

1. **conduction,**
2. **convection,**
3. **radiation and**
4. **evaporation**

Animal Form & Function II

Module 72

Some Solutions to Temperature Fluctuations in Animals

Part I

Some Solutions to Temp Fluctuations Part I

- There are three basic ways the animals cope with temperature fluctuations, either:
- 1. They can occupy a place in the environment where the temperature remains constant and compatible with their physiological processes.

Some Solutions to Temp Fluctuations Part I

- **2. Their physiological processes may have adapted to the range of temperatures in which the animals are capable of living, or**

Some Solutions to Temp Fluctuations Part I

- 3. They can generate and trap heat internally to maintain a constant body temperature, despite fluctuations in the temperature of the external environment.

Some Solutions to Temp Fluctuations Part I

- On this basis, animals can be categorized as:
- 1. Ectotherms and
- 2. Endotherms
- **Ectotherms** derive most of their body heat from the environment rather than from their own metabolism.
- They have low rates of metabolism and are poorly insulated.

Some Solutions to Temp Fluctuations Part I

- In general, invertebrates, fishes, amphibians and reptiles are ectotherms, although a few insects, fishes and reptiles, can raise their internal temperature.
- Ectotherms tend to move about the environment and find places that minimize heat or cold stress to their bodies.

Some Solutions to Temp Fluctuations Part I



Postures of a dragonfly to maximize and minimize heat gain

Some Solutions to Temp Fluctuations Part I



- A meerkat *(Suricata suricatta)* stands at attention, allowing the large surface area of its body to absorb radiation from the sun.

Some Solutions to Temp Fluctuations Part I

- **Endotherms** obtain their body heat from cellular processes.
- A constant source of internal heat allows them to maintain a nearly constant core temperature, despite the fluctuating environmental temperature.
- Examples are birds and mammals.

Some Solutions to Temp Fluctuations Part I

- Most endotherms have bodies insulated by feathers or fur and a relatively large amount of fat.
- This insulation enables them to retain heat more efficiently and to maintain a high core temperature.

Some Solutions to Temp Fluctuations Part I

- **Endothermy allows animals to stabilize their core temperature so that biochemical processes and nervous system functions can proceed at steady, high levels.**

END

Animal Form & Function II

Module 73

Some Solutions to Temperature Fluctuations in Animals

Part II

Some Solutions to Temp. Fluctuations Part II

- **Ectothermy and Heterothermy**
- Another way of categorizing animals is based on whether they maintain a constant or variable body temperature.
- Although most endotherms are homeotherms *(maintain a relatively constant body temperature)*, and most ectotherms are heterotherms.

Some Solutions to Temp. Fluctuations Part II

(have a variable body temperature)

- *There are many exceptions.* Some endotherms vary their body temperatures seasonally (e.g., hibernation); others vary it on a daily basis.

Some Solutions to Temp. Fluctuations Part II

- Examples:
- Hummingbirds and Shrews
- These animals can only maintain a high body temperature for a short period because they usually weigh less than 10 g and have a body mass so small that they cannot generate enough heat to compensate for the heat lost across their relatively large surface area.

Some Solutions to Temp. Fluctuations Part II





- **Hummingbirds must devote much of the day to locating and sipping nectar (a very high-calorie food source) as a constant energy source for metabolism.**

Some Solutions to Temp. Fluctuations Part II

- At night, when not feeding, hummingbirds enter a sleep-like state, called daily torpor and their body temperature approaches that of the cooler surroundings.
- Some bats also undergo daily torpor to conserve energy.

Some Solutions to Temp. Fluctuations Part II

- Some ectotherms can maintain fairly constant body temperatures.
- Among these are a number of reptiles that can maintain fairly constant body temperatures by changing position and location during the day to equalize heat gain and loss.

Some Solutions to Temp. Fluctuations Part II

- In general, ectotherms are more common in the tropics because they do not have to spend as much energy to maintain body temperature there, and they can devote more energy to food gathering and reproduction.
- Indeed, in the tropics, amphibians are far more abundant than mammals.

Some Solutions to Temp. Fluctuations Part II

- Conversely, in moderate to cool environments, endotherms have a selective advantage and are more abundant.
- High metabolic rates and insulation allow them to occupy even the polar regions (e.g., polar bears).

Some Solutions to Temp. Fluctuations Part II

- **In fact, the efficient circulatory systems of birds and mammals can be thought of as adaptations to endothermy and a high metabolic rate.**

END

Animal Form & Function II

Module 74

Temperature Regulations in Fishes

Temperature Regulations in Fishes

- The temperature of the surrounding water determines the body temperature of most fishes.
- How Fishes live in extremely cold water?
- 1. They have “antifreeze” materials in their blood in the form of Polyalcohols (e.g., sorbitol, glycerol) or water-soluble peptides and glycopeptides.
-

Temperature Regulations in Fishes

- Because these substances lower the freezing point of blood plasma and other body fluids.
- 2. These fishes also have proteins or protein-sugar compounds that inhibit the growth of ice crystals that begin to form.

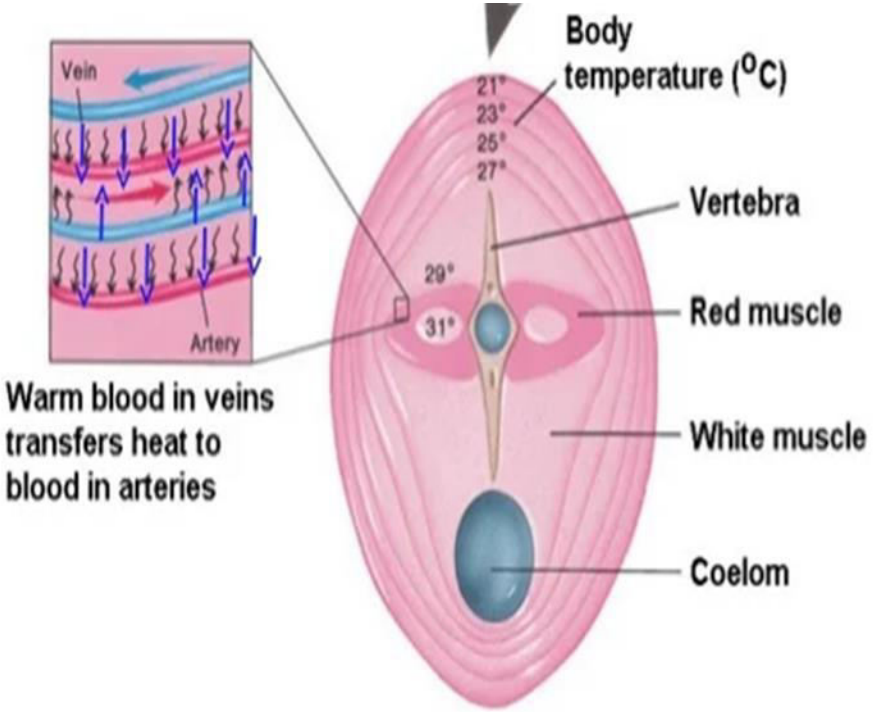
Temperature Regulations in Fishes

- 3. These adaptations enable these fishes to stay flexible and swim freely in a super-cooled state (i.e., at a temperature below the normal freezing temperature of a surrounding water).

Temperature Regulations in Fishes

- 4. These fishes have major blood vessels just under the skin.
- Branches deliver blood to the deeper, powerful, red swimming muscles, where smaller vessels are arranged in a countercurrent heat exchanger called the rete mirabile.

Temperature Regulations in Fishes



- A rete mirabile is a network of arteries and veins lying very close to each other, found in some vertebrates, including some fishes.
- The rete mirabile provides countercurrent blood flow within the net to act as a countercurrent exchanger.

Temperature Regulations in Fishes

- The heat generated by red muscles is not lost but it is transferred in the rete mirabile from venous warm blood to cold arterial blood.
- Some active fishes maintain a core temperature significantly above the temperature of the water.
- For example: bluefin tuna and the great white shark.

Temperature Regulations in Fishes

- This arrangement of blood vessels enhances vigorous activity by keeping the swimming muscles several degrees warmer than the tissue near the surface of the fish.
- This system has been adaptive for these fishes.

End

Animal Form & Function II

Module 75

Temperature Regulation in Amphibians

Temperature Regulations in Amphibians

- Surrounding medium of most the amphibians is air as well as water.
- They face marked daily and seasonal temperature changes.
- Most of these animals derive heat from their environment.
- Their body temperatures vary with external temperatures, i.e. they are ectothems.

Temperature Regulations in Amphibians

- Most amphibians have difficulty in controlling body temperature because they produce little heat metabolically and rapidly lose most of it from their body surfaces.
- However, behavioral adaptations enable them to maintain their body temp. within a homeostatic range most of the time.

Temperature Regulations in Amphibians

- Amphibians have an additional thermoregulatory problem because they must exchange oxygen and carbon dioxide across their skin surface, and this moisture layer acts as a natural evaporative cooling system.
- This problem of heat loss through evaporation limits the habitats and activities of amphibians to warm, moist areas.

Temperature Regulations in Amphibians

- Some amphibians, such as bullfrogs, can vary the amount of mucus they secrete from their body surface.
- This is as physiological response that helps regulate evaporative cooling.

Temperature Regulations in Amphibians





Temperature Regulations in Amphibians

Animal Form & Function II

Module 76

Temperature Regulations in Reptiles

Temperature Regulations in Reptiles

- Reptiles have dry rather than moist skin, which reduces the loss of body heat through evaporative cooling of the skin.
- They also have an expandable rib cage, which allows for more powerful and efficient ventilation
- Reptiles have a low metabolic rate and warm themselves by behavioral adaptations.

Temperature Regulations in Reptiles

- In addition, some of the more sophisticated regulatory mechanisms found in mammals are first found in reptiles.
- For example, diving reptiles conserve body heat by routing blood through circulatory shunts (an alternative path for the passage of the blood) into the center of the body. (e.g., sea turtles, sea snakes)

Temperature Regulations in Reptiles

- These animals can also increase heat production in response to the hormones thyroxine and epinephrine.
- In addition, tortoises and land turtles can cool themselves through salivating and frothing at the mouth, urinating on the back legs, moistening the eyes, and panting.

Temperature Regulations in Reptiles









Animal Form & Function II

Module 77

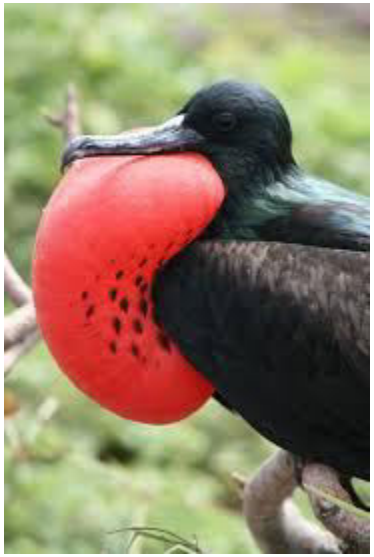
Animal Form & Function II

Temperature Regulation in Aves or Birds

Temperature Regulations in Birds

- **Birds** like mammals are the most active and behaviorally complex vertebrates.
- They can live in habitats all over the earth because they are homeothermic endotherms.
- They can maintain body temperatures between 35 and 42° C with metabolic heat.

Temperature Regulations in Birds







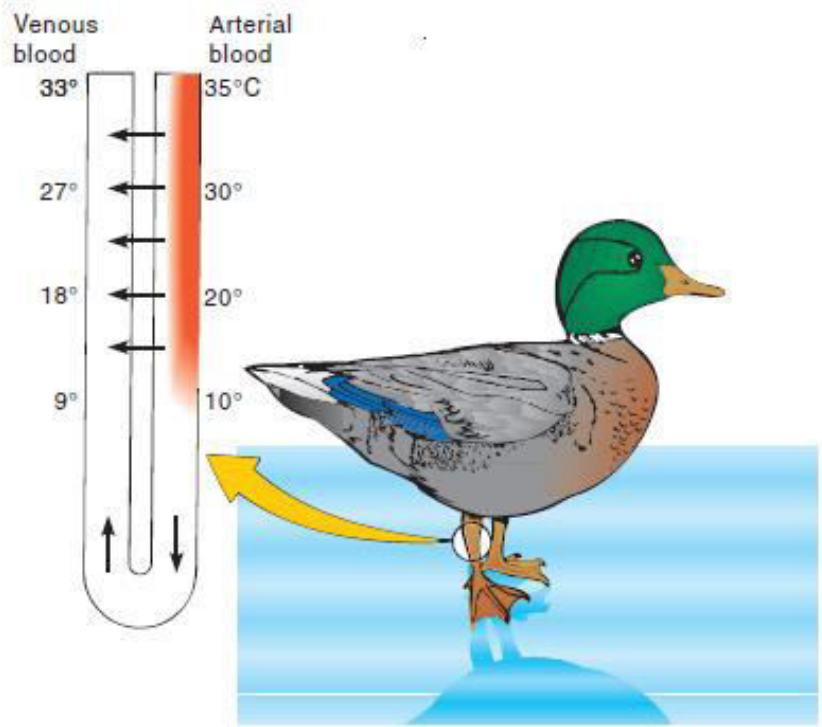
- Various cooling mechanisms prevent excessive warming in birds. Because they have no sweat glands, birds pant to lose heat through evaporative cooling.
- Some species have a highly vascularized pouch (gular pouch) in their throat that they can flutter to increase evaporation from the respiratory system.

Temperature Regulations in Birds



- Some birds possess mechanisms for preventing heat loss.
- Feathers are excellent insulators for the body, especially downy feathers that trap a layer of air next to the body to reduce heat loss from the skin.

Temperature Regulations in Birds



- Aquatic species, who lose heat from their legs and feet, have peripheral countercurrent heat exchange vessels called a rete mirabile in their legs to reduce heat loss.
- The arteries carry warm blood down the legs to warm the cooler blood in the veins, so that the heat is carried back to the body rather than lost through the feet that are in contact with a cold surface.

Animal Form & Function II

Module 78

Animal Form & Function II

Temperature Regulation in Mammals

Temperature Regulations in Mammals

- Mammals like birds are also active and behaviorally complex vertebrates.
- They can live in habitats all over the earth because they are homeothermic endotherms.
- They can maintain body temperatures between 35 and 42° C with metabolic heat.

Temperature Regulations in Mammals





- **Mammals that live in cold regions, such as the arctic fox and barren-ground caribou, also have these exchange vessels in their extremities (e.g., legs, tails, ears, nose).**

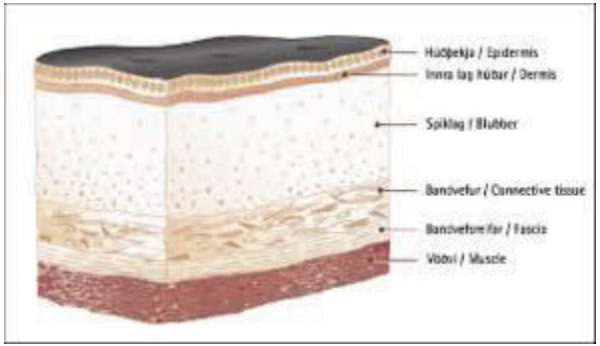
Temperature Regulations in Mammals



- **Animals in hot climates, such as jackrabbits, have mechanisms (e.g., large ears) to get rid of excess heat from the body surface.**

Temperature Regulations in Mammals





- Marine animals, such as seals and whales, thick pelts and a thick layer of insulating fat called blubber just under the skin help to maintain a body temperature around 36 to 38°C
- In the tail and flippers, which have no blubber, a countercurrent system of arteries and veins helps minimize heat loss.

Temperature Regulations in Mammals

- Birds and mammals also use behavioral mechanisms to cope with external temperature changes.
- Like ectotherms, they sun themselves or seek shade as the temperature fluctuates.
- Many animals cluster to keep warm; others share burrows for protection from temperature extremes.

Temperature Regulations in Mammals

- Migration to warm climates and hibernation enable many different birds and mammals to survive the harsh winter months.
- Others, such as the desert camel, have a multitude of evolutionary adaptations for surviving in some of the hottest and driest climates on earth.

Temperature Regulations in Mammals

- Camels are well adapted for survival in the desert. Their adaptations include
- large, flat feet - to spread their weight on the sand.
- thick fur on the top of the body for shade, and
- thin fur elsewhere to allow easy heat loss.
- It excrete very little of urine and dry dung. It has hump which stores fat.

End

Animal Form & Function II

Module 79

Heat Production in Birds

Heat Production in Birds

- In birds, heat generation can warm the body as it dissipates through the tissues and organs.
- Birds like mammals can generate heat by muscle contraction (thermogenesis).
- Every time a muscle cell contracts, the actin and myosin filaments sliding over each other and the hydrolysis of ATP molecules generate heat.

Heat Production in Birds

- Both voluntary muscular work (e.g., running, flying, jumping) and involuntary muscular work (e.g., shivering) generate heat.
- Heat generation by shivering is called **shivering thermogenesis.**

Heat Production in Birds

- When birds's body cools, the thyroid gland releases the hormone thyroxine.
- Thyroxine increases the permeability of many cells to sodium (Na) ions, which leak into the cells.
- The ATPase pump quickly pumps these ions out.
- In this process, ATP is hydrolyzed, releasing heat energy.

Heat Production in Birds

- This hormonal triggering of heat production is called non-shivering thermogenesis.
- The basal metabolic rate (BMR) of birds and mammals is high and also produces heat involuntarily but useful by-product.

End

Animal Form & Function II

Module 80

Heat Production in Mammals

Heat Production in Mammals

- In endotherms (birds and mammals), heat generation can warm the body as it dissipates through the tissues and organs.
- Mammals like birds can generate heat by muscle contraction (thermogenesis).
- Every time when a muscle cell contracts, the actin and myosin filaments sliding over each other the hydrolysis of ATP molecules generate heat.

Heat Production in Mammals

- Mammals that live in cold climates, and hibernate have a specialized type of fat called Brown fat found in newborn babies.
- The brown color of this fat comes from the large number of mitochondria with their iron-containing cytochromes.

Heat Production in Mammals

- Deposits of brown fat are beneath the ribs and in the shoulders.
- A large amount of heat is produced when brown fat cells oxidize fatty acids.
- Blood flowing past brown fat is heated and contributes to warming the body.

Heat Production in Mammals

- **The BMR of mammals is high and also produces heat involuntarily but useful by-product.**

Heat Production in Mammals





- During winter, various endotherms (e.g., bats, woodchucks, ground squirrels) go into hibernation .
- During hibernation the metabolic rate slows, as do the heart and breathing rates.
- Mammals prepare for hibernation by building up fat reserves and growing long winter pelts.

Heat Production in Mammals

- **Hibernation** is a state of inactivity and metabolic depression in endotherms.
- Hibernation is a seasonal heterothermy characterized by low body-temperature, slow breathing and heart-rate, and low metabolic rate.
- It occurs most commonly during winter months.

Heat Production in Mammals

- Aestivation is a state of animal dormancy, similar to hibernation, although taking place in the summer rather than the winter.
- Aestivation is characterized by inactivity and a lowered metabolic rate, that is entered in response to high temperatures and arid conditions.

Heat Production in Mammals

- **Aestivation takes place during times of heat and dryness, the hot dry season, which are often the summer months.**

End

Animal Form & Function II

Module 81

Hibernation and Aestivation

Hibernation and Aestivation

- **Hibernation** is a state of seasonal inactivity in animals characterized by low body-temperature, slow breathing and heart-rate, and low metabolic rate.
- It commonly occurs during winter months.

Hibernation and Aestivation

- Hibernation functions to conserve energy when sufficient food is unavailable.
- To achieve this energy saving, an endothermic animal decreases its metabolic rate and thereby its body temperature.

Hibernation and Aestivation

- Hibernation may last days, weeks, or months - depending on:
- the species,
- ambient temperature,
- time of year, and
- the individual's body-condition.

Hibernation and Aestivation

- Before entering hibernation, animals need to store enough energy to last through the duration of their dormant period, possibly as long as an entire winter.
- Larger species become hyperphagic, eating a large amount of food and storing the energy in fat deposits.

Hibernation and Aestivation

- Some species of mammals hibernate while pregnant.
- Babies are born either during the hibernation of the mother or shortly afterwards.
- For example, female polar-bears go into hibernation during the cold winter months in order to give birth to their offspring.

Hibernation and Aestivation

- Aestivation is a state of animal dormancy, similar to hibernation, although taking place in the summer rather than the winter.
- Aestivation is characterized by inactivity and a lower metabolic rate, in response to high temperatures and arid conditions.

Hibernation and Aestivation

- It takes place during times of heat and dryness, the hot dry season, which are often the summer months.
- The "baseline" metabolic rate of an animal is measured as the basal metabolic rate (BMR) for an endotherm or as the standard metabolic rate (SMR) for an ectotherm.

Hibernation and Aestivation

- **Torpor** is a state of decreased physiological activity in an animal, usually by a reduced body temperature and metabolic rate.

Hibernation and Aestivation

- Animals that undergo daily torpor include birds (even tiny hummingbirds, and some mammals, including many marsupial species, rodent species (such as mice), and bats.

Hibernation and Aestivation









Some animals, such as badgers, bears, opossums, raccoons, and skunks, enter a state of prolonged sleep in the winter. Since their body temperature remains near normal, this is not true hibernation.

Animal Form & Function II

Module 82

Osmoregulation and Excretion

(Control of Water and Solutes)

Osmoregulation and Excretion

Osmoregulation

- **Osmoregulation is the process by which an organism regulates the water balance in its body to maintain homeostasis.**
- **Osmoregulation is necessary for animals in all habitats.**

Osmoregulation and Excretion

- If the osmotic concentration of the body fluids of an animal equals that of the medium *(the animal's environment)*, the animal is an osmoconformer.
- This type of osmoregulation is not efficient and has limited the distribution of those animals using it.

Osmoregulation and Excretion



- Osmoconformers are marine organisms that maintain an internal environment which is isotonic to their external environment.
- This means that the osmotic pressure of the organism's cells is equal to the osmotic pressure of their surrounding environment.
- Examples: hagfish, skates and sharks.

Osmoregulation and Excretion

Osmoregulator.

- In contrast, an animal that maintains its body fluids at a different osmotic concentration from that of its surrounding environment is an omoregulator.
- Examples. Some fish have evolved osmoregulatory mechanisms to survive in all kinds of aquatic environments.

Osmoregulation and Excretion

- Land animals have a higher concentration of water in their fluids than in the surrounding air.
- They tend to lose water to the air through evaporation and may use considerable amounts of water to dispose of wastes.

Osmoregulation and Excretion

- **The form and function of organs or systems associated with osmoregulation are related both to environmental conditions (saltwater, freshwater, terrestrial) and to body size (especially the surface-to-volume ratio).**

END

Animal Form & Function II

Module 83

Animal Form & Function II

Excretion in Animals

Excretion in Animals

Excretion

- Excretion can be defined broadly as the elimination of metabolic waste products from an animal's body.
- These products include CO_2 and H_2O (which cellular respiration produces) and Nitrogen products.

Excretion in Animals

- Excess nitrogen is produced as:
- Ammonia from deamination of amino acids.
- Urea and uric acid and solutes (various ions).
- Some of Nitrogen is excreted as ceatinine and creatine .

Excretion in Animals

- Excretion of nitrogenous wastes is usually associated with the regulation/supply of water and solute (ionic) balance by a physiological process called osmoregulation.
- The nature of excretory product is determined mainly by the availability of water to the organism i.e. its habitat.

Excretion in Animals

This correlation is summarized as:

Ammonia	*aquatic*
Urea	*aquatic/terrestrial*
Uric acid	*terrestrial*

- Ammonia is extremely toxic and must be eliminated rapidly and safely. This applies only to freshwater organisms.
- Urea is produced by animals who have moderate water supply. Such as amphibians and human.

Excretion in Animals

- **Uric acid and its salts are ideal products for terrestrial organisms, for example insects and birds.**

END

Animal Form & Function II

Module 84

Animal Form & Function II

Invertebrate Excretory Systems

Invertebrate Excretory Systems

- Aquatic invertebrates occur in a wide range of media, from freshwater to markedly hypersaline water (e.g., salt lakes).
- Generally, marine invertebrates have about the same osmotic concentration as seawater (i.e., they are osmoconformers).
- It means they need not to osmoregulate.

Invertebrate Excretory Systems

Most water and ions are gained:

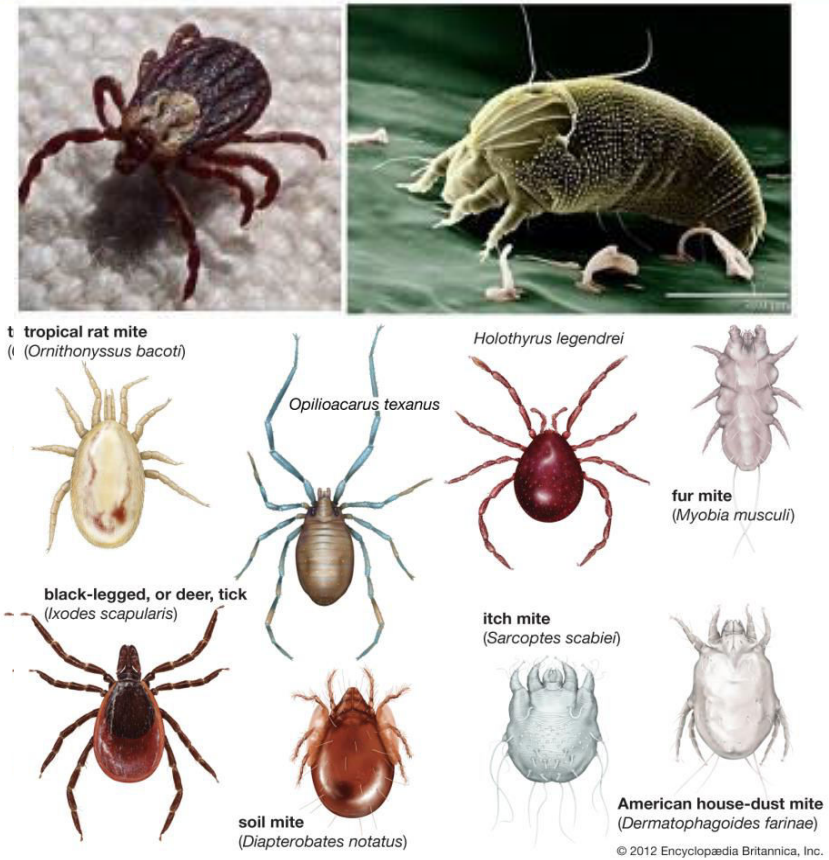
- across the integument,
- via gills,
- by drinking,
- and in food.

- **Ions and wastes are mostly lost by diffusion via the integument, gills, or urine.**

Invertebrate Excretory Systems

- Freshwater invertebrates are strong osmoregulators.
- Any water gain is usually eliminated as urine.

Invertebrate Excretory Systems



- A number of invertebrate taxa have more or less successfully invaded terrestrial habitats.
- The most successful terrestrial invertebrates are the Arthropods, particularly the insects, spiders, scorpions, ticks, mites, centipedes, and millipedes.

Invertebrate Excretory Systems

- Overall, the water and ion balance of terrestrial invertebrates is quite different from that of aquatic animals because terrestrial invertebrates face limited water supplies and water loss by evaporation from their integument.
- Some of the invertebrate excretory mechanisms and systems are now discussed.

Invertebrate Excretory Systems

- **Mites can feed on plants and animals while ticks strictly live on animal hosts.**
- **Both ticks and mites transmit disease to their hosts.**
- **Mites transmit two major diseases including scrub typhus, and scabies.**

END

Animal Form & Function II

Module 85

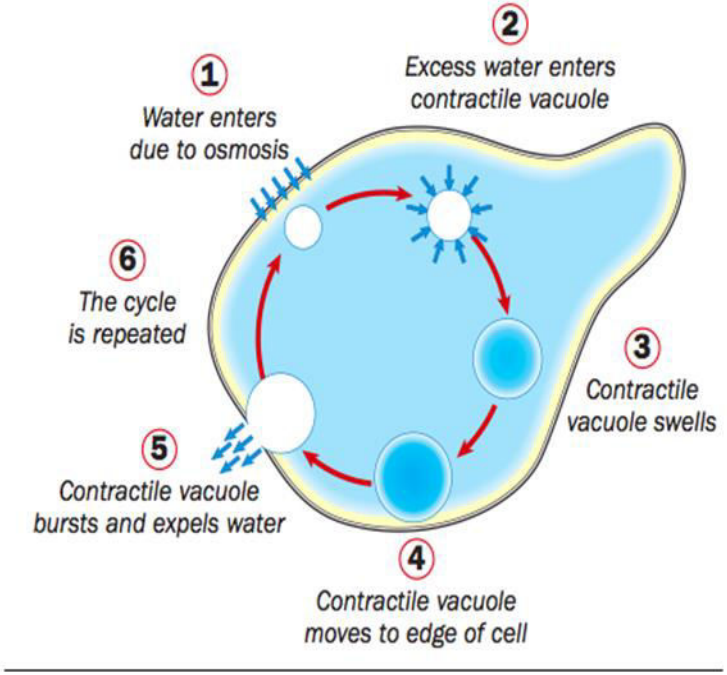
Animal Form & Function II

Contractile Vacuoles

Contractile Vacuoles

- **The contractile vacuole** is a specialized type of vacuole that regulates the quantity of water inside a cell.
- A contractile vacuole is a sub-cellular structure involved in osmoregulation.
- The contractile vacuole, as its name suggests, expels water out of the cell by contracting.

Contractile Vacuoles



- In freshwater environments, the concentration of solutes is hypotonic, higher inside than outside the cell.
- Many freshwater species (protozoa, sponges), have contractile vacuoles that pump out excess water.

Contractile Vacuoles

- Contractile vacuoles are energy-requiring devices that expel excess water from individual cells exposed to hypo-osmotic environments

Contractile Vacuoles

- **Some protists and marine invertebrates (e.g., protozoa, cnidarians, echinoderms, sponges) do not have specialized excretory structures because wastes simply diffuse into the surrounding isoosmotic water.**

END

Final Term Syllabus
Lecture 01 to 85

Efforts By

AARISH ALI (+96569002641)

Contact us to join our esteemed group

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Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

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